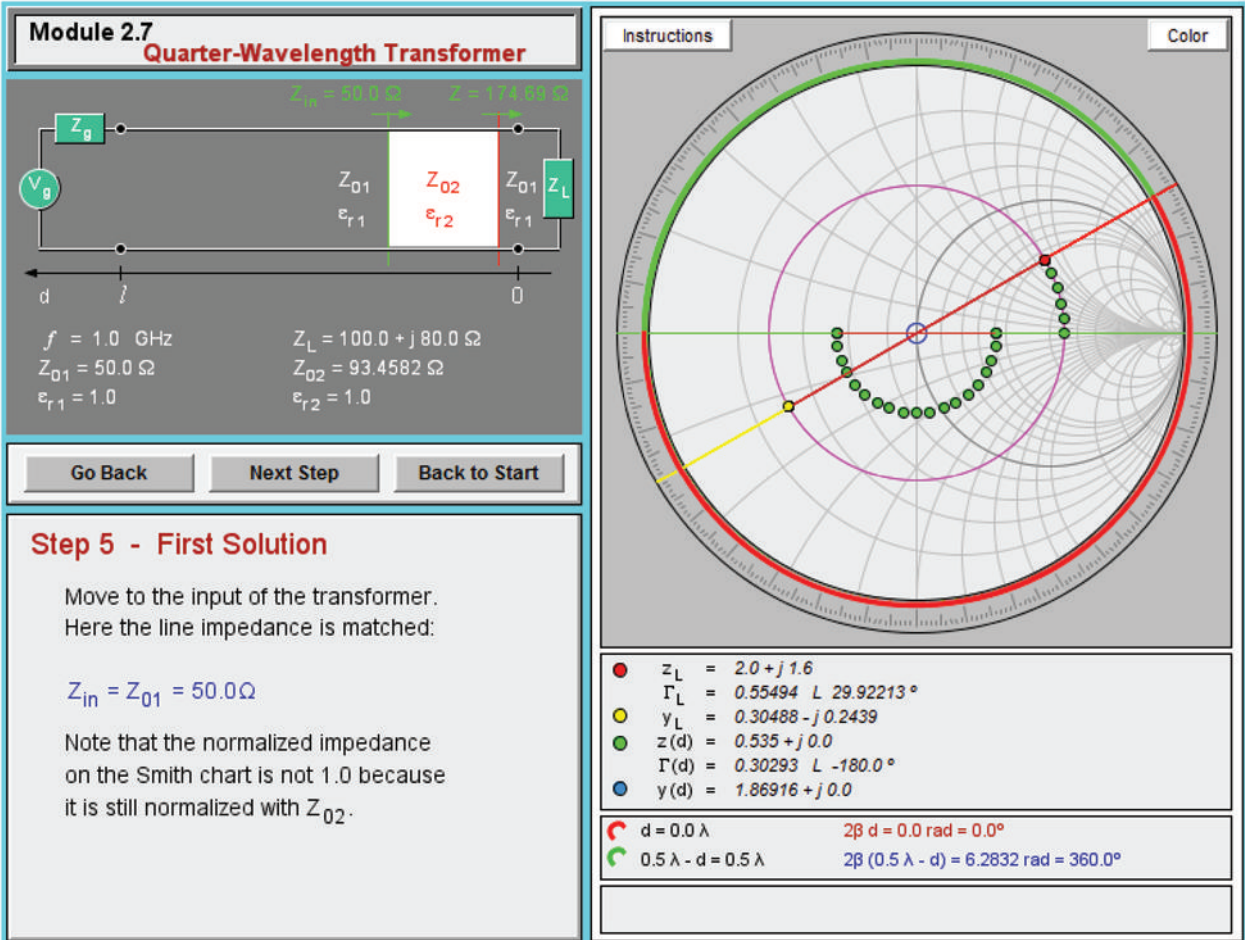
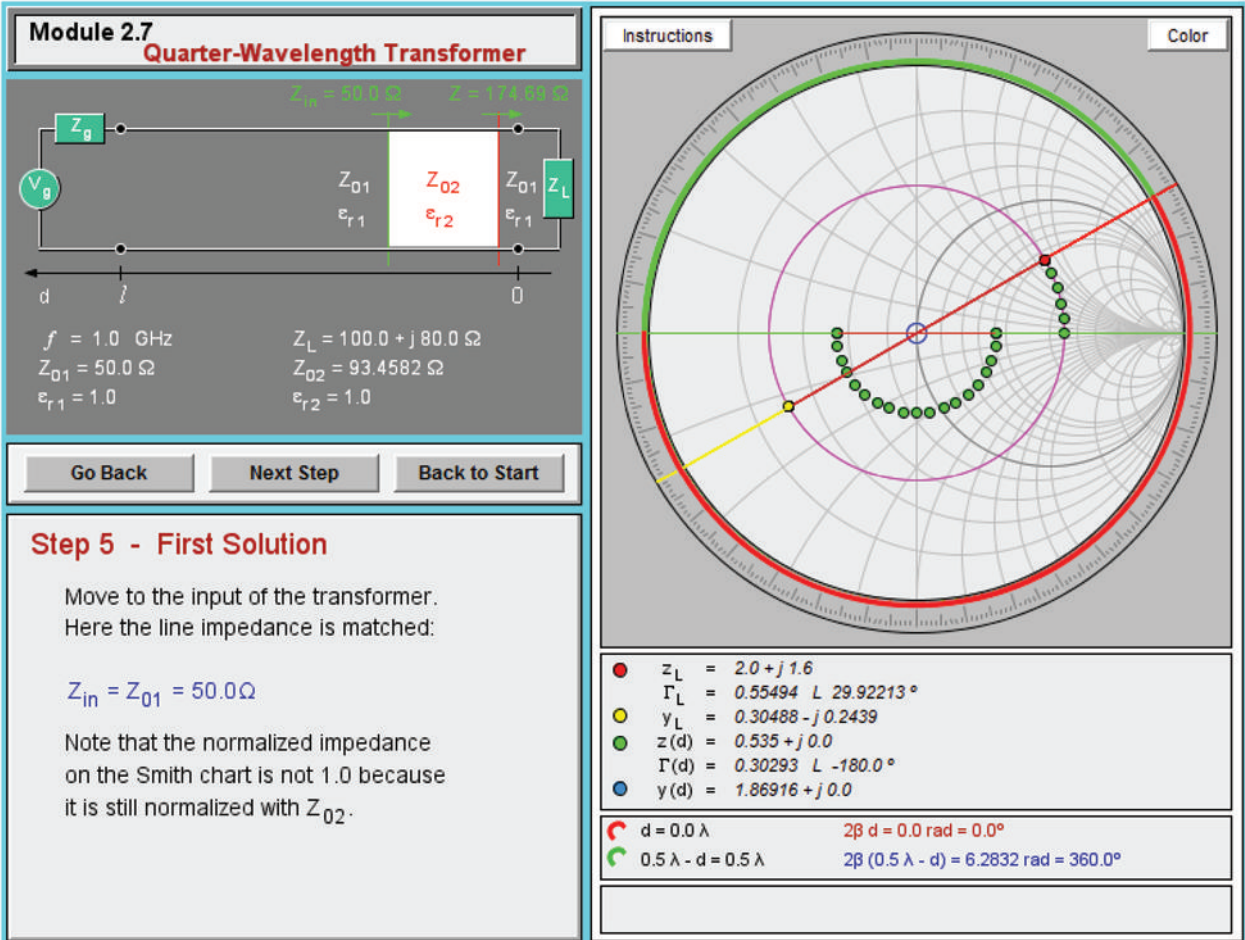


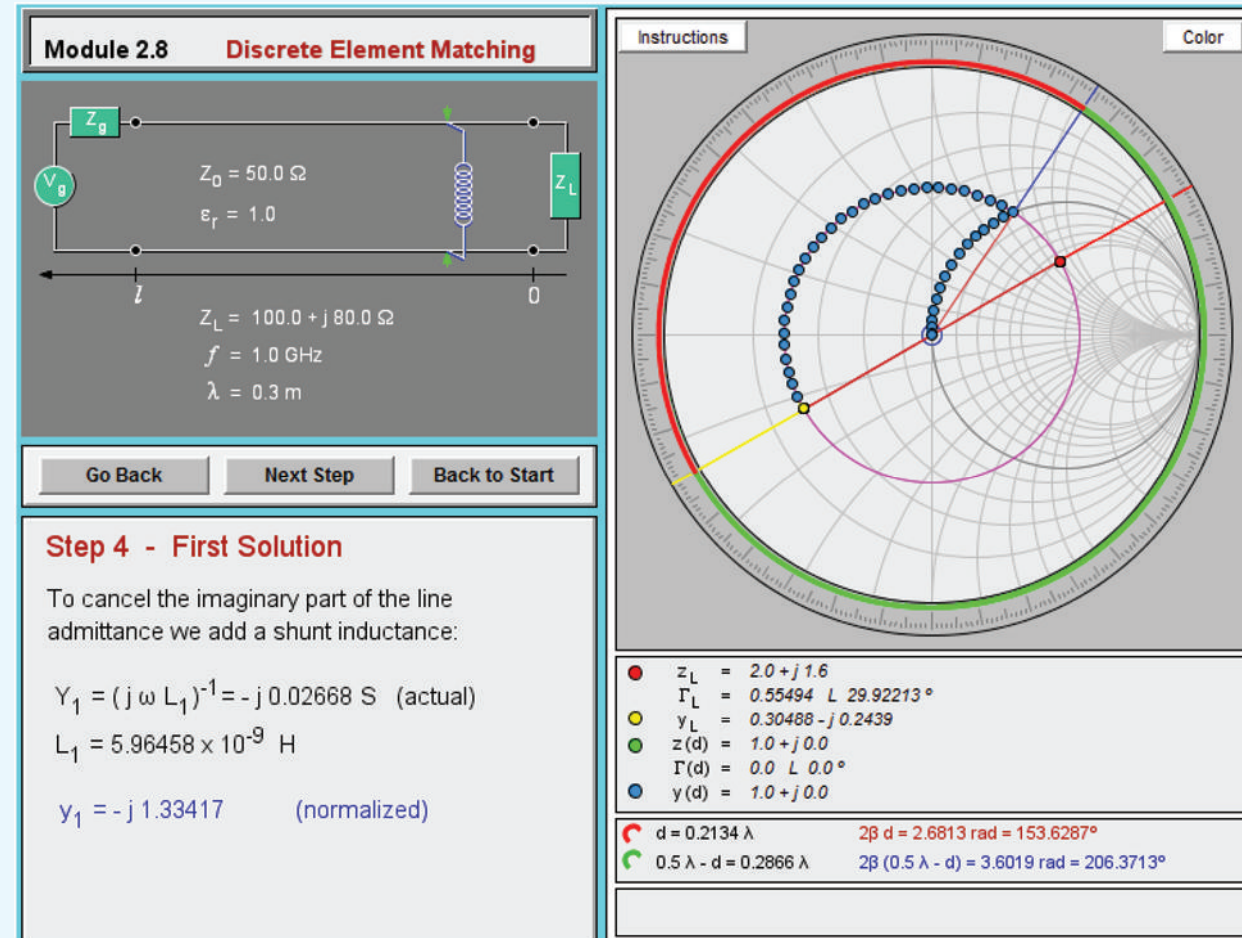
Module 2.7 Quarter-Wavelength Transformer This module allows you to go through a multi-step procedure to design a quarter-wavelength transmission line that, when inserted at the appropriate location on the original line, presents a matched load to the feedline.



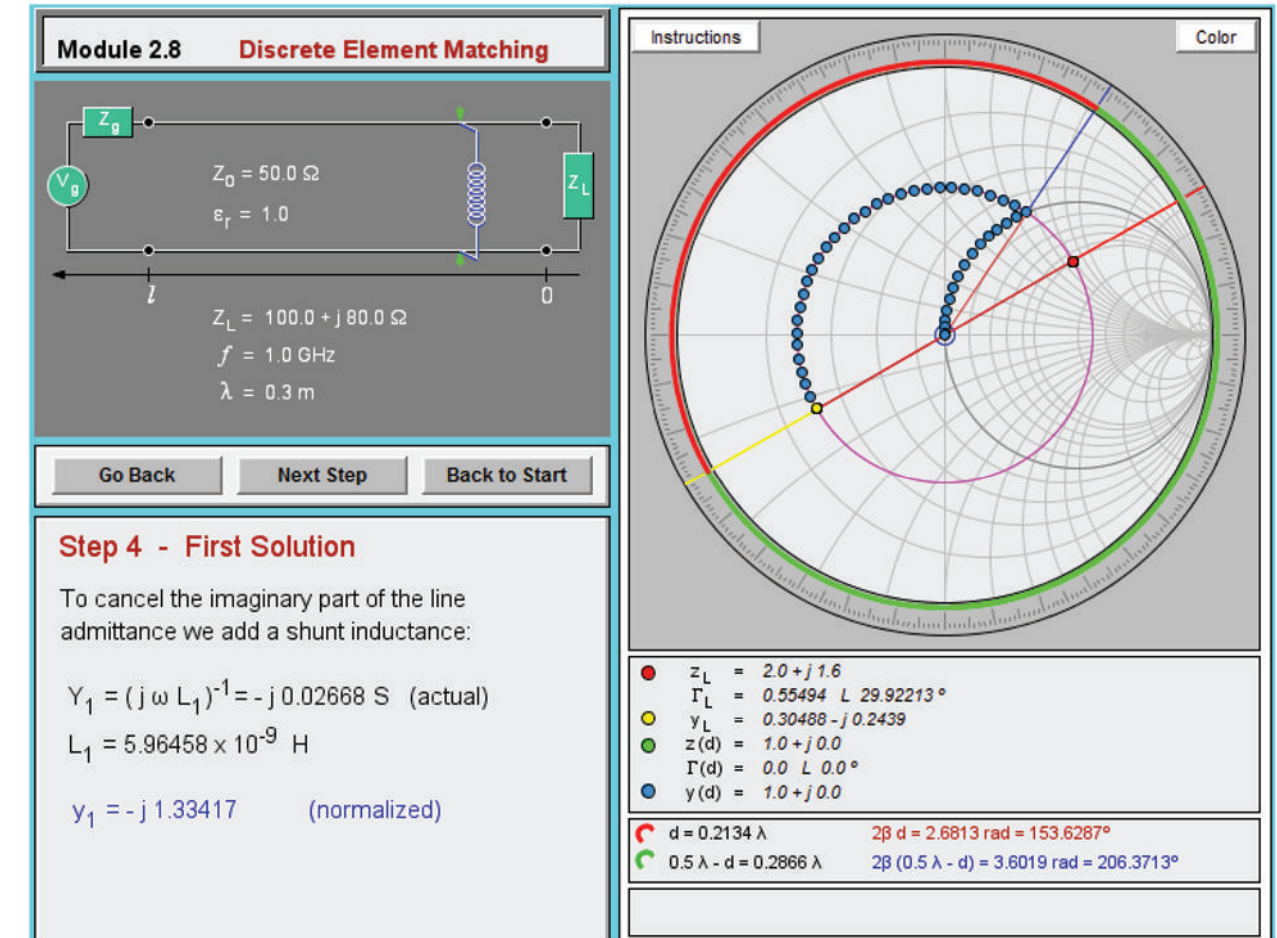
模块 2.7 四分之一波长变压器 该模块允许您通过多步骤程序来设计四分之一波长传输线，当将其插入到原始线路上的适当位置时，可为馈线提供匹配的负载。



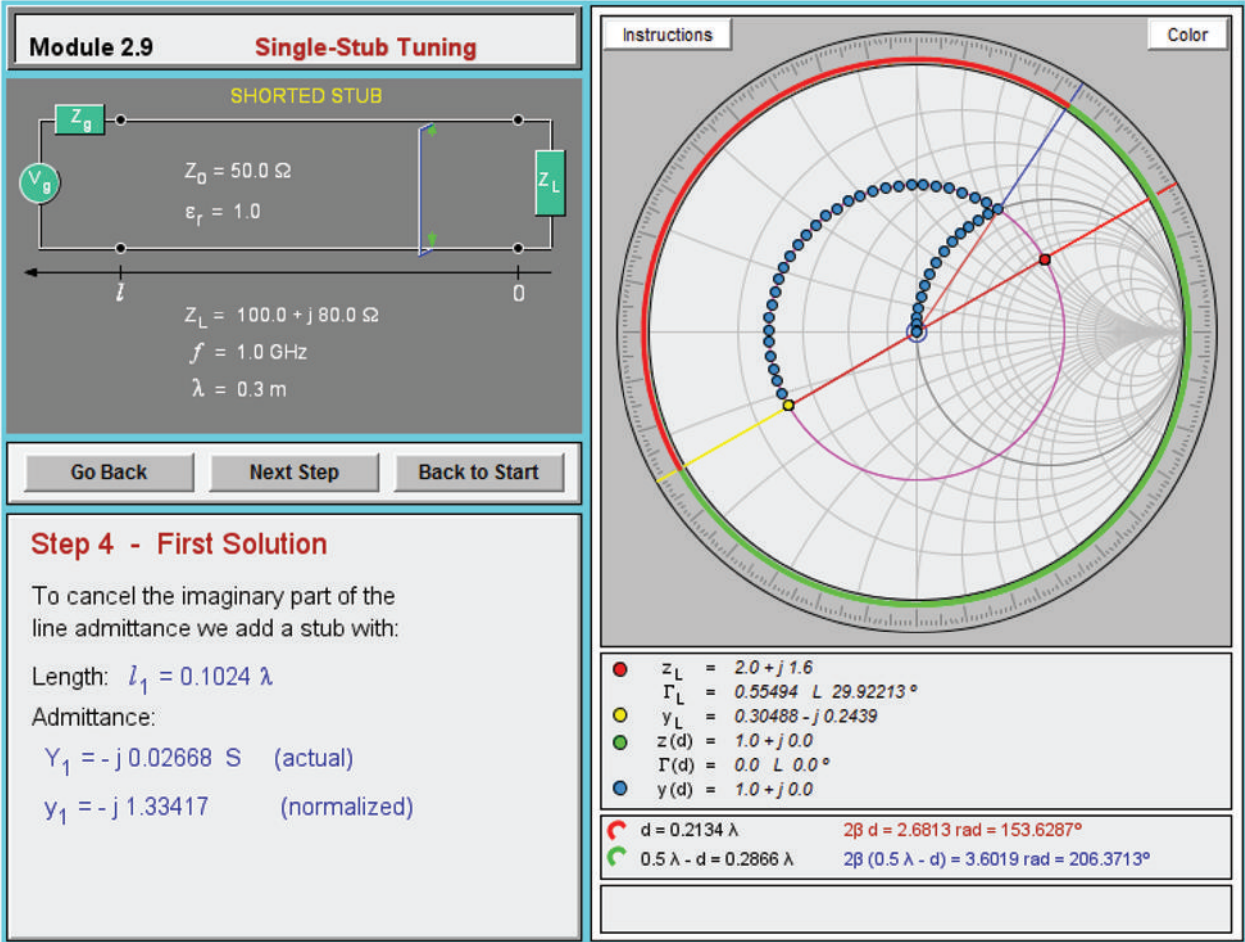
Module 2.8 Discrete Element Matching For each of two possible solutions, the module guides the user through a procedure to match the feedline to the load by inserting a capacitor or an inductor at an appropriate location along the line.



模块 2.8 离散元件匹配 对于两种可能的解决方案中的每一种，该模块都会引导用户完成一个程序，通过在馈线沿线的适当位置插入电容器或电感器来将馈线与负载进行匹配。



Module 2.9 Single-Stub Tuning Instead of inserting a lumped element to match the feedline to the load, this module determines the length of a shorted stub that can accomplish the same goal.



2-12 Transients on Transmission Lines

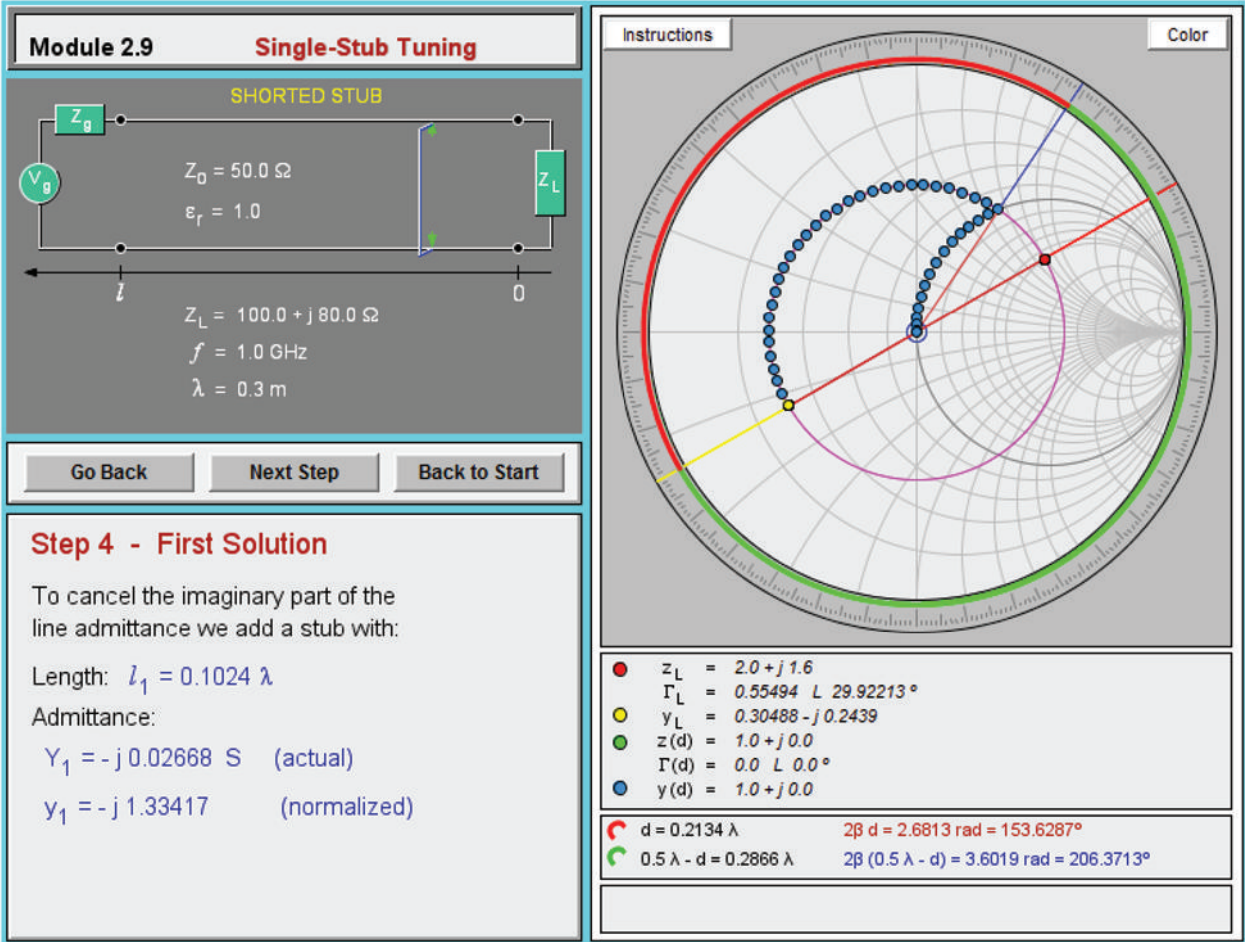
Thus far, our treatment of wave propagation on transmission lines has focused on the analysis of single-frequency, time-harmonic signals under steady-state conditions. The impedance-matching and Smith chart techniques we developed, while useful for a wide range of applications, are inappropriate for dealing with digital or wideband signals that exist in digital chips, circuits, and computer networks. For such signals, we need to examine the transient transmission line response instead.

► The **transient response** of a voltage pulse on a transmission line is a time record of its back-and-forth travel between the sending and receiving ends of the line, taking into account all the multiple reflections (echoes) at both ends. ◀

Let us start by considering the case of a single rectangular pulse of amplitude V_0 and duration τ , as shown in Fig. 2-41(a). The amplitude of the pulse is zero prior to $t = 0$, V_0 over the interval $0 \leq t \leq \tau$, and zero afterwards. The pulse can be described mathematically as the sum of two unit step functions:

$$V(t) = V_1(t) + V_2(t) = V_0 u(t) - V_0 u(t - \tau), \quad (2.143)$$

模块 2.9 单短截线调整 该模块不是插入集总元件来匹配馈线与负载，而是确定可以实现相同目标的短路短截线的长度。



2-12 传输线上的瞬变

到目前为止，我们对传输线上波传播的处理主要集中在稳态条件下单频时谐信号的分析。我们开发的阻抗匹配和史密斯圆图技术虽然可用于广泛的应用，但不适合处理数字芯片、电路和计算机网络中存在的数字或宽带信号。对于此类信号，我们需要检查瞬态传输线响应。

► 传输线上电压脉冲的瞬态响应是其在传输线的发送端和接收端之间来回传输的时间记录，同时考虑了两端的所有多次反射（回声）。 ◀

让我们首先考虑幅度为 V_0 和持续时间为 τ 的单个矩形脉冲的情况，如图 2-41(a) 所示。脉冲幅度在 $t = 0$ 之前为零，在间隔 $0 \leq t \leq \tau$ 内为 V_0 ，之后为零。脉冲可以在数学上描述为两个单位阶跃函数的总和：

$$V(t) = V_1(t) + V_2(t) = V_0 u(t) - V_0 u(t - \tau), \quad (2.143)$$

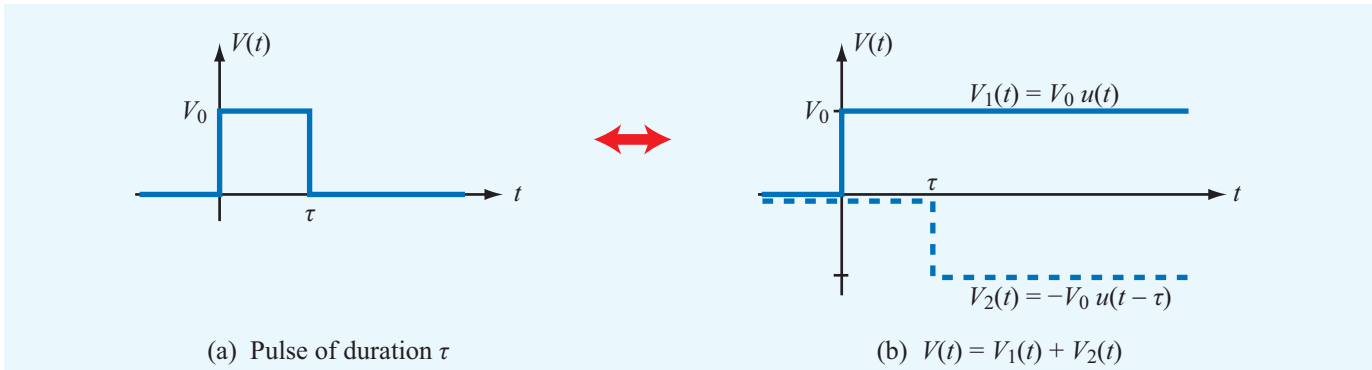


Figure 2-41 A rectangular pulse $V(t)$ of duration τ can be represented as the sum of two step functions of opposite polarities displaced by τ relative to each other.

where the unit step function $u(x)$ is

$$u(x) = \begin{cases} 1 & \text{for } x > 0, \\ 0 & \text{for } x < 0. \end{cases} \quad (2.144)$$

The first component, $V_1(t) = V_0 u(t)$, represents a dc voltage of amplitude V_0 that is switched on at $t = 0$ and then retains that value indefinitely. The second component, $V_2(t) = -V_0 u(t - \tau)$, represents a dc voltage of amplitude $-V_0$ that is switched on at $t = \tau$ and remains that way indefinitely. As can be seen from **Fig. 2-41(b)**, the sum $V_1(t) + V_2(t)$ is equal to V_0 for $0 < t < \tau$ and equal to zero for $t < 0$ and $t > \tau$. This representation of a pulse in terms of two step functions allows us to analyze the transient behavior of the pulse on a transmission line as the superposition of two dc signals. Hence, if we can develop basic tools for describing the transient behavior of a single step function, we can apply the same tools for each of the two components of the pulse and then add the results to obtain the response to $V(t)$.

2-12.1 Transient Response to a Step Function

The circuit shown in **Fig. 2-42(a)** consists of a generator, composed of a dc voltage source V_g and a series resistance R_g , connected to a lossless transmission line of length l and characteristic impedance Z_0 . The line is terminated in a purely resistive load R_L at $z = l$.

► Note that in previous sections $z = 0$ was defined as the location of the load; now it is more convenient to define it as the location of the source. ◀

The switch between the generator circuit and the transmission line is closed at $t = 0$. The instant the switch is closed,

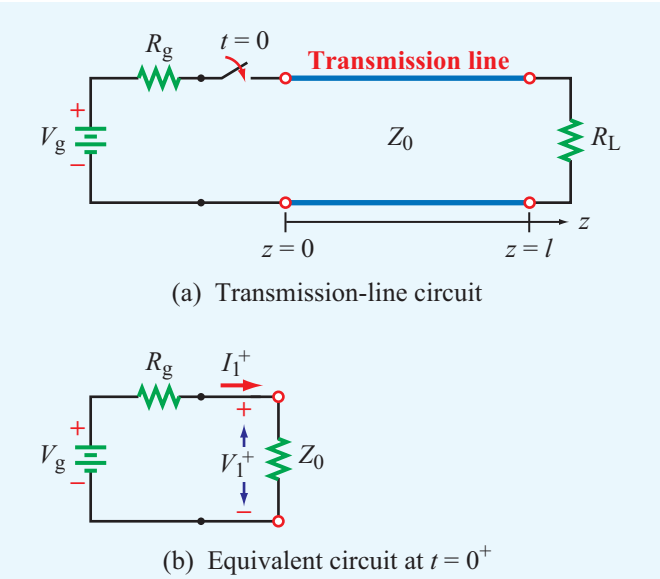


Figure 2-42 At $t = 0^+$, immediately after closing the switch in the circuit in part (a), the circuit can be represented by the equivalent circuit in part (b). This is because from that instant and until a reflection is received back from the load the generator circuit “sees” only an impedance Z_0 .

the transmission line appears to the generator circuit as a load with impedance Z_0 . This is because, in the absence of a signal on the line, the input impedance of the line is unaffected by the load impedance R_L . The circuit representing the **initial condition** is shown in **Fig. 2-42(b)**. The **initial current** I_1^+ and corresponding **initial voltage** V_1^+ at the sending end of the

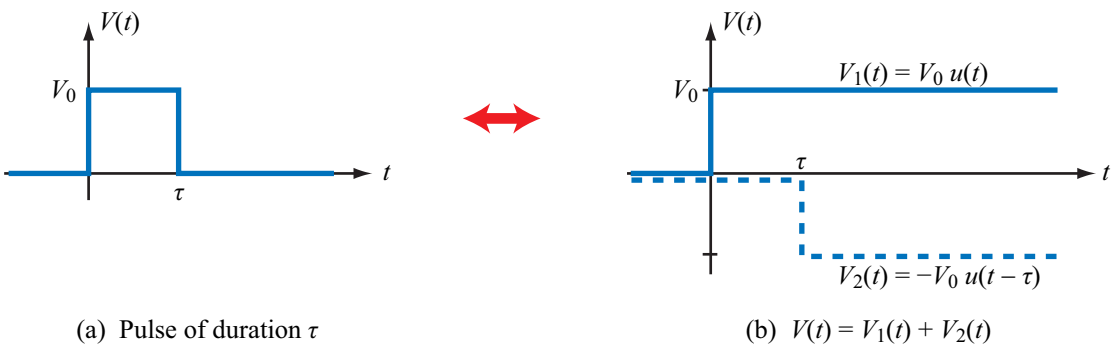


图 2-41 持续时间为 τ 的矩形脉冲 $V(t)$ 可以表示为两个极性相反的阶跃函数 $V_1(t)$ 和 $V_2(t)$ 的和。相对于彼此而言。

其中单位阶跃函数 $u(x)$ 为

$$u(x) = \begin{cases} 1 & \text{for } x > 0, \\ 0 & \text{for } x < 0. \end{cases} \quad (2.144)$$

第一个分量 $V_1(t) = V_0 u(t)$ 表示幅度为 V_0 的直流电压，该电压在 $t = 0$ 处接通，然后无限期地保留该值。第二个分量 $V_2(t) = -V_0 u(t - \tau)$ 表示幅度为 $-V_0$ 的直流电压，该电压在 $t = \tau$ 处接通并无限期地保持这种状态。从图2-41(b)可以看出，对于 $0 < t < \tau$ ，总和 $V_1(t) + V_2(t)$ 等于 V_0 ，对于 $t < 0$ 和 $t > \tau$ 等于 0。这种用两步函数表示脉冲的方式使我们能够将传输线上脉冲的瞬态行为分析为两个直流信号的叠加。因此，如果我们能够开发用于描述单阶函数瞬态行为的基本工具，我们就可以对脉冲的两个分量中的每一个应用相同的工具，然后将结果相加以获得对 $V(t)$ 的响应。

2-12.1 对阶跃函数的瞬态响应

图2-42(a)所示的电路由一个发生器组成，该发生器由直流电压源 V_g 和串联电阻 R_g 组成，连接到长度为 l 、特性阻抗为 Z_0 的无损传输线。该线路在 $z = l$ 处终止于纯电阻负载 R_L 。

► 请注意，在前面的章节中， $z = 0$ 被定义为负载的位置；现在将其定义为源的位置会更方便。◀

发电电路和输电线路之间的开关在 $t = 0$ 时闭合。开关闭合的瞬间，

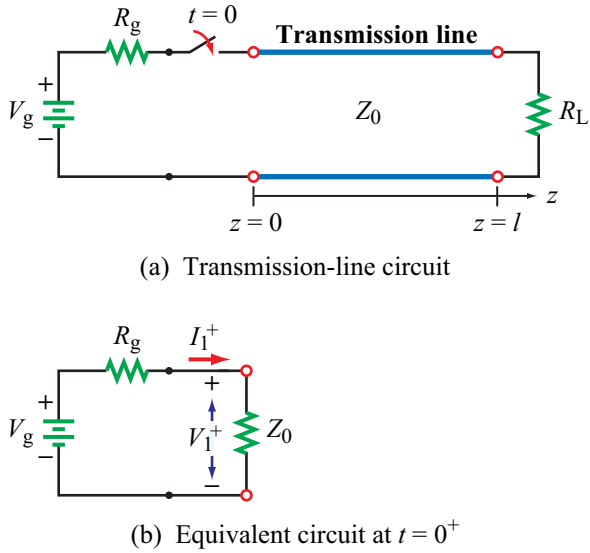
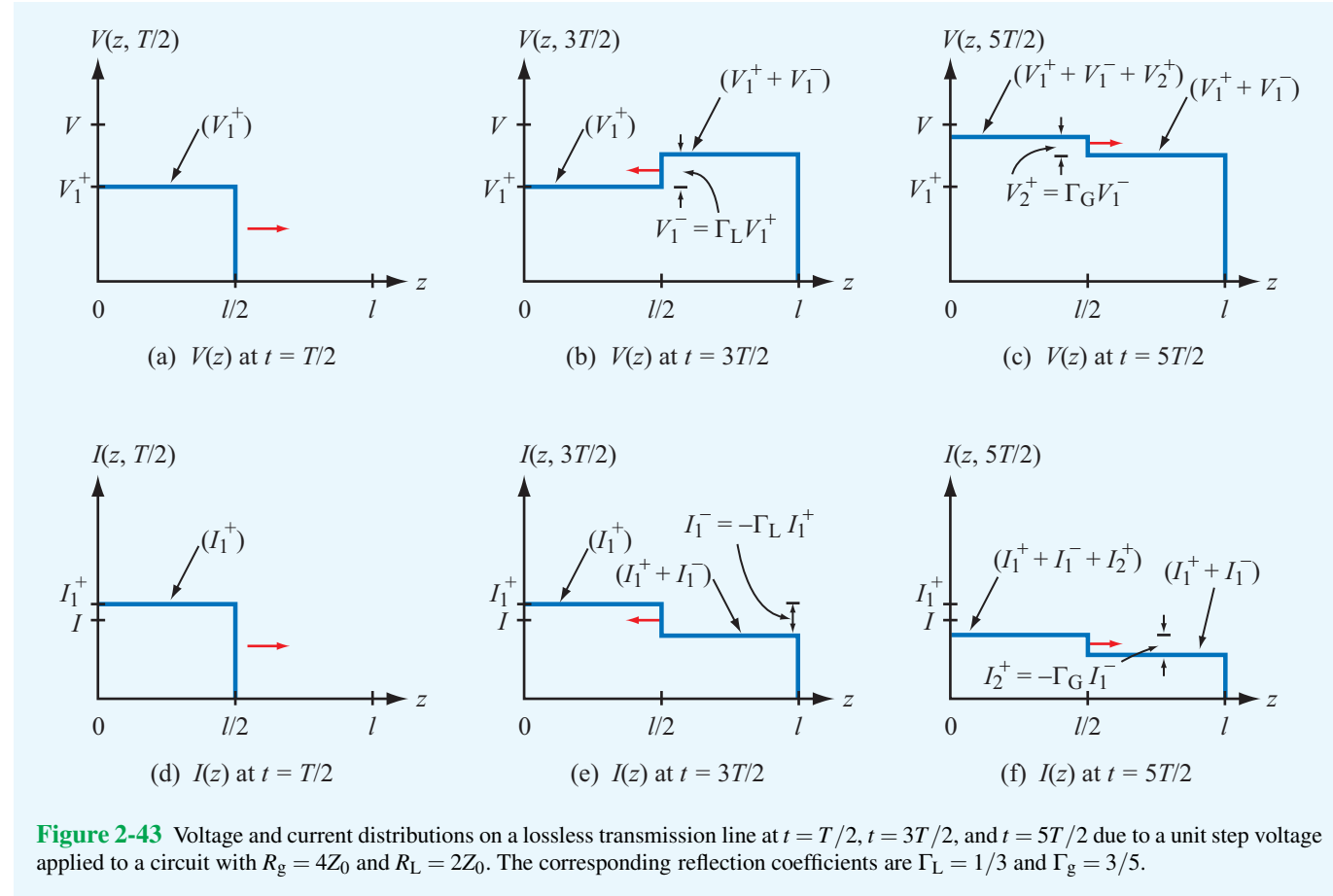


图2-42 在 $t = 0^+$ 时，(a)部分电路中的开关闭合后，该电路可以用(b)部分的等效电路表示。这是因为从那一刻起，直到从负载接收到反射为止，发生器电路仅“看到”阻抗 Z_0 。

传输线对于发生器电路来说是一个阻抗为 Z_0 的负载。这是因为，当线路上没有信号时，线路的输入阻抗不受负载阻抗 R_L 的影响。代表初始条件的电路如图2-42(b)所示。发送端的初始电流 I_1^+ 和对应的初始电压 V_1^+



transmission line are given by

$$I_1^+ = \frac{V_g}{R_g + Z_0}, \quad (2.145a)$$

$$V_1^+ = I_1^+ Z_0 = \frac{V_g Z_0}{R_g + Z_0}. \quad (2.145b)$$

The combination of V_1^+ and I_1^+ constitutes a wave that travels along the line with velocity $u_p = 1/\sqrt{\mu\epsilon}$ immediately after the switch is closed. The plus-sign superscript denotes the fact that the wave is traveling in the $+z$ direction. The transient response of the wave is shown in **Fig. 2-43** at each of three instances in time for a circuit with $R_g = 4Z_0$ and $R_L = 2Z_0$. The first response is at time $t_1 = T/2$, where $T = l/u_p$ is the time it takes the wave to travel the full length of the line. By time t_1 , the wave has traveled halfway down the line; consequently, the voltage on the first half of the line is equal to V_1^+ , while the voltage on the second half is still zero (**Fig. 2-43(a)**). At $t = T$, the wave reaches the load at $z = l$, and because $R_L \neq Z_0$, the

mismatch generates a reflected wave with amplitude

$$V_1^- = \Gamma_L V_1^+, \quad (2.146)$$

where

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} \quad (2.147)$$

is the reflection coefficient of the load. For the specific case illustrated in **Fig. 2-43**, $R_L = 2Z_0$, which leads to $\Gamma_L = 1/3$. After this first reflection, the voltage on the line consists of the sum of two waves: the initial wave V_1^+ and the reflected wave V_1^- . The voltage on the transmission line at $t_2 = 3T/2$ is shown in **Fig. 2-43(b)**; $V(z, 3T/2)$ equals V_1^+ on the first half of the line ($0 \leq z < l/2$) and $(V_1^+ + V_1^-)$ on the second half ($l/2 \leq z \leq l$).

At $t = 2T$, the reflected wave V_1^- arrives at the sending end of the line. If $R_g \neq Z_0$, the mismatch at the sending end generates a reflection at $z = 0$ in the form of a wave with

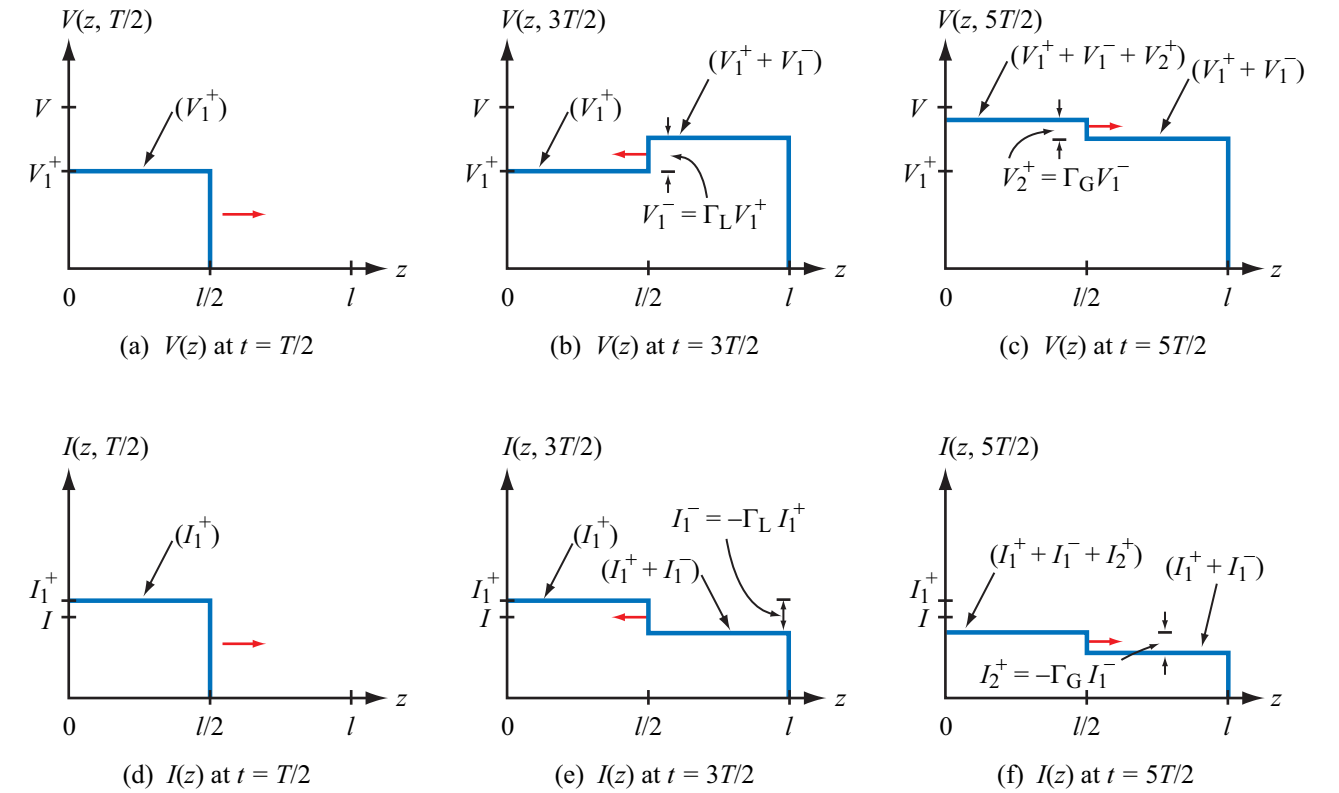


图 2-43 由于将单位阶跃电压施加到具有 $R_g = 4Z_0$ 和 $R_L = 2Z_0$ 的电路，无损传输线上在 $t = T/2$ 、 $t = 3T/2$ 和 $t = 5T/2$ 处的电压和电流分布。相应的反射系数为 $\Gamma_L = 1/3$ 和 $\Gamma_g = 3/5$ 。

传输线由下式给出

$$I_1^+ = \frac{V_g}{R_g + Z_0}, \quad (2.145a)$$

$$V_1^+ = I_1^+ Z_0 = \frac{V_g Z_0}{R_g + Z_0}. \quad (2.145b)$$

不匹配会产生具有振幅的反射波

$$V_1^- = \Gamma_L V_1^+, \quad (2.146)$$

在哪里

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} \quad (2.147)$$

V_1^+ 和 I_1^+ 的组合构成了一个波，在开关闭合后立即以速度 $u_p = 1/\sqrt{\mu\epsilon}$ 沿直线传播。加号上标表示波沿 $+z$ 方向传播。对于具有 $R_g = 4Z_0$ 和 $R_L = 2Z_0$ 的电路，波的瞬态响应如图 2-43 所示。第一个响应出现在时间 $t_1 = T/2$ ，其中 $T = l/u_p$ 是波走完全程所用的时间。到时间 t_1 时，波已沿线行进了一半；因此，线路前半部分的电压等于 V_1^+ ，而后半部分的电压仍然为零（图2-43 (a)）。在 $t = T$ 处，波到达 $z = l$ 处的负载，并且因为 $R_L \neq Z_0$ ，

是负载的反射系数。对于图 2-43 所示的具体情况， $R_L = 2Z_0$ ，导致 $\Gamma_L = 1/3$ 。经过第一次反射后，线路上的电压由两个波之和组成：初始波 V_1^+ 和反射波 V_1^- 。 $t_2 = 3T/2$ 时传输线上的电压如图2-43(b)所示； $V(z, 3T/2)$ 等于该行前半部分 ($0 \leq z < l/2$) 上的 V_1^+ 和后半部分 ($l/2 \leq z \leq l$) 上的 $(V_1^+ + V_1^-)$ 。

在 $t = 2T$ 时，反射波 V_1^- 到达线路的发送端。如果 $R_g \neq Z_0$ ，发送端的不匹配会在 $z = 0$ 处以波的形式产生反射

voltage amplitude V_2^+ given by

$$V_2^+ = \Gamma_g V_1^- = \Gamma_g \Gamma_L V_1^+, \quad (2.148)$$

where

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} \quad (2.149)$$

is the reflection coefficient of the generator resistance R_g . For $R_g = 4Z_0$, we have $\Gamma_g = 0.6$. As time progresses after $t = 2T$, the wave V_2^+ travels down the line toward the load and adds to the previously established voltage on the line. Hence, at $t = 5T/2$, the total voltage on the first half of the line is

$$V(z, 5T/2) = V_1^+ + V_1^- + V_2^+ = (1 + \Gamma_L + \Gamma_L \Gamma_g) V_1^+ \quad (0 \leq z < l/2), \quad (2.150)$$

while on the second half of the line the voltage is only

$$V(z, 5T/2) = V_1^+ + V_1^- = (1 + \Gamma_L) V_1^+ \quad (l/2 \leq z \leq l). \quad (2.151)$$

The voltage distribution is shown in **Fig. 2-43(c)**.

So far, we have examined the transient response of the voltage wave $V(z, t)$. The associated transient response of the current $I(z, t)$ is shown in **Figs. 2-43(d)–(f)**. The current behaves similarly to the voltage $V(z, t)$, except for one important difference. Whereas at either end of the line the reflected voltage is related to the incident voltage by the reflection coefficient at that end, the reflected current is related to the incident current by the negative of the reflection coefficient. This property of wave reflection is expressed by Eq. (2.61). Accordingly,

$$I_1^- = -\Gamma_L I_1^+, \quad (2.152a)$$

$$I_2^+ = -\Gamma_g I_1^- = \Gamma_g \Gamma_L I_1^+, \quad (2.152b)$$

and so on.

► The multiple-reflection process continues indefinitely, and the ultimate value that $V(z, t)$ reaches as t approaches $+\infty$ is the same at all locations on the transmission line. ◀

The final value of V is given by

$$\begin{aligned} V_\infty &= V_1^+ + V_1^- + V_2^+ + V_2^- + V_3^+ + V_3^- + \cdots \\ &= V_1^+ [1 + \Gamma_L + \Gamma_L \Gamma_g + \Gamma_L^2 \Gamma_g + \Gamma_L^2 \Gamma_g^2 + \Gamma_L^3 \Gamma_g^2 + \cdots] \\ &= V_1^+ [(1 + \Gamma_L)(1 + \Gamma_L \Gamma_g + \Gamma_L^2 \Gamma_g^2 + \cdots)] \\ &= V_1^+ (1 + \Gamma_L) [1 + x + x^2 + \cdots], \end{aligned} \quad (2.153)$$

where $x = \Gamma_L \Gamma_g$. The series inside the square bracket is the geometric series of the function

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots \quad \text{for } |x| < 1. \quad (2.154)$$

Hence, Eq. (2.153) can be rewritten in the compact form

$$V_\infty = V_1^+ \frac{1 + \Gamma_L}{1 - \Gamma_L \Gamma_g}. \quad (2.155)$$

Upon replacing V_1^+ , Γ_L , and Γ_g with Eqs. (2.145b), (2.147), and (2.149) and simplifying the resulting expression, we obtain

$$V_\infty = \frac{V_g R_L}{R_g + R_L}. \quad (2.156)$$

The voltage V_∞ is called the **steady-state voltage** on the line, and its expression is exactly what we should expect on the basis of dc analysis of the circuit in **Fig. 2-43(a)**, where we treat the transmission line as simply a connecting wire between the generator circuit and the load. The corresponding **steady-state current** is

$$I_\infty = \frac{V_\infty}{R_L} = \frac{V_g}{R_g + R_L}. \quad (2.157)$$

2-12.2 Bounce Diagrams

Keeping track of the voltage and current waves as they bounce back and forth on the line is a rather tedious process. The **bounce diagram** is a graphical presentation that allows us to accomplish the same goal but with relative ease. The horizontal axes in **Figs. 2-44(a)** and **(b)** represent the position along the transmission line, while the vertical axes denote time. **Figures 2-44(a)** and **(b)** pertain to $V(z, t)$ and $I(z, t)$, respectively. The bounce diagram in **Fig. 2-44(a)** consists of a zigzag line indicating the progress of the voltage wave on the line. The incident wave V_1^+ starts at $z = t = 0$ and travels in the $+z$ direction until it reaches the load at $z = l$ at time $t = T$. At the very top of the bounce diagram, the reflection coefficients are indicated by $\Gamma = \Gamma_g$ at the generator end and by $\Gamma = \Gamma_L$ at the load end. At the end of the first straight-line segment of the zigzag line, a second line is drawn to represent the reflected voltage wave $V_1^- = \Gamma_L V_1^+$. The amplitude of each new straight-line segment equals the product of the amplitude of the preceding straight-line segment and the reflection coefficient at that end of the line. The bounce diagram for the current $I(z, t)$ in **Fig. 2-44(b)** adheres to the same principle except for the reversal of the signs of Γ_L and Γ_g at the top of the bounce diagram.

电压幅度 V_2^+ 由下式给出

$$V_2^+ = \Gamma_g V_1^- = \Gamma_g \Gamma_L V_1^+, \quad (2.148)$$

在哪里

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} \quad (2.149)$$

是发电机电阻 R_g 的反射系数。对于 $R_g = 4Z_0$, 我们有 $\Gamma_g = 0.6$ 。随着时间在 $t = 2T$ 之后进行, 波 V_2^+ 沿着线路向负载传播, 并添加到线路上先前建立的电压。因此, 在 $t = 5T/2$ 时, 线路前半部分的总电压为

$$V(z, 5T/2) = V_1^+ + V_1^- + V_2^+ = (1 + \Gamma_L + \Gamma_L \Gamma_g) V_1^+ \quad (0 \leq z < l/2), \quad (2.150)$$

而线路的后半部分电压仅为

$$V(z, 5T/2) = V_1^+ + V_1^- = (1 + \Gamma_L) V_1^+ \quad (l/2 \leq z \leq l). \quad (2.151)$$

电压分布如图2-43(c)所示。

到目前为止, 我们已经研究了电压波 $V(z, t)$ 的瞬态响应。电流 $I(z, t)$ 的相关瞬态响应如图 2 和 3 所示。2-43(d)–(f)。除了一个重要差异外, 电流的行为与电压 $V(z, t)$ 类似。在线路的任一端, 反射电压与入射电压通过该端的反射系数相关, 而反射电流与入射电流通过反射系数的负值相关。波反射的这一特性可以用方程表示: (2.61)。因此,

$$I_1^- = -\Gamma_L I_1^+, \quad (2.152a)$$

$$I_2^+ = -\Gamma_g I_1^- = \Gamma_g \Gamma_L I_1^+, \quad (2.152b)$$

等等。

► The multiple-reflection process continues indefinitely, and the ultimate value that $V(z, t)$ reaches as t approaches $+\infty$ is the same at all locations on the transmission line. ◀

V 的最终值由下式给出

$$\begin{aligned} V_\infty &= V_1^+ + V_1^- + V_2^+ + V_2^- + V_3^+ + V_3^- + \cdots \\ &= V_1^+ [1 + \Gamma_L + \Gamma_L \Gamma_g + \Gamma_L^2 \Gamma_g + \Gamma_L^2 \Gamma_g^2 + \Gamma_L^3 \Gamma_g^2 + \cdots] \\ &= V_1^+ [(1 + \Gamma_L)(1 + \Gamma_L \Gamma_g + \Gamma_L^2 \Gamma_g^2 + \cdots)] \\ &= V_1^+ (1 + \Gamma_L) [1 + x + x^2 + \cdots], \end{aligned} \quad (2.153)$$

其中 $x = \Gamma_L \Gamma_g$ 。方括号内的级数是函数的几何级数

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots \quad \text{for } |x| < 1. \quad (2.154)$$

因此, 等式。 (2.153) 可以重写为紧凑形式

$$V_\infty = V_1^+ \frac{1 + \Gamma_L}{1 - \Gamma_L \Gamma_g}. \quad (2.155)$$

将 V_1^+ 、 Γ_L 和 Γ_g 替换为等式后: (2.145b)、(2.147) 和 (2.149) 并简化结果表达式, 我们得到

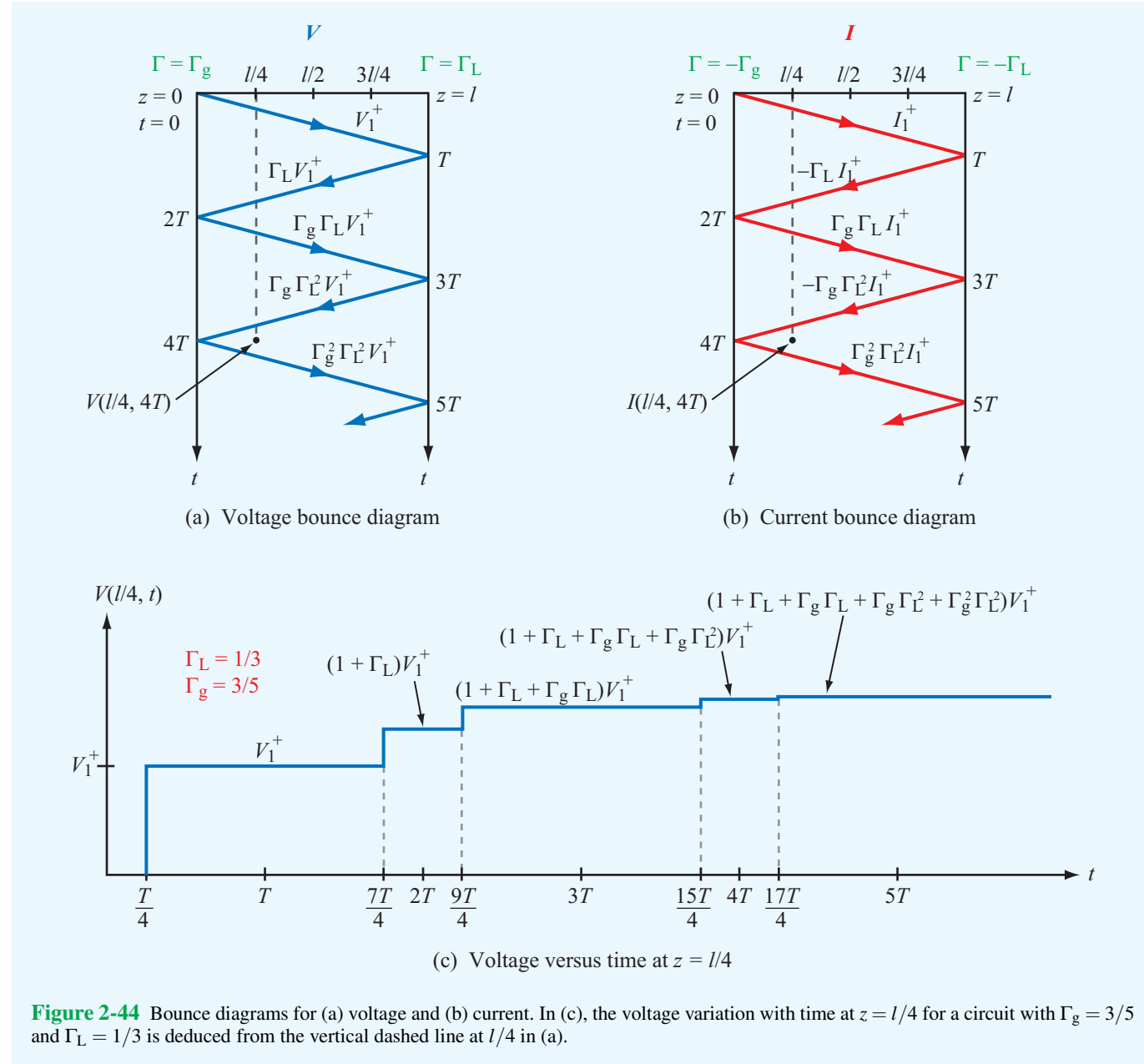
$$V_\infty = \frac{V_g R_L}{R_g + R_L}. \quad (2.156)$$

电压 V_∞ 被称为线路上的稳态电压, 其表达式正是我们根据图2-43(a)中电路的直流分析所期望的, 其中我们将传输线视为发电机电路和负载之间的简单连接线。相应的稳态电流为

$$I_\infty = \frac{V_\infty}{R_L} = \frac{V_g}{R_g + R_L}. \quad (2.157)$$

2-12.2 弹跳图

跟踪在线上来回反弹的电压和电流波是一个相当乏味的过程。反弹图是一种图形表示, 它使我们能够相对轻松地实现相同的目标。图中的水平轴。图2-44(a)和(b)表示沿着传输线的位置, 而纵轴表示时间。图 2-44(a) 和 (b) 分别涉及 $V(z, t)$ 和 $I(z, t)$ 。图2-44(a)中的反弹图由一条锯齿线组成, 指示线上电压波的进展。入射波 V_1^+ 从 $z = t = 0$ 开始, 沿 $+z$ 方向行进, 直到在时间 $t = T$ 到达 $z = l$ 处的载荷。在弹跳图的最顶部, 反射系数在发电末端用 $\Gamma = \Gamma_g$ 表示, 在负载端用 $\Gamma = \Gamma_L$ 表示。在锯齿形线的第一条直线段的末端, 绘制第二条线来表示反射电压波 $V_1^- = \Gamma_L V_1^+$ 。每个新直线段的振幅等于前一个直线段的振幅与该直线末端的反射系数的乘积。图2-44(b)中的电流 $I(z, t)$ 的反弹图遵循相同的原理, 除了反弹图顶部的 Γ_L 和 Γ_g 的符号相反。



Using the bounce diagram, the total voltage (or current) at any point z_1 and time t_1 can be determined by first drawing a vertical line through point z_1 , and then adding the voltages (or currents) of all the zigzag segments intersected by that line between $t = 0$ and $t = t_1$. To find the voltage at $z = l/4$ and $T = 4T$, for example, we draw a dashed vertical line in **Fig. 2-44(a)** through $z = l/4$, and we extend it from $t = 0$ to

$t = 4T$. The dashed line intersects four line segments. The total voltage at $z = l/4$ and $t = 4T$ therefore is

$$\begin{aligned} V(l/4, 4T) &= V_1^+ + \Gamma_L V_1^+ + \Gamma_g \Gamma_L V_1^+ + \Gamma_g \Gamma_L^2 V_1^+ \\ &= V_1^+ (1 + \Gamma_L + \Gamma_g \Gamma_L + \Gamma_g \Gamma_L^2). \end{aligned}$$

The time variation of $V(z, t)$ at a specific location z can be obtained by plotting the values of $V(z, t)$ along the (dashed)

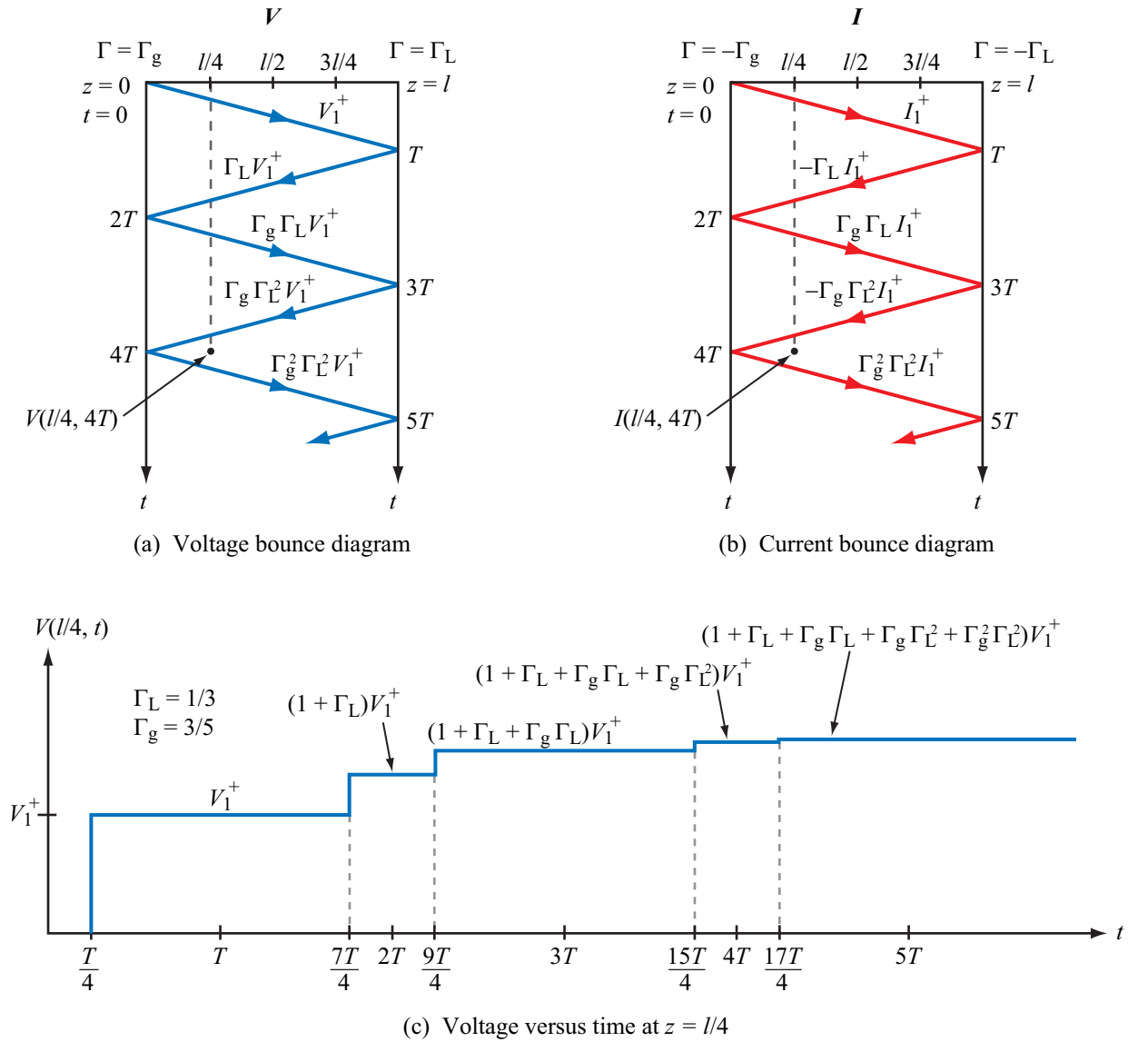


图 2-44 (a) 电压和 (b) 电流的跳动图。在 (c) 中，对于具有 $\Gamma_g = 3/5$ 和 $\Gamma_L = 1/3$ 的电路， $z = l/4$ 处的电压随时间的变化是从 (a) 中 $l/4$ 处的垂直虚线推导出来的。

使用反弹图，可以通过首先画一条穿过点 z_1 的垂直线，然后将 $t = 0$ 和 $t = t_1$ 之间与该线相交的所有锯齿形段的电压（或电流）相加来确定任意点 z_1 和时间 t_1 处的总电压（或电流）。例如，为了求 $z = l/4$ 和 $T = 4T$ 处的电压，我们在图 2-44(a) 中通过 $z = l/4$ 画一条垂直虚线，并将其从 $t = 0$ 延伸到

$t = 4T$ 。虚线与四个线段相交。因此 $z = l/4$ 和 $t = 4T$ 处的总电压为

$$\begin{aligned} V(l/4, 4T) &= V_1^+ + \Gamma_L V_1^+ + \Gamma_g \Gamma_L V_1^+ + \Gamma_g \Gamma_L^2 V_1^+ \\ &= V_1^+ (1 + \Gamma_L + \Gamma_g \Gamma_L + \Gamma_g \Gamma_L^2). \end{aligned}$$

通过沿（虚线）绘制 $V(z, t)$ 值，可以获得特定位置 z 处 $V(z, t)$ 的时间变化。

2-12 TRANSIENTS ON TRANSMISSION LINES

113

vertical line passing through z . **Figure 2-44(c)** shows the variation of V as a function of time at $z = l/4$ for a circuit with $\Gamma_g = 3/5$ and $\Gamma_L = 1/3$.

Example 2-16: Pulse Propagation

The transmission-line circuit of **Fig. 2-45(a)** is excited by a rectangular pulse of duration $\tau = 1$ ns that starts at $t = 0$. Establish the waveform of the voltage response at the load given that the pulse amplitude is 5 V, the phase velocity is c , and the length of the line is 0.6 m.

Solution: The one-way propagation time is

$$T = \frac{l}{c} = \frac{0.6}{3 \times 10^8} = 2 \text{ ns}.$$

The reflection coefficients at the load and the sending end are

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} = \frac{150 - 50}{150 + 50} = 0.5,$$

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} = \frac{12.5 - 50}{12.5 + 50} = -0.6.$$

By Eq. (2.143), the pulse is treated as the sum of two step functions: one that starts at $t = 0$ with an amplitude $V_{10} = 5$ V and a second one that starts at $t = 1$ ns with an amplitude $V_{20} = -5$ V. Except for the time delay of 1 ns and the sign reversal of all voltage values, the two step functions generate identical bounce diagrams, as shown in **Fig. 2-45(b)**. For the first step function, the initial voltage is given by

$$V_1^+ = \frac{V_{01}Z_0}{R_g + Z_0} = \frac{5 \times 50}{12.5 + 50} = 4 \text{ V}.$$

Using the information displayed in the bounce diagram, it is straightforward to generate the voltage response shown in **Fig. 2-45(c)**. Note that at $t = 2$ ns, the voltage at the load consists of both $V_1^+ = 4$ V, and its reflection $V_1^- = \Gamma_L V_1^+ = 2$ V. Thus, the total voltage at the load at $t = 2$ ns is 6 V. A similar process occurs at 3 ns due to the negative step function, which generates -6 V. Consequently, the sum of the two step-function contributions adds up to zero and stays that way until $t = 6$ ns.

Example 2-17: Time-Domain Reflectometer

A time-domain reflectometer (TDR) is an instrument used to locate faults on a transmission line. Consider, for example, a (formerly matched) long underground or undersea cable that

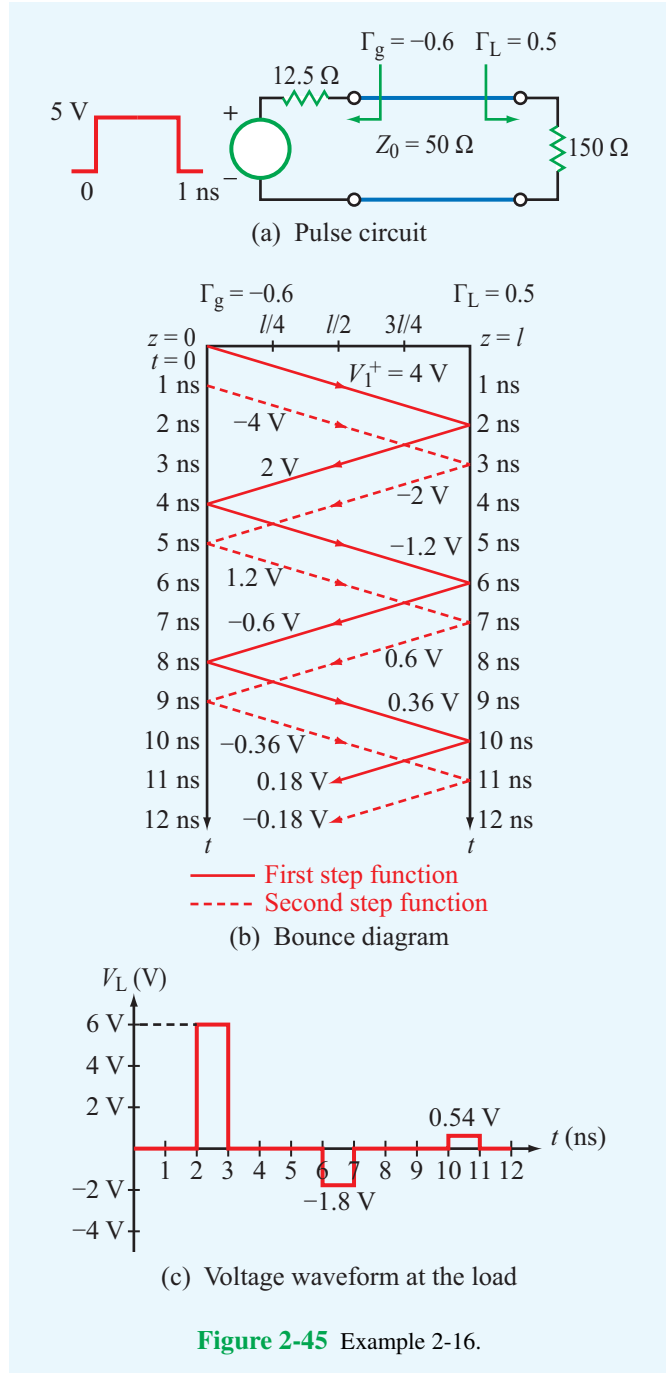


Figure 2-45 Example 2-16.

gets damaged at some distance d from the sending end of the line. The damage may alter the electrical properties or the shape of the cable, causing it to exhibit at the fault location an **effective resistance** R_{Lf} . A TDR sends a step voltage down

2-12 TRANSIENTS ON TRANSMISSION LINES

113

通过 z 的垂直线。图 2-44(c) 显示了具有 $\Gamma_g = 3/5$ 和 $\Gamma_L = 1/3$ 的电路在 $z = l/4$ 处 V 随时间的变化。

例 2-16: 脉冲传播

图 2-45(a) 的传输线电路由从 $t = 0$ 开始、持续时间为 $\tau = 1$ ns 的矩形脉冲激励。假设脉冲幅度为 5 V、相速度为 c 、线路长度为 0.6 m，则建立负载处的电压响应波形。

解：单向传播时间为

$$T = \frac{l}{c} = \frac{0.6}{3 \times 10^8} = 2 \text{ ns}.$$

负载端和发送端的反射系数为

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} = \frac{150 - 50}{150 + 50} = 0.5,$$

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} = \frac{12.5 - 50}{12.5 + 50} = -0.6.$$

由方程式在方程 (2.143) 中，脉冲被视为两个阶跃函数的和：一个从 $t = 0$ 开始，幅度为 $V_{10} = 5$ V，第二个从 $t = 1$ ns 开始，幅度为 $V_{20} = -5$ V。除了 1 ns 的时间延迟和所有电压值的符号反之外，这两个阶跃函数生成相同的反弹图，如图 1 所示。2-45(b)。对于第一步函数，初始电压由下式给出

$$V_1^+ = \frac{V_{01}Z_0}{R_g + Z_0} = \frac{5 \times 50}{12.5 + 50} = 4 \text{ V}.$$

使用反弹图中显示的信息，可以直接生成图 2-45(c) 中所示的电压响应。请注意，在 $t = 2$ ns 时，负载上的电压由 $V_1^+ = 4$ V 及其反射 $V_1^- = \Gamma_L V_1^+ = 2$ V 组成。因此，在 $t = 2$ ns 时负载上的总电压为 6 V。由于负阶跃函数，在 3 ns 时会发生类似的过程，生成 -6 V。因此，两个阶跃函数贡献的总和加上直到 0 并保持这种状态直到 $t = 6$ ns。

例 2-17: 时域反射计

时域反射计 (TDR) 是一种用于定位传输线路故障的仪器。例如，考虑一条 (以前匹配的) 长地下或海底电缆

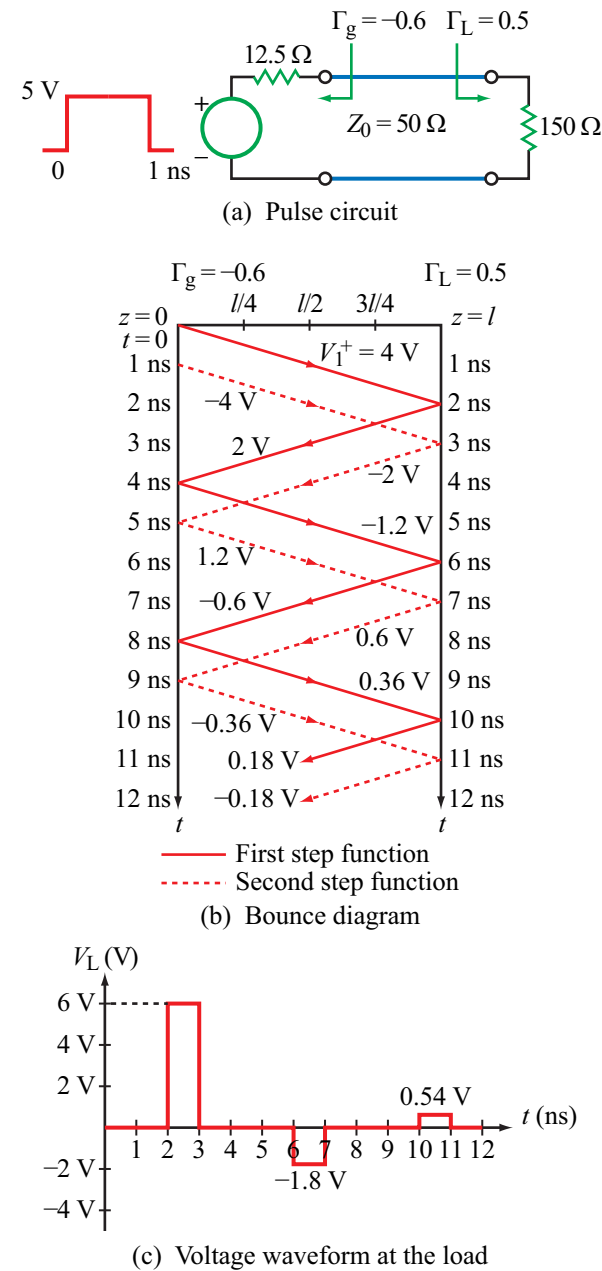


图 2-45 示例 2-16。

在距线路发送端一定距离 d 处损坏。损坏可能会改变电缆的电气特性或形状，导致其在故障位置表现出有效电阻 R_{Lf} 。TDR 发送阶跃电压下降

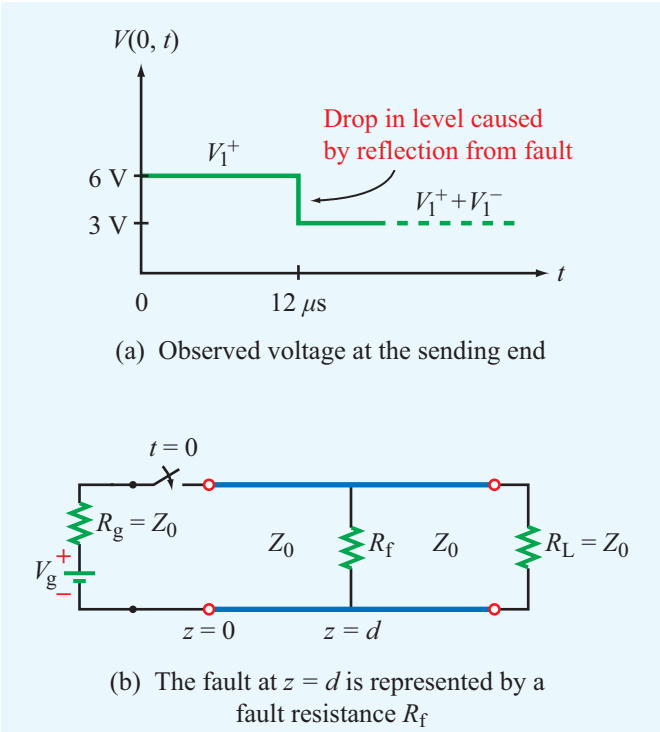


Figure 2-46 Time-domain reflectometer of Example 2-17.

the line, and by observing the voltage at the sending end as a function of time, it is possible to determine the location of the fault and its severity.

If the voltage waveform shown in Fig. 2-46(a) is seen on an oscilloscope connected to the input of a 75 Ω matched transmission line, determine (a) the generator voltage, (b) the location of the fault, and (c) the fault shunt resistance. The line’s insulating material is Teflon with $\epsilon_r = 2.1$.

Solution: (a) Since the line is properly matched, $R_g = R_L = Z_0$. In Fig. 2-46(b), the fault located a distance d from the sending end is represented by a shunt resistance R_f . For a matched line, Eq. (2.145b) gives

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0} = \frac{V_g Z_0}{2Z_0} = \frac{V_g}{2}.$$

According to Fig. 2-46(a), $V_1^+ = 6\text{ V}$. Hence,

$$V_g = 2V_1^+ = 12\text{ V}.$$

(b) The propagation velocity on the line is

$$u_p = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.1}} = 2.07 \times 10^8\text{ m/s}.$$

For a fault at a distance d , the round-trip time delay of the echo is

$$\Delta t = \frac{2d}{u_p}.$$

From Fig. 2-46(a), $\Delta t = 12\text{ }\mu\text{s}$. Hence,

$$d = \frac{\Delta t}{2} u_p = \frac{12 \times 10^{-6}}{2} \times 2.07 \times 10^8 = 1,242\text{ m}.$$

(c) The drop in level of $V(0, t)$ shown in Fig. 2-46(a) represents V_1^- . Thus,

$$V_1^- = \Gamma_f V_1^+ = -3\text{ V},$$

or

$$\Gamma_f = \frac{-3}{6} = -0.5,$$

where Γ_f is the reflection coefficient due to the effective fault resistance R_{Lf} that appears at $z = d$.

From Eq. (2.59),

$$\Gamma_f = \frac{R_{Lf} - Z_0}{R_{Lf} + Z_0},$$

which leads to $R_{Lf} = 25\text{ }\Omega$. This fault load is composed of the parallel combination of the fault shunt resistance R_f and the characteristic impedance Z_0 of the line to the right of the fault:

$$\frac{1}{R_{Lf}} = \frac{1}{R_f} + \frac{1}{Z_0},$$

so the shunt resistance must be 37.5 Ω.

Example 2-18: Pulse Transmission

When excited by a pulse source of amplitude 1 V and duration 5 μs, a transmission line—with unknown characteristic impedance Z_0 and unknown load resistance R_L —exhibits the voltage response shown in Fig. 2-47(b) when observed at the middle of the line. The line relative permittivity is $\epsilon_r = 2.25$. Determine (a) the length of the line, (b) Z_0 , and (c) R_L .

Solution: (a) The phase velocity is

$$u_p = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.25}} = 2 \times 10^8\text{ m/s}.$$

It takes 10 μs for the initial part of the pulse to reach the middle of the line, so it would take it $T = 20\text{ }\mu\text{s}$ to reach the end of the line. Hence,

$$l = u_p T = 2 \times 10^8 \times 20 \times 10^{-6} = 4000\text{ m}.$$

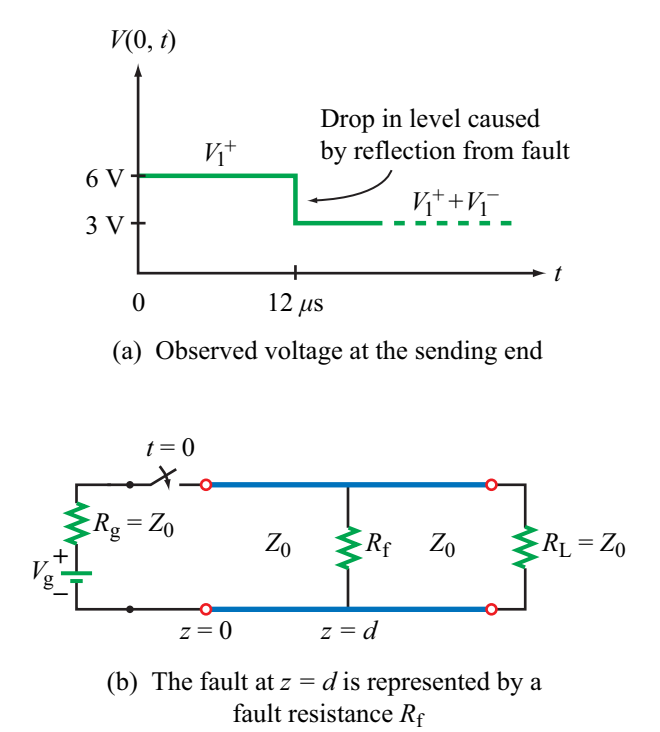


图 2-46 例 2-17 的时域反射计。

线路，并通过观察发送端的电压随时间的变化，可以确定故障的位置及其严重程度。

如果在连接到 75 Ω 匹配传输线输入的示波器上看到如图 2-46(a) 所示的电压波形，则确定 (a) 发电电压、(b) 故障位置和 (c) 故障分流电阻。该线路的绝缘材料为 $\epsilon_r = 2.1$ 的聚四氟乙烯。

解：(a) 由于线路正确匹配，因此 $R_g = R_L = Z_0$ 。图2-46(b)中，距发送端距离d处的故障用分流电阻 R_f 表示。对于匹配线，方程。(2.145b) 给出

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0} = \frac{V_g Z_0}{2Z_0} = \frac{V_g}{2}.$$

根据图2-46(a), $V_1^+ = 6\text{ V}$ 。因此，

$$V_g = 2V_1^+ = 12\text{ V}.$$

(b) 线上的传播速度为

$$u_p = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.1}} = 2.07 \times 10^8\text{ m/s}.$$

对于距离 d 处的故障，回波的往返时间延迟为

$$\Delta t = \frac{2d}{u_p}.$$

由图2-46(a)可知， $\Delta t = 12\text{ }\mu\text{s}$ 。因此，

$$d = \frac{\Delta t}{2} u_p = \frac{12 \times 10^{-6}}{2} \times 2.07 \times 10^8 = 1,242\text{ m}.$$

(c) 图2-46(a)所示的 $V(0, t)$ 电平下降代表 V_1^- 。因此，

$$V_1^- = \Gamma_f V_1^+ = -3\text{ V},$$

或者

$$\Gamma_f = \frac{-3}{6} = -0.5,$$

其中， Γ_f 是由出现在 $z = d$ 处的有效故障电阻 R_{Lf} 产生的反射系数。

从方程。(2.59),

$$\Gamma_f = \frac{R_{Lf} - Z_0}{R_{Lf} + Z_0},$$

这导致 $R_{Lf} = 25\text{ }\Omega$ 。该故障负载由故障分流电阻 R_f 和故障右侧线路的特性阻抗 Z_0 并联组合而成：

$$\frac{1}{R_{Lf}} = \frac{1}{R_f} + \frac{1}{Z_0},$$

因此分流电阻必须为 37.5 Ω。

例 2-18：脉冲传输

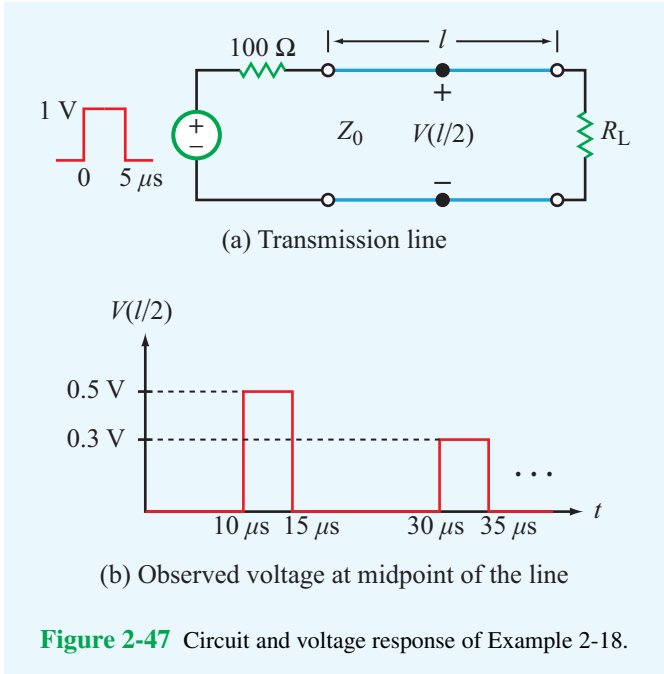
当由幅度为 1 V、持续时间为 5 μs 的脉冲源激励时，特性阻抗 Z_0 和负载电阻 R_L 未知的传输线在传输线中部观察时会表现出如图 2-47(b) 所示的电压响应。线相对介电常数为 $\epsilon_r = 2.25$ 。确定 (a) 线的长度、(b) Z_0 和 (c) R_L 。

解：(a) 相速度为

$$u_p = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.25}} = 2 \times 10^8\text{ m/s}.$$

脉冲的初始部分需要 10 μ s 才能到达线的中间，因此需要 $T = 20\text{ }\mu\text{ s}$ 才能到达线的末尾。因此，

$$l = u_p T = 2 \times 10^8 \times 20 \times 10^{-6} = 4000\text{ m}.$$



(b) From Eq. (2.145b),

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}.$$

In the present case, $V_g = 1\text{ V}$, $R_g = 100\ \Omega$, and the waveform in **Fig. 2-47(b)** indicates that $V_1^+ = 0.5\text{ V}$. Solving for Z_0 leads to $Z_0 = 100\ \Omega$.

(c) The pulse that appears between $30\ \mu\text{s}$ and $35\ \mu\text{s}$ is due to reflection by the load and its amplitude $V_1^- = 0.3\text{ V}$. Hence,

$$\Gamma_L = \frac{V_1^-}{V_1^+} = \frac{0.3}{0.5} = 0.6.$$

Also,

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}.$$

Using $\Gamma_L = 0.6$ and $Z_0 = 100\ \Omega$ leads to

$$R_L = 400\ \Omega.$$

Concept Question 2-27: What is transient analysis used for?

Concept Question 2-28: The transient analysis presented in this section was for a step voltage. How does one use it for analyzing the response to a pulse?

Concept Question 2-29: What is the difference between the bounce diagram for voltage and the bounce diagram for current?

Concept Question 2-30: What is a TDR and what is it used for?

Concept Question 2-31: What do V_∞ and I_∞ represent?

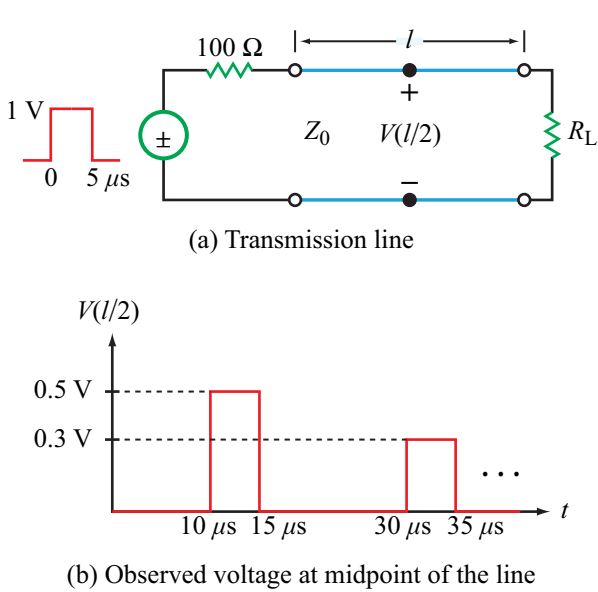


图 2-47 例 2-18 的电路和电压响应。

(b) 从方程式(2.145b),

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}.$$

在本例中, $V_g = 1\text{ V}$, $R_g = 100\ \Omega$, 图2-47(b)中的波形表明 $V_1^+ = 0.5\text{ V}$ 。求解 Z_0 导致 $Z_0 = 100\ \Omega$ 。

(c) 在 $30\ \mu\text{s}$ 和 $35\ \mu\text{s}$ 之间出现的脉冲是由于负载的反射而产生的, 其幅度 $V_1^- = 0.3\text{ V}$ 。因此,

$$\Gamma_L = \frac{V_1^-}{V_1^+} = \frac{0.3}{0.5} = 0.6.$$

还,

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}.$$

使用 $\Gamma_L = 0.6$ 和 $Z_0 = 100\ \Omega$ 会导致

$$R_L = 400\ \Omega.$$

Concept Question 2-27: What is transient analysis used for?

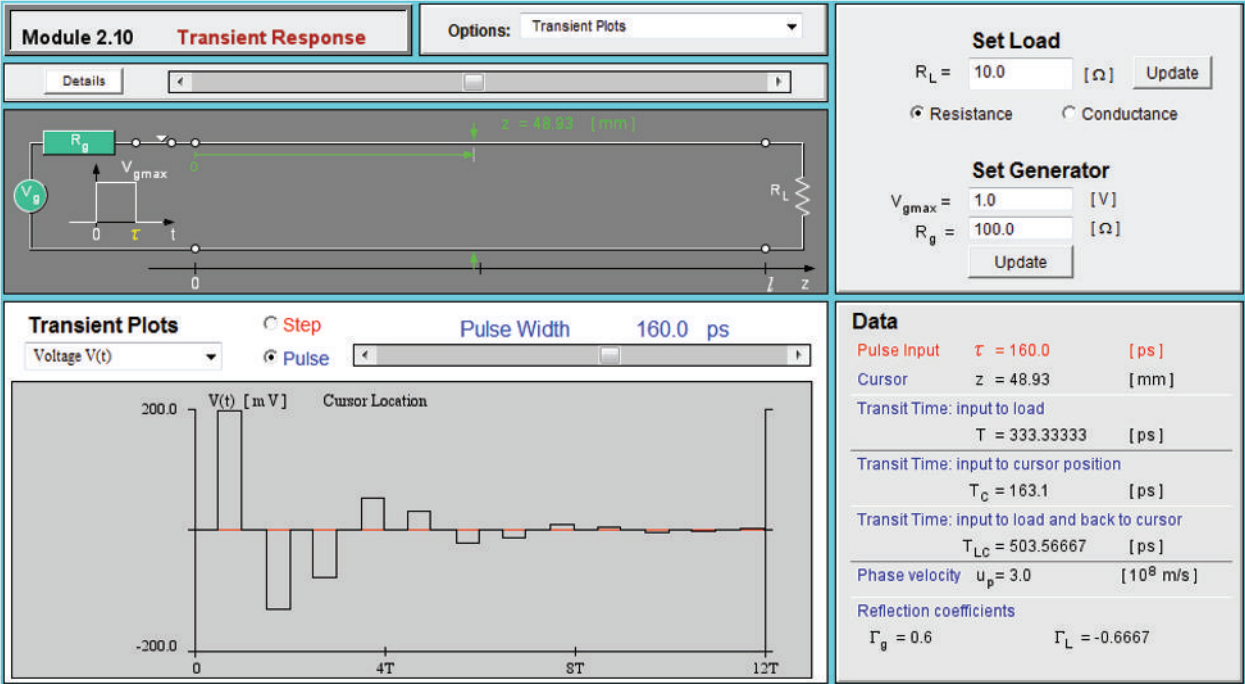
Concept Question 2-28: The transient analysis presented in this section was for a step voltage. How does one use it for analyzing the response to a pulse?

Concept Question 2-29: What is the difference between the bounce diagram for voltage and the bounce diagram for current?

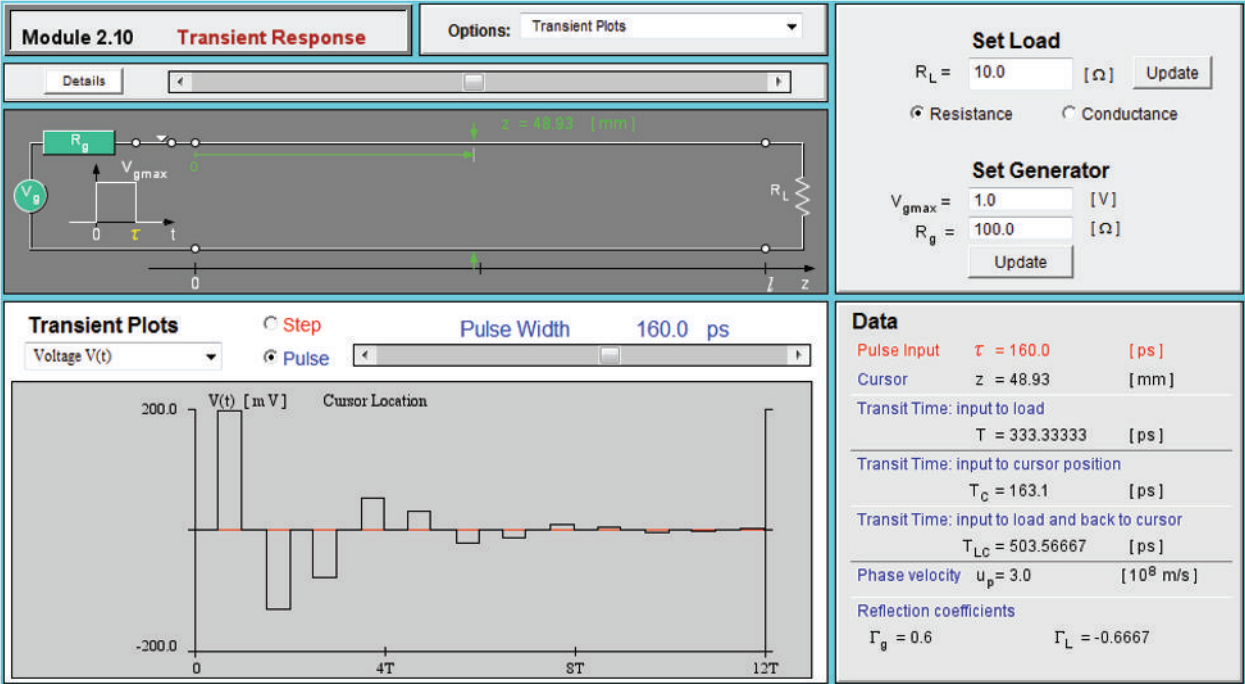
Concept Question 2-30: What is a TDR and what is it used for?

Concept Question 2-31: What do V_∞ and I_∞ represent?

Module 2.10 Transient Response For a lossless line terminated in a resistive load, the module simulates the dynamic response—at any location on the line—to either a step or pulse waveform sent by the generator.



模块 2.10 瞬态响应 对于端接电阻负载的无损线路，该模块模拟线路上任何位置对发生器发送的阶跃波形或脉冲波形的动态响应。



Technology Brief 4:
EM Cancer Zappers

From laser eye surgery to 3-D X-ray imaging, EM sources and sensors have been used as medical diagnostic and treatment tools for many decades. Future advances in information processing and other relevant technologies will undoubtedly lead to the greater performance and utility of EM devices, as well as to the introduction of entirely new types of devices. This Technology Brief introduces two recent EM technologies that are still in their infancy but are quickly developing into serious techniques for the surgical treatment of cancer tumors.

Microwave Ablation

In medicine, **ablation** is defined as the “surgical removal of body tissue,” usually through the direct application of chemical or thermal therapies.

► Microwave ablation applies the same heat-conversion process used in a microwave oven (see TB3), but instead of using microwave energy to cook food, it is used instead to destroy cancerous tumors by exposing them to a focused beam of microwaves. ◀

The technique can be used **percutaneously** (through the skin), **laparoscopically** (via an incision), or **intraoperatively** (open surgical access). Guided by an imaging system, such as a CT scanner or an ultrasound imager, the surgeon can localize the tumor and then insert a thin coaxial transmission line (~1.5 mm in diameter) directly through the body to position the tip of the transmission line (a probe-like antenna) inside the tumor (Fig. TF4-1). The transmission line is connected to a generator capable of delivering 60 W of power at 915 MHz (Fig. TF4-2). The rise in temperature of the tumor is related to the amount of microwave energy it receives, which is equal to the product of the generator's power level and the duration of the ablation treatment. Microwave ablation is a promising new technique for the treatment of liver, lung, and adrenal tumors.

High-Power Nanosecond Pulses

Bioelectrics is an emerging field focused on the study of how electric fields behave in biological systems. Of particular recent interest is the desire to understand

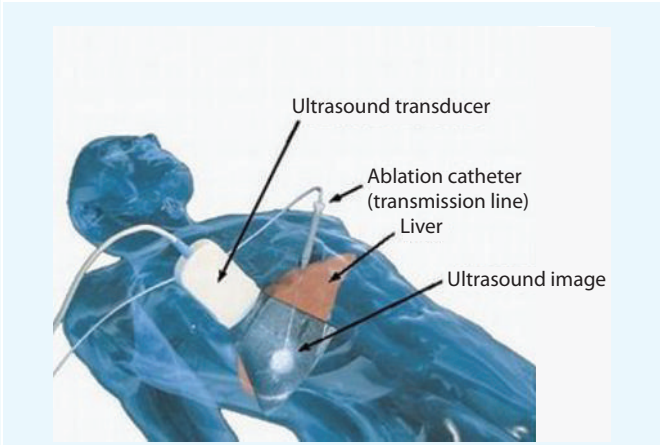


Figure TF4-1 Microwave ablation for liver cancer treatment.

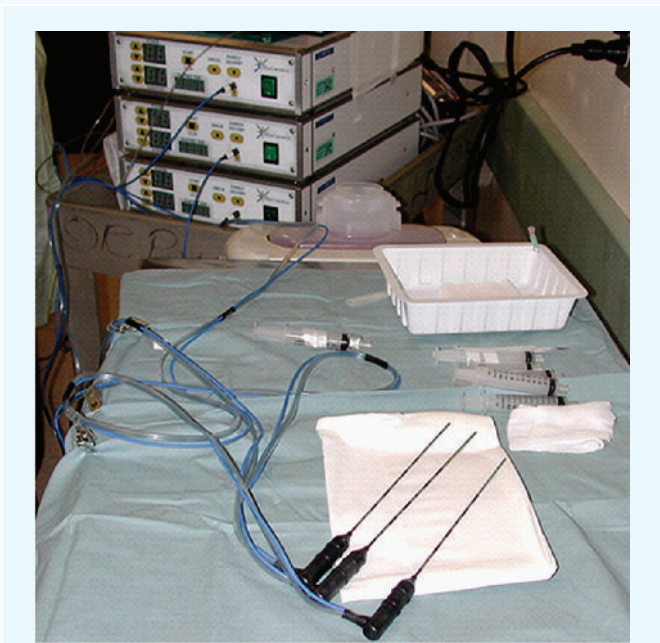


Figure TF4-2 Photograph of the setup for a percutaneous microwave ablation procedure in which three single microwave applicators are connected to three microwave generators. (Courtesy of RadioGraphics, October 2005, pp. 569–583.)

how living cells might respond to the application of extremely short pulses (on the order of nanoseconds (10^{-9} s) and even as short as picoseconds (10^{-12} s)) with exceptionally high voltage and current amplitudes.

技术简介 4：电磁癌
症消灭器

从激光眼科手术到 3D X 射线成像，数十年来，电磁源和传感器一直被用作医疗诊断和治疗工具。信息处理和其他相关技术的未来进步无疑将提高电磁设备的性能和实用性，并引入全新类型的设备。本技术简介介绍了两项最新的 EM 技术，它们仍处于起步阶段，但正在迅速发展成为用于癌症肿瘤外科治疗的重要技术。

微波消融

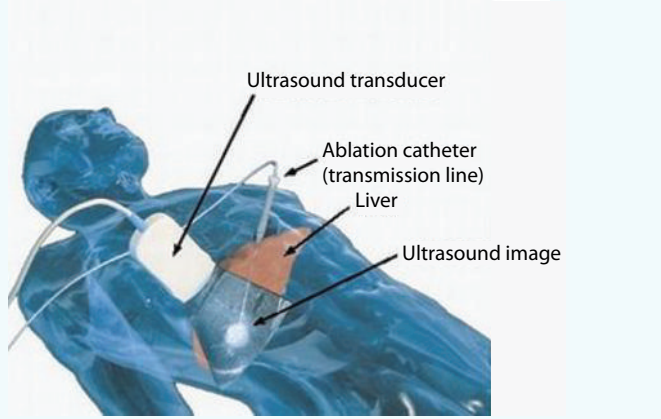
在医学上，消融被定义为“手术切除身体组织”，通常通过直接应用化学或热疗法。

► 微波消融采用与微波炉相同的热转换过程（见TB 3），但不是使用微波能量来烹饪食物，而是通过将恶性肿瘤暴露在聚焦的微波束下来摧毁它们。 ◀

该技术可经皮（通过皮肤）、腹腔镜（通过切口）或术中（开放手术通道）使用。在 CT 扫描仪或超声成像仪等成像系统的引导下，外科医生可以定位肿瘤，然后插入细同轴传输线（~直径 1.5 毫米）直接穿过身体，将传输线（探针状天线）的尖端定位在肿瘤内部（图 TF4-1）。传输线连接到能够以 915 MHz 提供 60 W 功率的发电机（图 TF4-2）。肿瘤温度的升高与其接收的微波能量有关，该能量等于发生器的功率水平与消融治疗持续时间的乘积。微波消融是治疗肝、肺和肾上腺肿瘤的一种有前途的新技术。

高功率纳秒脉冲

生物电学是一个新兴领域，专注于研究电场在生物系统中的行为方式。最近特别感兴趣的是了解的愿望



图TF4-1 微波消融治疗肝癌。

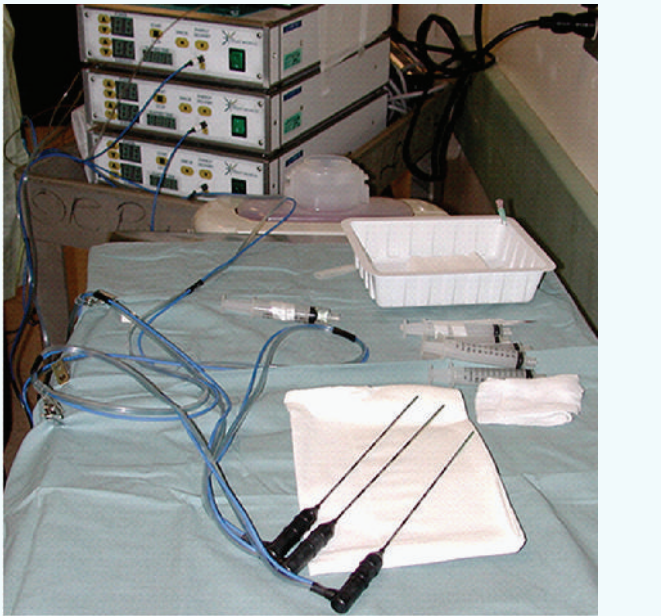


图 TF4-2 经皮微波消融手术的设置照片，其中三个单个微波发生器连接到三个微波发生器。（RadioGraphics 提供，2005 年 10 月，第 569-583 页。）

活细胞如何响应具有极高电压和电流幅度的极短脉冲（纳秒 (10^{-9} s) 量级，甚至短至皮秒 (10^{-12} s)）。

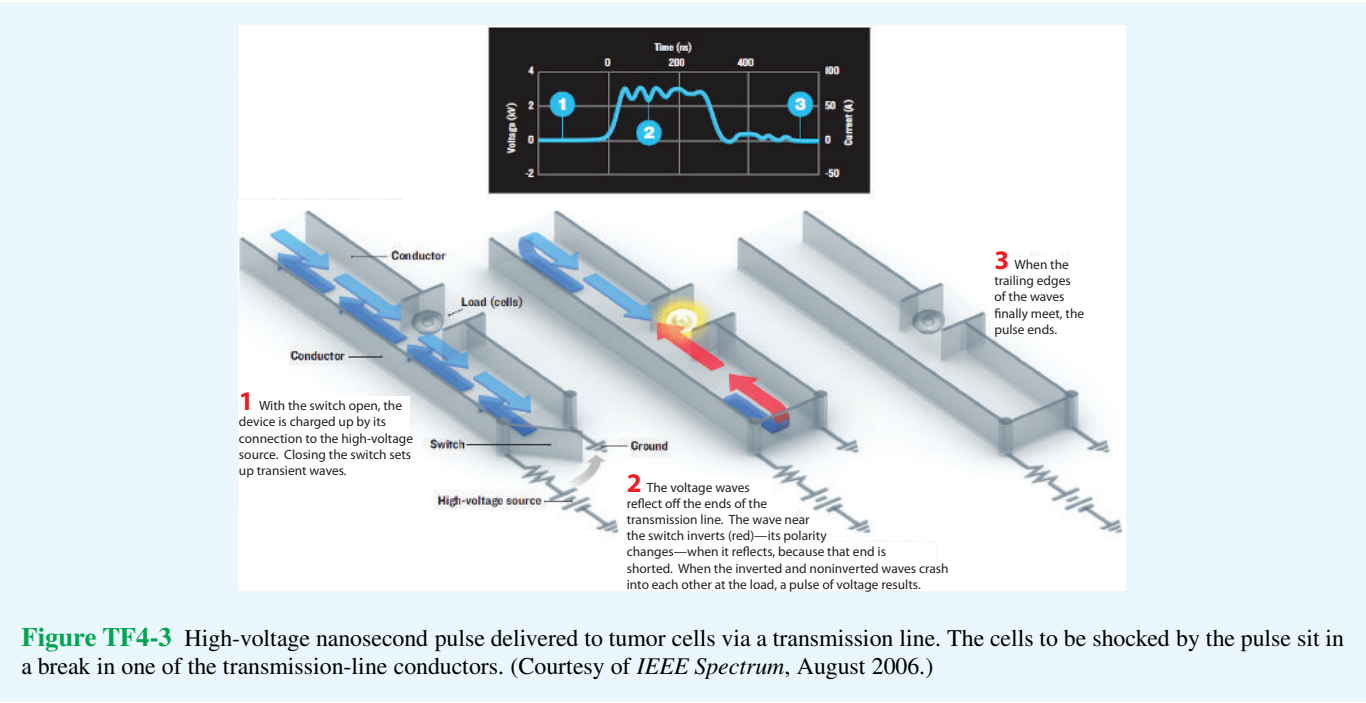


Figure TF4-3 High-voltage nanosecond pulse delivered to tumor cells via a transmission line. The cells to be shocked by the pulse sit in a break in one of the transmission-line conductors. (Courtesy of *IEEE Spectrum*, August 2006.)

► The motivation is to treat cancerous cells by **zapping** them with high-power pulses. The pulse power is delivered to the cell via a transmission line, as illustrated by the example in **Fig. TF4-3**. ◀

Note that the pulse is about 200 ns long, and its voltage and current amplitudes are approximately 3,000 V and 60 A, respectively. Thus, the peak power level is about 180,000 W! However, the total energy carried by the pulse is only $(1.8 \times 10^5) \times (2 \times 10^{-7}) = 0.0036$ joules. Despite the low energy content, the very high voltage appears to be very effective in destroying malignant tumors (in mice, so far), with no regrowth.

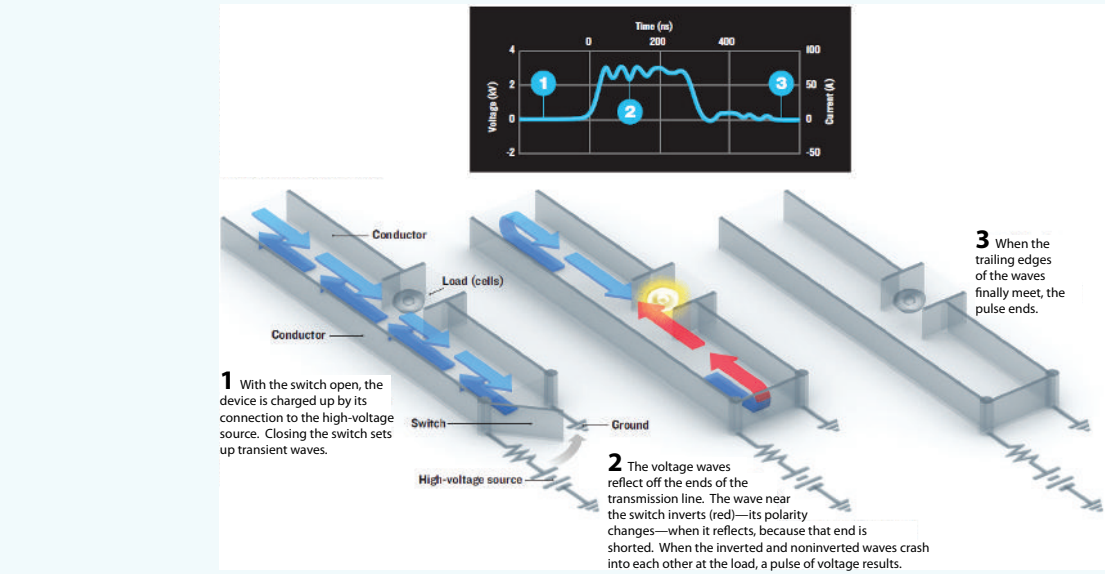


图 TF4-3 通过传输线传送至肿瘤细胞的高压纳秒脉冲。受到脉冲冲击的细胞位于一根传输线导体的断口处。（由 IEEE Spectrum 提供，2006 年 8 月。）

► 其动机是通过用高功率脉冲攻击癌细胞来治疗它们。脉冲功率通过传输线传送到细胞，如图 TF4-3 中的示例所示。 ◀

请注意，脉冲长约 200 ns，其电压和电流幅度分别约为 3,000 V 和 60 A。因此，峰值功率约为 180,000 W！然而，脉冲携带的总能量仅为 $(1.8 \times 10^5) \times (2 \times 10^{-7}) = 0.0036$ 焦耳。尽管能量含量较低，但非常高的电压似乎可以非常有效地摧毁恶性肿瘤（迄今为止在小鼠中），并且不会再生。

Chapter 2 Summary

Concepts

- A transmission line is a two-port network connecting a generator to a load. EM waves traveling on the line may experience ohmic power losses, dispersive effects, and reflections at the generator and load ends of the line. These transmission-line effects may be ignored if the line length is much shorter than λ .
- TEM transmission lines consist of two conductors that can support the propagation of transverse electromagnetic waves characterized by electric and magnetic fields that are transverse to the direction of propagation. TEM lines may be represented by a lumped-element model consisting of four line parameters (R' , L' , G' , and C') whose values are specified by the specific line geometry, the constitutive parameters of the conductors and of the insulating material between them, and the angular frequency ω .
- Wave propagation on a transmission line, which is represented by the phasor voltage $\tilde{V}(z)$ and associated current $\tilde{I}(z)$, is governed by the propagation constant of the line, $\gamma = \alpha + j\beta$, and its characteristic impedance Z_0 . Both γ and Z_0 are specified by ω and the four line parameters.
- If $R' = G' = 0$, the line becomes lossless ($\alpha = 0$). A lossless line is generally nondispersive, meaning that the phase velocity of a wave is independent of the frequency.
- In general, a line supports two waves, an incident wave supplied by the generator and another wave reflected by the load. The sum of the two waves generates a standing-wave pattern with a period of $\lambda/2$. The voltage standing-wave ratio S , which is equal to the ratio of the maximum to minimum voltage magnitude on the line, varies between 1 for a matched load ($Z_L = Z_0$) to ∞ for a line terminated in an open circuit, a short circuit, or a purely reactive load.
- The input impedance of a line terminated in a short circuit or open circuit is purely reactive. This property can be used to design equivalent inductors and capacitors.
- The fraction of the incident power delivered to the load by a lossless line is equal to $(1 - |\Gamma|^2)$.
- The Smith chart is a useful graphical tool for analyzing transmission-line problems and for designing impedance-matching networks.
- Matching networks are placed between the load and the feed transmission line for the purpose of eliminating reflections toward the generator. A matching network may consist of lumped elements in the form of capacitors and/or inductors, or it may consist of sections of transmission lines with appropriate lengths and terminations.
- Transient analysis of pulses on transmission lines can be performed using a bounce-diagram graphical technique that tracks reflections at both the load and generator ends of the transmission line.

Chapter 2 Summary

概念

- 传输线是将发电机连接到负载的二端口网络。在线路上传播的电磁波可能会经历欧姆功率损耗、色散效应以及线路发电机和负载端的反射。如果线路长度比 λ 短得多，则可以忽略这些传输线效应。
 - TEM 传输线由两个导体组成，可以支持横向电磁波的传播，其特征是电场和磁场垂直于传播方向。TEM 线可以用集总元件模型表示，该模型由四个线参数 (R' 、 L' 、 G' 和 C') 组成，其值由特定的线几何形状、导体及其之间的绝缘材料的本构参数以及角频率 ω 指定。
 - 传输线上的波传播由相量电压 $\tilde{V}(z)$ 和相关电流 $\tilde{I}(z)$ 表示，由传输线的传播常数 $\gamma = \alpha + j\beta$ 及其特征阻抗 Z_0 控制。 γ 和 Z_0 均由 ω 和四个行参数指定。
 - 如果 $R' = G' = 0$ ，则线路变为无损 ($\alpha = 0$)。无损线通常是非色散的，这意味着波的相速度与频率无关。
 - 一般来说，一条线支持两种波，一种是由发生器提供的入射波，另一种是反射波
- 由负载。两个波的总和产生周期为 $\lambda/2$ 的驻波图案。电压驻波比 S 等于线路上最大与最小电压幅度之比，在匹配负载 ($Z_L = Z_0$) 的 1 到开路、短路或纯无功负载端接的线路的 ∞ 之间变化。
 - 短路或开路端接线路的输入阻抗是纯电抗性的。该特性可用于设计等效电感器和电容器。
 - 通过无损线路传送到负载的入射功率的部分等于 $(1 - |\Gamma|^2)$ 。
 - 史密斯圆图是分析传输线问题和设计阻抗匹配网络的有用图形工具。
 - 匹配网络放置在负载和馈电传输线之间，目的是消除对发电机的反射。匹配网络可以由电容器和/或电感器形式的集总元件组成，或者它可以由具有适当长度和终端的传输线部分组成。
 - 传输线上的脉冲瞬态分析可以使用反弹图图形技术来执行，该技术跟踪传输线负载端和发电机端的反射。

Mathematical and Physical Models

TEM Transmission Lines

$$L'C' = \mu \varepsilon$$
$$\frac{G'}{C'} = \frac{\sigma}{\varepsilon}$$
$$\alpha = \Re(\gamma) = \Re\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{Np/m})$$
$$\beta = \Im(\gamma) = \Im\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{rad/m})$$
$$Z_0 = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \quad (\Omega)$$
$$\Gamma = \frac{z_L - 1}{z_L + 1}$$

Step Function Transient Response

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}$$
$$V_\infty = \frac{V_g R_L}{R_g + R_L}$$
$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0}$$
$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$$

Lossless Line

$$\alpha = 0$$
$$\beta = \omega \sqrt{L'C'}$$
$$Z_0 = \sqrt{\frac{L'}{C'}}$$
$$u_p = \frac{1}{\sqrt{\mu \varepsilon}} \quad (\text{m/s})$$
$$\lambda = \frac{u_p}{f} = \frac{c}{f} \frac{1}{\sqrt{\varepsilon_r}} = \frac{\lambda_0}{\sqrt{\varepsilon_r}}$$
$$d_{\max} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}$$
$$d_{\min} = \frac{\theta_r \lambda}{4\pi} + \frac{(2n+1)\lambda}{4}$$
$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$
$$P_{\text{av}} = \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2]$$

数学与物理模型

TEM传输线

$$L'C' = \mu \varepsilon$$
$$\frac{G'}{C'} = \frac{\sigma}{\varepsilon}$$
$$\alpha = \Re(\gamma) = \Re\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{Np/m})$$
$$\beta = \Im(\gamma) = \Im\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{rad/m})$$
$$Z_0 = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \quad (\Omega)$$
$$\Gamma = \frac{z_L - 1}{z_L + 1}$$

阶跃激励的暂态响应

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}$$
$$V_\infty = \frac{V_g R_L}{R_g + R_L}$$
$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0}$$
$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$$

无损传输线

$$\alpha = 0$$
$$\beta = \omega \sqrt{L'C'}$$
$$Z_0 = \sqrt{\frac{L'}{C'}}$$
$$u_p = \frac{1}{\sqrt{\mu \varepsilon}} \quad (\text{m/s})$$
$$\lambda = \frac{u_p}{f} = \frac{c}{f} \frac{1}{\sqrt{\varepsilon_r}} = \frac{\lambda_0}{\sqrt{\varepsilon_r}}$$
$$d_{\max} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}$$
$$d_{\min} = \frac{\theta_r \lambda}{4\pi} + \frac{(2n+1)\lambda}{4}$$
$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$
$$P_{\text{av}} = \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2]$$

使用条件：波长式 $\lambda=\lambda_0/\sqrt{\varepsilon_r}$ 假设 $\mu_r=1$ 。驻波极大、极小位置须取整数 n ，使距离落在传输线实际范围内； $\Gamma=0$ 时无驻波极大、极小之分。

Important Terms

Provide definitions or explain the meaning of the following terms:

admittance Y	matched transmission line	standing wave
air line	matching network	standing-wave pattern
attenuation constant α	microstrip line	surface resistance R_s
bounce diagram	normalized impedance	susceptance B
characteristic impedance Z_0	normalized load reactance x_L	SWR circle
coaxial line	normalized load resistance r_L	telegrapher's equations
complex propagation constant γ	open-circuited line	TEM transmission lines
conductance G	optical fiber	time-average power P_{av}
current maxima and minima	parallel-plate line	transient response
dispersive transmission line	perfect conductor	transmission-line parameters
distortionless line	perfect dielectric	two-wire line
effective relative permittivity ϵ_{eff}	phase constant β	unit circle
guide wavelength λ	phase opposition	voltage maxima and minima
higher-order transmission lines	phase-shifted reflection coefficient Γ_d	voltage reflection coefficient Γ
impedance matching	quarter-wave transformer	voltage standing-wave ratio (VSWR or SWR) S
in-phase	short-circuited line	wave equations
input impedance Z_{in}	single-stub matching	wave impedance $Z(d)$
load impedance Z_L	slotted line	waveguide
lossless line	Smith chart	WTG and WTL
lumped-element model		

PROBLEMS

Sections 2-1 to 2-4: Transmission-Line Model

2.1 A transmission line of length l connects a load to a sinusoidal voltage source with an oscillation frequency f . Assuming that the velocity of wave propagation on the line is c , for which of the following situations is it reasonable to ignore the presence of the transmission line in the solution of the circuit?

- (a) $l = 20\text{ cm}$, $f = 20\text{ kHz}$
- (b) $l = 50\text{ km}$, $f = 60\text{ Hz}$
- (c) $l = 20\text{ cm}$, $f = 600\text{ MHz}$
- (d) $l = 1\text{ mm}$, $f = 100\text{ GHz}$

2.2 A two-wire copper transmission line is embedded in a dielectric material with $\epsilon_r = 2.6$ and $\sigma = 2 \times 10^{-6}\text{ S/m}$. Its wires are separated by 3 cm and their radii are 1 mm each.

- (a) Calculate the line parameters R' , L' , G' , and C' at 2 GHz.
- (b) Compare your results with those based on Module 2.1. Include a printout of the screen display.

2.3 Show that the transmission-line model shown in Fig. P2.3 yields the same telegrapher's equations given by Eqs. (2.14) and (2.16).

* Answer(s) available in Appendix E.

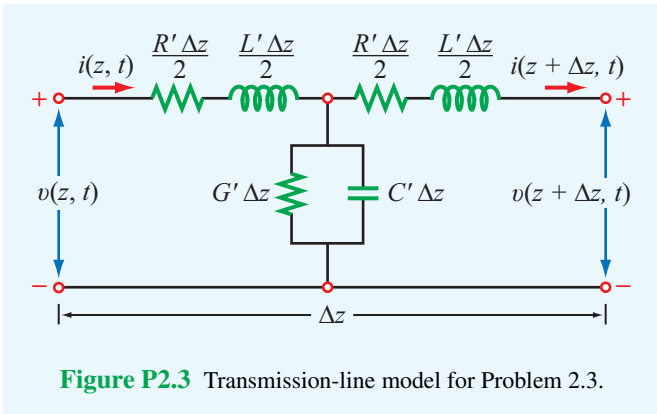


Figure P2.3 Transmission-line model for Problem 2.3.

2.4 A 1 GHz parallel-plate transmission line consists of 1.2-cm wide copper strips separated by a 0.15-cm thick layer of polystyrene. Appendix B gives $\mu_c = \mu_0 = 4\pi \times 10^{-7}\text{ H/m}$ and $\sigma_c = 5.8 \times 10^7\text{ (S/m)}$ for copper, and $\epsilon_r = 2.6$ for polystyrene. Use Table 2-1 to determine the line parameters of the transmission line. Assume that $\mu = \mu_0$ and $\sigma \approx 0$ for polystyrene.

2.5 For the parallel-plate transmission line of Problem 2.4, the line parameters are given by $R' = 1\text{ }\Omega/\text{m}$, $L' = 167\text{ nH/m}$, $G' = 0$, and $C' = 172\text{ pF/m}$. Find α , β , u_p , and Z_0 at 1 GHz.

重要术语 提供以下术语的定义或解释其含义：

导纳 Y 空气线路衰减常数 α 弹跳图 特性阻抗 Z_0 同轴线复传播常数 γ 电导 G 电流最大值和最小值色散传输线 无失真线路 有效相对介电常数 ϵ_{eff} 波导波长 λ 高阶传输线阻抗匹配 同相输入阻抗 Z_{in} 负载阻抗 Z_L 无损线路集总元件模型	匹配传输线 匹配网络 微带线 归一化阻抗 归一化负载电抗 x_L 归一化负载电阻 r_L 开路线 光纤 平行板线 理想导体 理想介电相位常数 β 反相移相反射系数 Γ_d 四分之一波变压器 短路线 单截线匹配 开槽线 史密斯圆图 驻波 驻波图 表面电阻 R_s 电纳 B SWR 圆电报方程 TEM 传输线 时间平均功率 P_{av} 瞬态响应 传输线参数 两线传输线单位圆 电压最大值和最小值 电压反射系数 Γ 电压驻波比 (VSWR 或 SWR) S 波方程 波阻抗 $Z(d)$ 波导 WTG 和 WTL
--	---

问题

第 2-1 节到 2-4 节：传输线模型

2.1 长度为 l 的传输线将负载连接到振荡频率为 f 的正弦电压源。假设波在线路上的传播速度为 c ，对于以下哪种情况，在电路解中忽略传输线的存在是合理的？

- (a) $l = 20\text{ 厘米}$, $f = 20\text{ kHz}$
- (b) $l = 50\text{ 公里}$, $f = 60\text{ Hz}$
- (c) $l = 20\text{ 厘米}$, $f = 600\text{ MHz}$
- (d) $l = 1\text{ 毫米}$, $f = 100\text{ GHz}$

2.2 将两线铜传输线嵌入 $\epsilon_r = 2.6$ 和 $\sigma = 2 \times 10^{-6}\text{ S/m}$ 的介电材料中。其电线间隔 3 厘米，半径各 1 毫米。

- (a) 计算 2 GHz 下的线路参数 R' 、 L' 、 G' 和 C' 。(b) 将您的结果与基于模块 2.1 的结果进行比较。包括屏幕显示的打印输出。

2.3 表明图 P2.3 中所示的传输线模型产生与方程 1 给出的相同的电报员方程。(2.14) 和 (2.16)。

* Answer(s) available in Appendix E.

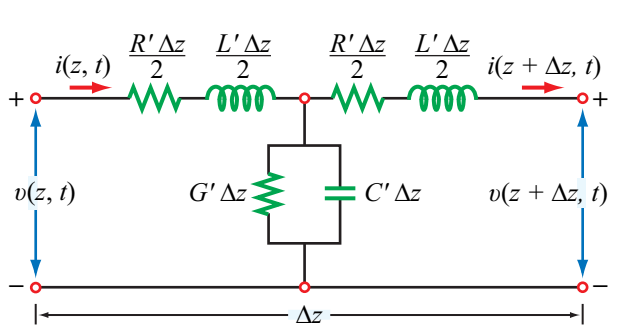


图 P2.3 问题 2.3 的传输线模型。

2.4 1 GHz 平行板传输线由 1.2 厘米宽的铜带组成，铜带之间被 0.15 厘米厚的聚苯乙烯层隔开。附录 B 给出了铜的 $\mu_c = \mu_0 = 4\pi \times 10^{-7}\text{ (H/m)}$ 和 $\sigma_c = 5.8 \times 10^7\text{ (S/m)}$ ，以及聚苯乙烯的 $\epsilon_r = 2.6$ 。使用表 2-1 确定传输线的线路参数。假设聚苯乙烯的 $\mu = \mu_0$ 和 $\sigma \approx 0$ 。

2.5 对于问题 2.4 的平行板传输线，线参数由 $R' = 1\text{ }\Omega/\text{m}$ 、 $L' = 167\text{ nH/m}$ 、 $G' = 0$ 和 $C' = 172\text{ pF/m}$ 给出。查找 1 GHz 处的 α 、 β 、 u_p 和 Z_0 。

2.6 A coaxial line with inner and outer conductor diameters of 0.5 cm and 1 cm, respectively, is filled with an insulating material with $\epsilon_r = 4.5$ and $\sigma = 10^{-3}$ S/m. The conductors are made of copper.

- (a) Calculate the line parameters at 1 GHz.
- (b) Compare your results with those based on Module 2.2. Include a printout of the screen display.

2.7 Find α , β , u_p , and Z_0 for the coaxial line of Problem 2.6. Verify your results by applying Module 2.2. Include a printout of the screen display.

***2.8** Find α , β , u_p , and Z_0 for the two-wire line of Problem 2.2. Compare results with those based on Module 2.1. Include a printout of the screen display.

Section 2-5: Microstrip

2.9 A lossless microstrip line uses a 1-mm wide conducting strip over a 1-cm thick substrate with $\epsilon_r = 2.5$. Determine the line parameters, ϵ_{eff} , Z_0 , and β at 10 GHz. Compare your results with those obtained by using Module 2.3. Include a printout of the screen display.

***2.10** Use Module 2.3 to design a 100 Ω microstrip transmission line. The substrate thickness is 1.8 mm and its $\epsilon_r = 2.3$. Select the strip width w , and determine the guide wavelength λ at $f = 5$ GHz. Include a printout of the screen display.

2.11 A 50 Ω microstrip line uses a 0.6 mm alumina substrate with $\epsilon_r = 9$. Use Module 2.3 to determine the required strip width w . Include a printout of the screen display.

2.12 Generate a plot of Z_0 as a function of strip width w over the range from 0.05 mm to 5 mm for a microstrip line fabricated on a 0.7-mm thick substrate with $\epsilon_r = 9.8$.

Section 2-6: Lossless Line

2.13 In addition to not dissipating power, a lossless line has two important features: (1) it is dispersionless (u_p is independent of frequency), and (2) its characteristic impedance Z_0 is purely real. Sometimes, it is not possible to design a transmission line such that $R' \ll \omega L'$ and $G' \ll \omega C'$, but it is possible to choose the dimensions of the line and its material properties to satisfy the condition

$$R'C' = L'G' \qquad (\text{distortionless line})$$

Such a line is called a **distortionless** line, because despite the fact that it is not lossless, it nonetheless possesses the

previously mentioned features of the lossless line. Show that for a distortionless line,

$$\alpha = R'\sqrt{\frac{C'}{L'}} = \sqrt{R'G'},$$

$$\beta = \omega\sqrt{L'C'},$$

$$Z_0 = \sqrt{\frac{L'}{C'}}.$$

***2.14** For a distortionless line (see Problem 2.13) with $Z_0 = 50\ \Omega$, $\alpha = 20$ (mNp/m), and $u_p = 2.5 \times 10^8$ (m/s), find the line parameters and λ at 100 MHz.

2.15 Find α and Z_0 of a distortionless line whose $R' = 2\ \Omega/\text{m}$ and $G' = 2 \times 10^{-4}$ S/m.

***2.16** A transmission line operating at 125 MHz has $Z_0 = 40\ \Omega$, $\alpha = 0.02$ (Np/m), and $\beta = 0.75$ rad/m. Find the line parameters R' , L' , G' , and C' .

2.17 Using a slotted line, the voltage on a lossless transmission line was found to have a maximum magnitude of 1.5 V and a minimum magnitude of 0.6 V. Find the magnitude of the load's reflection coefficient.

***2.18** Polyethylene with $\epsilon_r = 2.25$ is used as the insulating material in a lossless coaxial line with a characteristic impedance of 50 Ω . The radius of the inner conductor is 1.2 mm.

(a) What is the radius of the outer conductor?

(b) What is the phase velocity of the line?

2.19 A 50 Ω lossless transmission line is terminated in a load with impedance $Z_L = (30 - j50)\ \Omega$. The wavelength is 8 cm. Determine:

(a) The reflection coefficient at the load.

(b) The standing-wave ratio on the line.

(c) The position of the voltage maximum nearest the load.

(d) The position of the current maximum nearest the load.

(e) Verify quantities in parts (a)–(d) using Module 2.4. Include a printout of the screen display.

2.20 A 300 Ω lossless air transmission line is connected to a complex load composed of a resistor in series with an inductor, as shown in **Fig. P2.20**. At 5 MHz, determine: (a) Γ , (b) S , (c) location of voltage maximum nearest to the load, and (d) location of current maximum nearest to the load.

2.6 内、外导体直径分别为 0.5 cm 和 1 cm 的同轴线，填充有 $\epsilon_r = 4.5$ 和 $\sigma = 10^{-3}$ S/m 的绝缘材料。导体由铜制成。

(a) 计算 1 GHz 下的线路参数。

(b) 将您的结果与基于模块 2.2 的结果进行比较。包括屏幕显示的打印输出。

2.7 求问题2.6的同轴线的 α 、 β 、 u_p 和 Z_0 。通过应用模块 2.2 验证您的结果。包括屏幕显示的打印输出。

***2.8** 求问题 2.2 的两线线路的 α 、 β 、 u_p 和 Z_0 。将结果与基于模块 2.1 的结果进行比较。包括屏幕显示的打印输出。

第 2-5 节：微带线

2.9 无损微带线在 1 厘米厚的基板上使用 1 毫米宽的导电带， $\epsilon_r = 2.5$ 。确定 10 GHz 下的线路参数 ϵ_{eff} 、 Z_0 和 β 。将您的结果与使用模块 2.3 获得的结果进行比较。包括屏幕显示的打印输出。

***2.10** 使用模块 2.3 设计 100 Ω 微带传输线。基板厚度为 1.8 mm，其 $\epsilon_r = 2.3$ 。选择条带宽度 w ，并确定 $f = 5$ GHz 处的波导波长 λ 。包括屏幕显示的打印输出。

2.11 50 Ω 微带线使用带有 $\epsilon_r = 9$ 的 0.6 mm 氧化铝基板。使用模块 2.3 确定所需的带宽度 w 。包括屏幕显示的打印输出。

2.12 对于在 0.7 毫米厚的基板上制作的微带线， $\epsilon_r =$ 为 9.8，生成 Z_0 作为带宽 w 的函数在 0.05 毫米到 5 毫米范围内的图。

第 2-6 节：无损线路

2.13 除了不耗散功率之外，无损线路还有两个重要特征：(1) 无色散 (u_p 与频率无关)，(2) 其特征阻抗 Z_0 是纯实数。有时，不可能设计出满足 $R' \ll \omega L'$ 和 $G' \ll \omega C'$ 的传输线，但可以选择线的尺寸及其材料属性来满足条件

$$R'C' = L'G' \qquad (\text{distortionless line})$$

这样的线被称为无畸变线，因为尽管它不是无损的，但它仍然具有

前面提到了无损线路的特点。证明对于无畸变线，

$$\alpha = R'\sqrt{\frac{C'}{L'}} = \sqrt{R'G'},$$

$$\beta = \omega\sqrt{L'C'},$$

$$Z_0 = \sqrt{\frac{L'}{C'}}.$$

***2.14** 对于 $Z_0 = 50\ \Omega$ 、 $\alpha = 20$ (mNp/m) 和 $u_p = 2.5 \times 10^8$ (m/s) 的无失真线路（参见问题 2.13），求 100 MHz 处的线路参数和 λ 。

2.15 求一条无畸变线的 α 和 Z_0 ，其 $R' = 2\ \Omega/\text{m}$ 和 $G' = 2 \times 10^{-4}$ S/m。

***2.16** 工作在 125 MHz 的传输线的 $Z_0 = 40\ \Omega$ 、 $\alpha = 0.02$ (Np/m) 和 $\beta = 0.75$ rad/m。找到线参数 R' 、 L' 、 G' 和 C' 。

2.17 使用开槽线，发现无损传输线上的电压的最大值为 1.5 V，最小值为 0.6 V。求负载反射系数的大小。

***2.18** $\epsilon_r = 2.25$ 的聚乙烯用作特性阻抗为 50 Ω 的无损同轴线中的绝缘材料。内导体的半径为 1.2 毫米。

(a) 外导体的半径是多少？(b) 线的相速度是多少？

2.19 50 Ω 无损传输线的端接阻抗为 $Z_L = (30 - j50)\ \Omega$ 的负载。波长为 8 厘米。决定：

(a) 负载处的反射系数。

(b) 线上的驻波比。

(c) 最接近负载的电压最大值位置。

(d) 最接近负载的电流最大值的位置。(e) 使用模块 2.4 验证 (a)–(d) 部分的数量。包括屏幕显示的打印输出。

2.20 将一条 300 Ω 无损空气传输线连接到由电阻与电感串联组成的复杂负载上，如图 P2.20 所示。在 5 MHz 处，确定：(a) Γ 、(b) S 、(c) 最接近负载的电压最大值位置，以及 (d) 最接近负载的电流最大值位置。

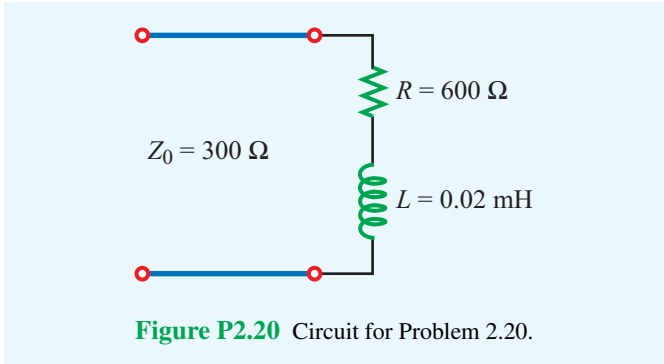


Figure P2.20 Circuit for Problem 2.20.

*2.21 On a 150 Ω lossless transmission line, the following observations were noted: distance of first voltage minimum from the load = 3 cm; distance of first voltage maximum from the load = 9 cm; and $S = 3$. Find Z_L .

2.22 Using a slotted line, the following results were obtained: distance of first minimum from the load = 4 cm; distance of second minimum from the load = 14 cm; and voltage standing-wave ratio = 1.5. If the line is lossless and $Z_0 = 50\ \Omega$, find the load impedance.

*2.23 A load with impedance $Z_L = (25 - j50)\ \Omega$ is to be connected to a lossless transmission line with the characteristic impedance Z_0 . Z_0 is chosen such that the standing-wave ratio is the smallest possible. What should Z_0 be?

2.24 A 50 Ω lossless line terminated in a purely resistive load has a voltage standing-wave ratio of 3. Find all possible values of Z_L .

2.25 Apply Module 2.4 to generate plots of the voltage standing-wave pattern for a 50 Ω line terminated in a load impedance $Z_L = (100 - j50)\ \Omega$. Set $V_g = 1\ \text{V}$, $Z_g = 50\ \Omega$, $\epsilon_r = 2.25$, $l = 40\ \text{cm}$, and $f = 1\ \text{GHz}$. Also determine S , $d_{\max}$, and $d_{\min}$.

2.26 A 50 Ω lossless transmission line is connected to a load composed of a 75 Ω resistor in series with a capacitor of unknown capacitance (Fig. P2.26). If at 10 MHz the voltage standing-wave ratio on the line was measured to be 3, determine the capacitance C .

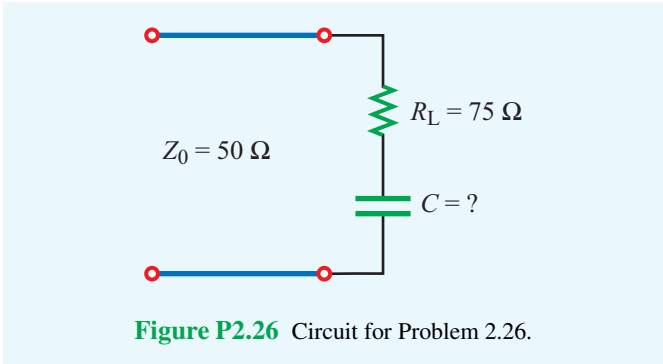


Figure P2.26 Circuit for Problem 2.26.

Section 2-7: Wave and Input Impedance

*2.27 At an operating frequency of 300 MHz, a lossless 50 Ω air-spaced transmission line 2.5 m in length is terminated with an impedance $Z_L = (40 + j20)\ \Omega$. Find the input impedance.

2.28 A lossless transmission line of electrical length $l = 0.35\lambda$ is terminated in a load impedance as shown in Fig. P2.28. Find Γ , S , and Z_{in} . Verify your results using Modules 2.4 or 2.5. Include a printout of the screen's output display.

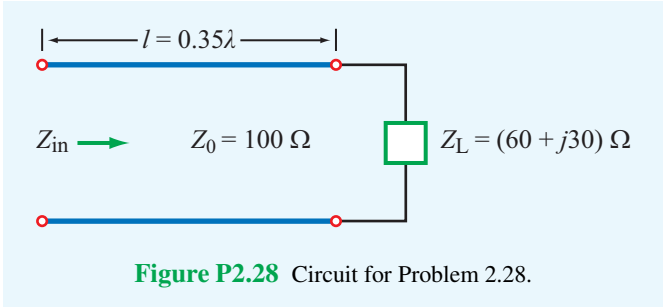


Figure P2.28 Circuit for Problem 2.28.

2.29 Show that the input impedance of a quarter-wavelength lossless line terminated in a short circuit appears as an open circuit.

2.30 Show that, at the position where the magnitude of the voltage on the line is a maximum, the input impedance is purely real.

2.31 A voltage generator with

$$v_g(t) = 5 \cos(2\pi \times 10^9 t) \text{ V}$$

and internal impedance $Z_g = 50\ \Omega$ is connected to a 50 Ω lossless air-spaced transmission line. The line length is 5 cm and the line is terminated in a load with impedance $Z_L = (100 - j100)\ \Omega$. Determine:

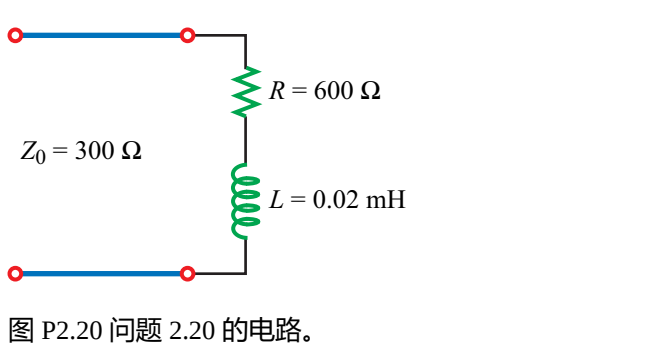


图 P2.20 问题 2.20 的电路。

*2.21 在 150 Ω 无损传输线上，注意到以下观察结果：第一电压最小值与负载 = 的距离为 3 cm；第一个电压最大值距负载 = 的距离 9 cm；和 $S = 3$ 。找到 Z_L 。

2.22 使用开槽线，获得以下结果：距负载 = 的第一个最小值的距离为 4 厘米；第二个最小值与负载的距离 = 14 cm；电压驻波比 = 1.5。如果线路是无损的并且 $Z_0 = 50\ \Omega$ ，则找到负载阻抗。

*2.23 阻抗为 $Z_L = (25 - j50)\ \Omega$ 的负载应连接到特性阻抗为 Z_0 的无损传输线上。选择 Z_0 以使驻波比尽可能最小。 Z_0 应该是什么？

2.24 端接于纯电阻负载的 50 Ω 无损线路的电压驻波比为 3。找出 Z_L 的所有可能值。

2.25 应用模块 2.4 生成终止于负载阻抗 $Z_L = (100 - j50)\ \Omega$ 的 50 Ω 线路的电压驻波模式图。设置 $V_g = 1\ \text{V}$ 、 $Z_g = 50\ \Omega$ 、 $\epsilon_r = 2.25$ 、 $l = 40\ \text{cm}$ 和 $f = 1\ \text{GHz}$ 。还要确定 S 、 $d_{\max}$ 和 $d_{\min}$ 。

2.26 一条 50 Ω 无损传输线连接到一个由 75 Ω 电阻器与未知电容的电容器串联组成的负载（图 P2.26）。如果在 10 MHz 时测得线路上的电压驻波比为 3，则确定电容 C 。

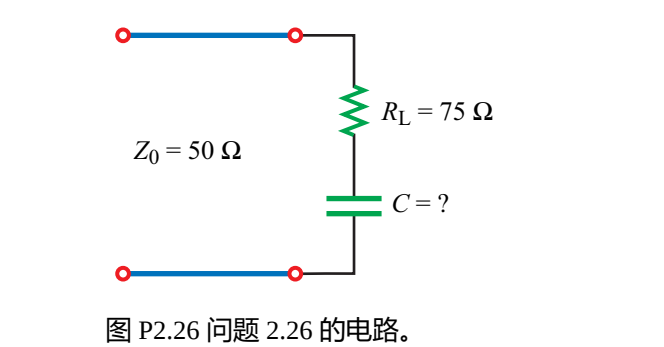


图 P2.26 问题 2.26 的电路。

第 2-7 节：波形和输入阻抗

*2.27 在 300 MHz 的工作频率下，长度为 2.5 m 的无损 50 Ω 空气间隔传输线的端接阻抗为 $Z_L = (40 + j20)\ \Omega$ 。求输入阻抗。

2.28 电长度 $l = 0.35\lambda$ 的无损传输线的负载阻抗如图 P2.28 所示。找到 Γ 、 S 和 Z_{in} 。使用模块 2.4 或 2.5 验证您的结果。包括屏幕输出显示的打印输出。

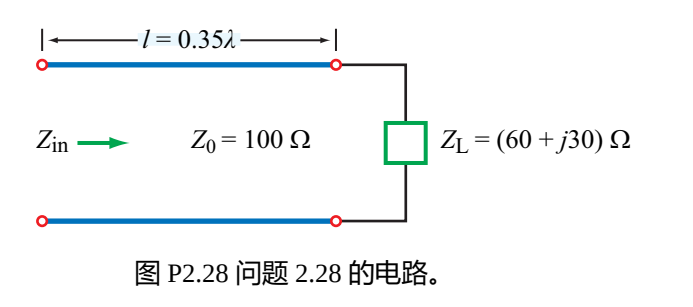


图 P2.28 问题 2.28 的电路。

2.29 证明以短路方式端接的四分之一波长无损线路的输入阻抗表现为开路。

2.30 证明，在线路上电压幅值最大的位置处，输入阻抗是纯实数。

2.31 电压发生器

$$v_g(t) = 5 \cos(2\pi \times 10^9 t) \text{ V}$$

内部阻抗 $Z_g = 50\ \Omega$ 连接至 50 Ω 无损空气间隔传输线。线路长度为 5 cm，线路终止于阻抗为 $Z_L = (100 - j100)\ \Omega$ 的负载。决定：

- *(a) Γ at the load.
- (b) Z_{in} at the input to the transmission line.
- (c) The input voltage $\tilde{V}_i$ and input current $\tilde{I}_i$.
- (d) The quantities in parts (a)–(c) using Modules 2.4 or 2.5.

2.32 A 6 m section of $150\ \Omega$ lossless line is driven by a source with

$$v_g(t) = 5 \cos(8\pi \times 10^7 t - 30^\circ) \quad (\text{V})$$

and $Z_g = 150\ \Omega$. If the line, which has a relative permittivity $\epsilon_r = 2.25$, is terminated in a load $Z_L = (150 - j50)\ \Omega$, determine:

- (a) λ on the line,
- *(b) the reflection coefficient at the load,
- (c) the input impedance,
- (d) the input voltage $\tilde{V}_i$,
- (e) the time-domain input voltage $v_i(t)$, and
- (f) quantities in parts (a)–(d) using Modules 2.4 or 2.5.

2.33 Two half-wave dipole antennas, each with an impedance of $75\ \Omega$, are connected in parallel through a pair of transmission lines, and the combination is connected to a feed transmission line, as shown in Fig. P2.33.

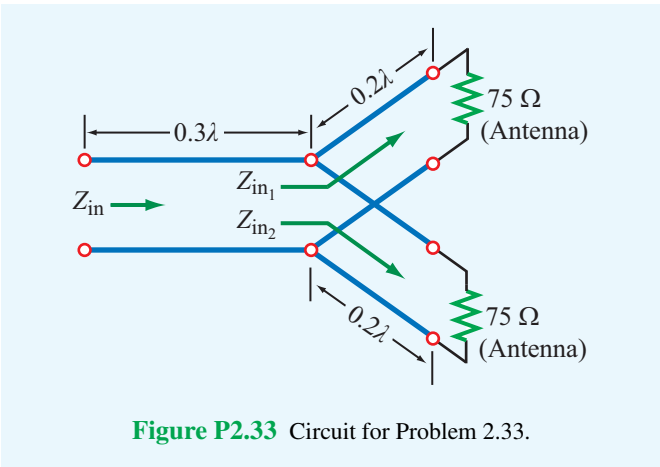


Figure P2.33 Circuit for Problem 2.33.

All lines are $50\ \Omega$ and lossless.

- *(a) Calculate Z_{in1} , the input impedance of the antenna-terminated line, at the parallel juncture.
- (b) Combine Z_{in1} and Z_{in2} in parallel to obtain Z'_L , the effective load impedance of the feedline.
- (c) Calculate Z_{in} of the feedline.

2.34 A $50\ \Omega$ lossless line is terminated in a load impedance $Z_L = (30 - j20)\ \Omega$.

- (a) Calculate Γ and S .
- (b) It has been proposed that by placing an appropriately selected resistor across the line at a distance d_{max} from the load (as shown in Fig. P2.34(b)), where d_{max} is the distance from the load of a voltage maximum, then it is possible to render $Z_i = Z_0$, thereby eliminating reflection back to the end. Show that the proposed approach is valid and find the value of the shunt resistance.

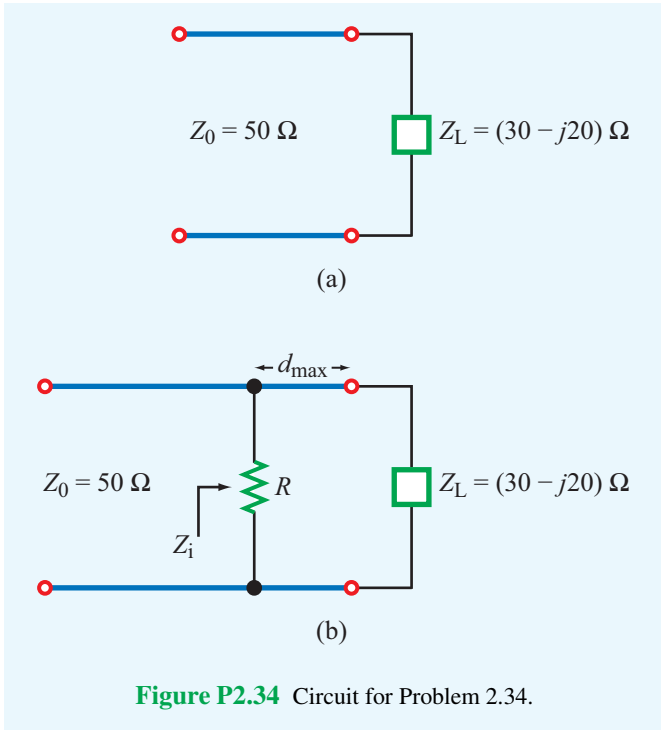


Figure P2.34 Circuit for Problem 2.34.

- *(b) For the lossless transmission line circuit shown in Fig. P2.35, determine the equivalent series lumped-element circuit at 400 MHz at the input to the line. The line has a characteristic impedance of $50\ \Omega$ and the insulating layer has $\epsilon_r = 2.25$.

- *(一个在负载处。
- (b) Z_{in} 位于传输线的输入端。
- (c) 输入电压 $\tilde{V}_i$ 和输入电流 $\tilde{I}_i$ 。
- (d) 使用模块 2.4 或 2.5 计算 (a)–(c) 部分的数量。

2.32 一段 6 m 的 $150\ \Omega$ 无损线路由具有

$$v_g(t) = 5 \cos(8\pi \times 10^7 t - 30^\circ) \quad (\text{V})$$

和 $Z_g = 150\ \Omega$ 。如果相对介电常数 $\epsilon_r = 2.25$ 的线路终止于负载 $Z_L = (150 - j50)\ \Omega$ ，则确定：

- (a) λ 上线,
- *(乙)负载下的反射系数,
- (c) 输入阻抗,
- (d) 输入电压 $\tilde{V}_i$,
- (e) 时域输入电压 $v_i(t)$, 以及 (f) 使用模块 2.4 或 2.5 计算 (a)–(d) 部分中的量。

2.33 两个阻抗为 $75\ \Omega$ 的半波偶极子天线通过一对传输线并联，并连接到馈电传输线，如图 P2.33 所示。

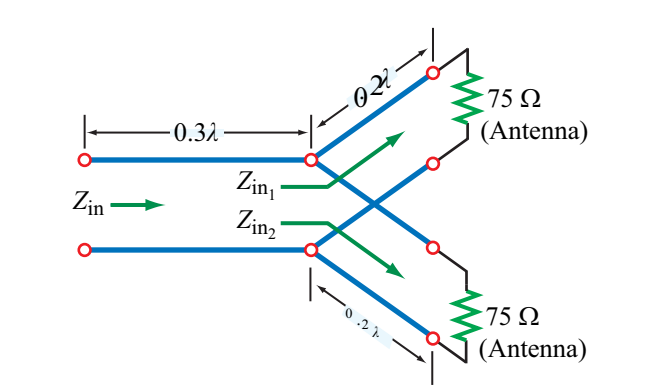


图 P2.33 问题 2.33 的电路。

所有线条均为 $50\ \Omega$ 且无损。

- *(a) 计算 Z_{in1} ，即天线端接线在平行接点处的输入阻抗。
- (b) 并联组合 Z_{in1} 和 Z_{in2} 以获得 Z'_L ，即馈线的有效负载阻抗。
- (c) 计算馈线的 Z_{in} 。

2.34 $50\ \Omega$ 无损线路的端接负载阻抗为 $Z_L = (30 - j20)\ \Omega$ 。

- (a) 计算 Γ 和 S 。
- (b) 有人建议，通过在距负载 d_{max} 的距离处在线上放置一个适当选择的电阻器（如图 P2.34(b) 所示），其中 d_{max} 是距电压最大值负载的距离，则可以呈现 $Z_i = Z_0$ ，从而消除回到末端的反射。证明所提出的方法是有效的并找出分流电阻的值。

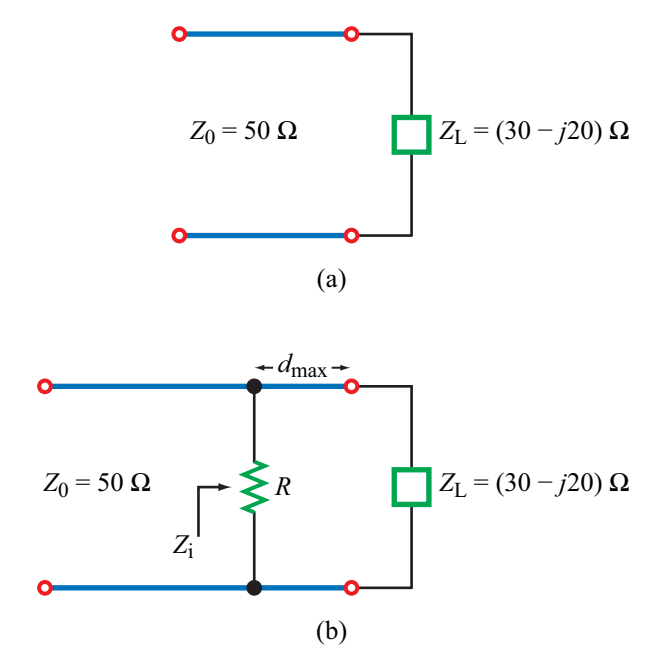
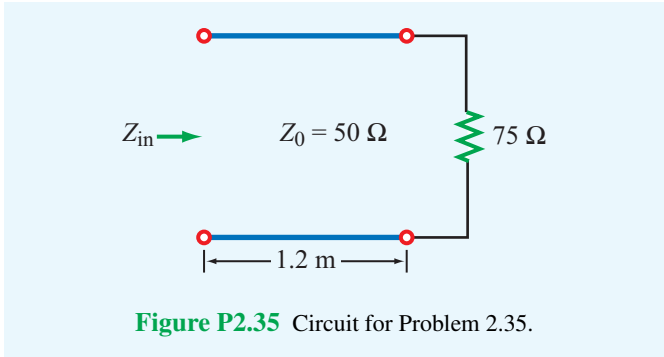


图 P2.34 问题 2.34 的电路。

- *(2.35 对于图 P2.35 所示的无损传输线电路，确定线路输入端 400 MHz 时的等效串联集总元件电路。线路的特性阻抗为 $50\ \Omega$ ，绝缘层的特性阻抗为 $\epsilon_r = 2.25$ 。



Section 2-8: Special Cases of the Lossless Line

2.36 At an operating frequency of 300 MHz, it is desired to use a section of a lossless 50 Ω transmission line terminated in a short circuit to construct an equivalent load with reactance $X = 40 \Omega$. If the phase velocity of the line is $0.75c$, what is the shortest possible line length that would exhibit the desired reactance at its input? Verify your result using Module 2.5.

***2.37** A lossless transmission line is terminated in a short circuit. How long (in wavelengths) should the line be for it to appear as an open circuit at its input terminals?

2.38 The input impedance of a 31-cm long lossless transmission line of unknown characteristic impedance was measured at 1 MHz. With the line terminated in a short circuit, the measurement yielded an input impedance equivalent to an inductor with inductance of $0.064 \mu\text{H}$, and when the line was open-circuited, the measurement yielded an input impedance equivalent to a capacitor with capacitance of 40 pF . Find Z_0 of the line, the phase velocity, and the relative permittivity of the insulating material.

***2.39** A 75Ω resistive load is preceded by a $\lambda/4$ section of a 50Ω lossless line, which itself is preceded by another $\lambda/4$ section of a 100Ω line. What is the input impedance? Compare your result with that obtained through two successive applications of Module 2.5.

2.40 A 100 MHz FM broadcast station uses a 300Ω transmission line between the transmitter and a tower-mounted half-wave dipole antenna. The antenna impedance is 73Ω . You are asked to design a quarter-wave transformer to match the antenna to the line.

- (a) Determine the electrical length and characteristic impedance of the quarter-wave section.
- (b) If the quarter-wave section is a two-wire line with $D = 2.5 \text{ cm}$ and the wires are embedded in polystyrene with $\epsilon_r = 2.6$, determine the physical length of the quarter-wave section and the radius of the two wire conductors.

2.41 A 50Ω lossless line of length $l = 0.375\lambda$ connects a 300 MHz generator with $\tilde{V}_g = 300 \text{ V}$ and $Z_g = 50 \Omega$ to a load Z_L . Determine the time-domain current through the load for:

- (a) $Z_L = (50 - j50) \Omega$
- * (b)** $Z_L = 50 \Omega$
- (c) $Z_L = 0$ (short circuit)

For part (a), verify your results by deducing the information you need from the output products generated by Module 2.4.

Section 2-9: Power Flow on Lossless Line

2.42 A generator with $\tilde{V}_g = 300 \text{ V}$ and $Z_g = 50 \Omega$ is connected to a load $Z_L = 75 \Omega$ through a 50Ω lossless line of length $l = 0.15\lambda$.

- * (a)** Compute Z_{in} , the input impedance of the line at the generator end.
- (b) Compute $\tilde{I}_i$ and $\tilde{V}_i$.
- (c) Compute the time-average power delivered to the line, $P_{in} = \frac{1}{2} \Re[\tilde{V}_i \tilde{I}_i^*]$.
- (d) Compute $\tilde{V}_L$, $\tilde{I}_L$, and the time-average power delivered to the load, $P_L = \frac{1}{2} \Re[\tilde{V}_L \tilde{I}_L^*]$. How does P_{in} compare with P_L ? Explain.
- (e) Compute the time-average power delivered by the generator, P_g , and the time-average power dissipated in Z_g . Is conservation of power satisfied?

2.43 If the two-antenna configuration shown in **Fig. P2.43** is connected to a generator with $\tilde{V}_g = 250 \text{ V}$ and $Z_g = 50 \Omega$, how much average power is delivered to each antenna?

***2.44** For the circuit shown in **Fig. P2.44**, calculate the average incident power, the average reflected power, and the average power transmitted into the infinite 100Ω line. The $\lambda/2$ line is lossless and the infinitely long line is slightly lossy. (*Hint:* The input impedance of an infinitely long line is equal to its characteristic impedance so long as $\alpha \neq 0$.)

2.45 The circuit shown in **Fig. P2.45** consists of a 100Ω lossless transmission line terminated in a load with $Z_L = (50 + j100) \Omega$. If the peak value of the load voltage was measured to be $|\tilde{V}_L| = 12 \text{ V}$, determine:

- * (a)** the time-average power dissipated in the load,
- (b) the time-average power incident on the line, and
- (c) the time-average power reflected by the load.

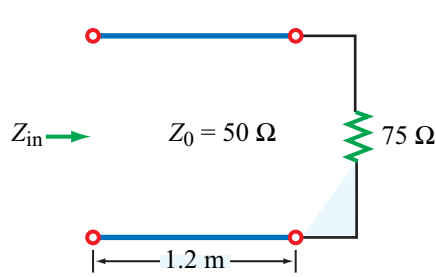


图 P2.35 问题 2.35 的电路。

第 2-8 节：无损线路的特殊情况

2.36 在 300 MHz 的工作频率下，希望使用一段以短路方式端接的无损 50 Ω 传输线来构建电抗 $X = 40 \Omega$ 的等效负载。如果线路的相速度为 $0.75c$ ，那么在其输入端表现出所需电抗的最短可能线路长度是多少？使用模块 2.5 验证您的结果。

***2.37** 无损传输线以短路终止。线路应该有多长（以波长为单位）才能在其输入端子处显示为开路？

2.38 在 1 MHz 下测量了 31 厘米长、未知特性阻抗的无损传输线的输入阻抗。当线路端接短路时，测量产生的输入阻抗相当于电感为 $0.064 \mu\text{H}$ 的电感器，当线路开路时，测量产生的输入阻抗相当于电容为 40 pF 的电容器。求线的 Z_0 、相速度和绝缘材料的相对介电常数。

***2.39** 75Ω 电阻负载之前是 50Ω 无损线路的 $\lambda/4$ 部分，而该线路本身又是 100Ω 线路的另一个 $\lambda/4$ 部分。输入阻抗是多少？将您的结果与通过两次连续应用模块 2.5 获得的结果进行比较。

2.40 100 MHz FM 广播电台在发射机和塔上安装的半波偶极天线之间使用 300Ω 传输线。天线阻抗为 73Ω 。您需要设计一个四分之一波长变压器以使天线与线路相匹配。

(a) 确定四分之一波长部分的电气长度和特性阻抗。(b) 如果四分之一波截面是 $D = 2.5 \text{ cm}$ 的两线线路，并且导线嵌入 $\epsilon_r = 2.6$ 的聚苯乙烯中，则确定四分之一波截面的物理长度和两根导线的半径。

2.41 长度为 $l = 0.375\lambda$ 的 50Ω 无损线路将具有 $\tilde{V}_g = 300 \text{ V}$ 和 $Z_g = 50 \Omega$ 的 300 MHz 发生器连接到负载 Z_L 。确定通过负载的时域电流：

- (a) $Z_L = (50 - j50) \Omega$
- * (b)** $Z_L = 50 \Omega$
- (c) $Z_L = 0$ (short circuit)

对于 (a) 部分，通过从模块 2.4 生成的输出结果中推导出所需的信息来验证您的结果。

第 2-9 节：无损线路上的功率流

2.42 $\tilde{V}_g = 300 \text{ V}$ 和 $Z_g = 50 \Omega$ 的发电机通过长度为 $l = 0.15\lambda$ 的 50Ω 无损线路连接到负载 $Z_L = 75 \Omega$ 。

*** (a)** 计算 Z_{in} ，即发电机端线路的输入阻抗。(b) 计算 $\tilde{I}_i$ 和 $\tilde{V}_i$ 。(c) 计算传输到线路的时间平均功率 $P_{in} = \frac{1}{2} \Re[\tilde{V}_i \tilde{I}_i^*]$ 。(d) 计算 $\tilde{V}_L$ 、 $\tilde{I}_L$ 以及输送到负载的时间平均功率 $P_L = \frac{1}{2} \Re[\tilde{V}_L \tilde{I}_L^*]$ 。 P_{in} 与 P_L 相比如何？解释。(e) 计算发电机提供的时间平均功率 P_g 和 Z_g 中消耗的时间平均功率。能量守恒定律是否得到满足？

2.43 如果图 P2.43 所示的双天线配置连接到 $\tilde{V}_g = 250 \text{ V}$ 和 $Z_g = 50 \Omega$ 的发电机，则向每个天线传输的平均功率是多少？

***2.44** 对于图 P2.44 所示的电路，计算平均入射功率、平均反射功率和传输到无限 100Ω 线的平均功率。 $\lambda/2$ 线是无损的，而无限长的线是轻微有损的。（提示：只要 $\alpha \neq 0$ ，无限长线路的输入阻抗就等于其特性阻抗。）

2.45 图 P2.45 所示的电路由一条 100Ω 无损传输线组成，该传输线终止于 $Z_L = (50 + j100) \Omega$ 的负载。如果测量负载电压的峰值为 $|\tilde{V}_L| = 12 \text{ V}$ ，则确定：

*** (a)** 负载中消耗的时间平均功率，(b) 线路上入射的时间平均功率，以及 (c) 负载反射的时间平均功率。

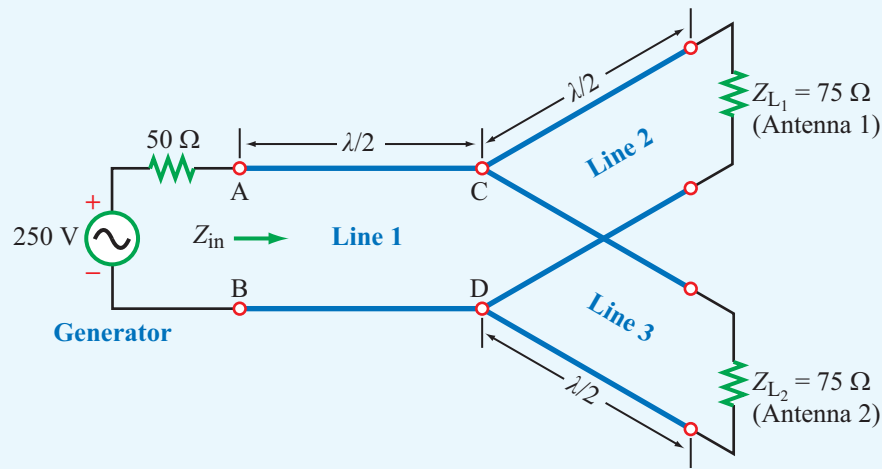


Figure P2.43 Antenna configuration for Problem 2.43.

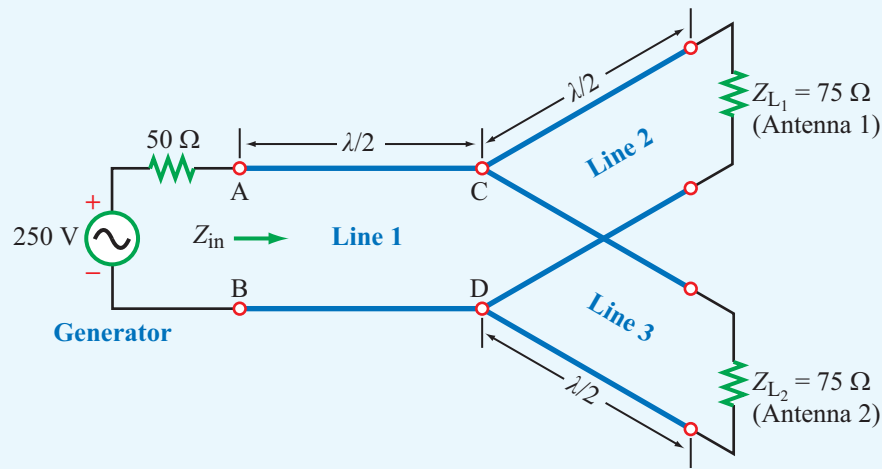


图 P2.43 习题2.43的天线连接。

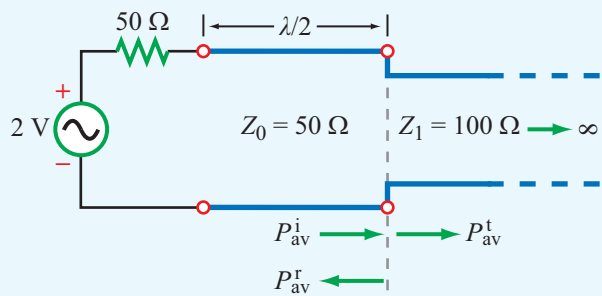


Figure P2.44 Circuit for Problem 2.44.

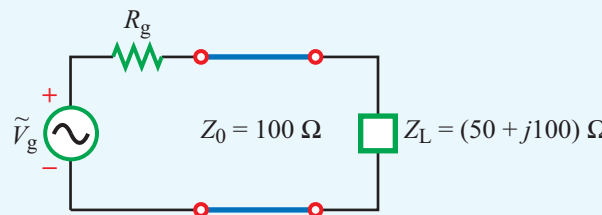


Figure P2.45 Circuit for Problem 2.45.

2.46 An antenna with a load impedance

$$Z_L = (75 + j25) \, \Omega$$

is connected to a transmitter through a $50 \, \Omega$ lossless transmission line. If under matched conditions ($50 \, \Omega$ load) the transmitter can deliver $20 \, \text{W}$ to the load, how much power can it deliver to the antenna? Assume that $Z_g = Z_0$.

Section 2-10: Smith Chart

2.47 Use the Smith chart to find the reflection coefficient corresponding to a load impedance of

- (a) $Z_L = 3Z_0$
- (b) $Z_L = (2 - j2)Z_0$
- (c) $Z_L = -j2Z_0$
- (d) $Z_L = 0$ (short circuit)

2.48 Repeat Problem 2.47 using Module 2.6.

2.49 Use the Smith chart to find the normalized load impedance corresponding to a reflection coefficient of

- (a) $\Gamma = 0.5$
- (b) $\Gamma = 0.5/\underline{60^\circ}$
- (c) $\Gamma = -1$
- (d) $\Gamma = 0.3/\underline{-30^\circ}$
- (e) $\Gamma = 0$
- (f) $\Gamma = j$

2.46 天线的负载阻抗为

$$Z_L = (75 + j25) \, \Omega$$

天线通过 $50 \, \Omega$ 无损传输线连接发射机。若匹配负载 ($50 \, \Omega$) 时发射机可向负载提供 $20 \, \text{W}$ ，接入该天线时可提供多少功率？设 $Z_g = Z_0$ 。

第2-10节 史密斯圆图

2.47 用史密斯圆图求下列负载阻抗对应的反射系数。

- (a) $Z_L = 3Z_0$
- (b) $Z_L = (2 - j2)Z_0$
- (c) $Z_L = -j2Z_0$
- (d) $Z_L = 0$ (短路)

2.48 用模块2.6重做习题2.47。

2.49 用史密斯圆图求下列反射系数对应的归一化负载阻抗。

- (a) $\Gamma = 0.5$
- (b) $\Gamma = 0.5/\underline{60^\circ}$
- (c) $\Gamma = -1$
- (d) $\Gamma = 0.3/\underline{-30^\circ}$
- (e) $\Gamma = 0$
- (f) $\Gamma = j$

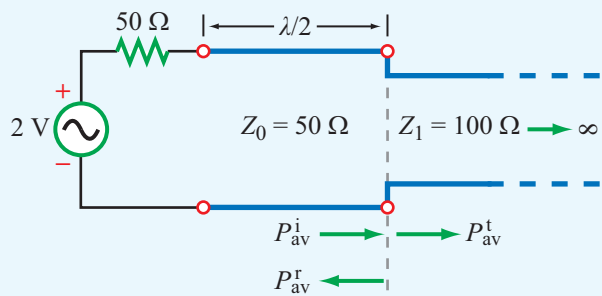


图 P2.44 习题2.44的电路。

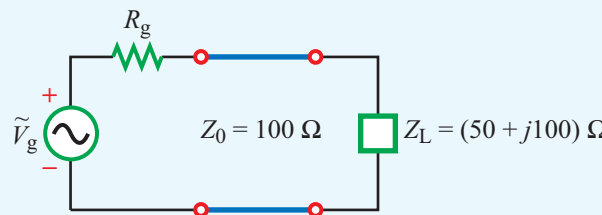


图 P2.45 习题2.45的电路。

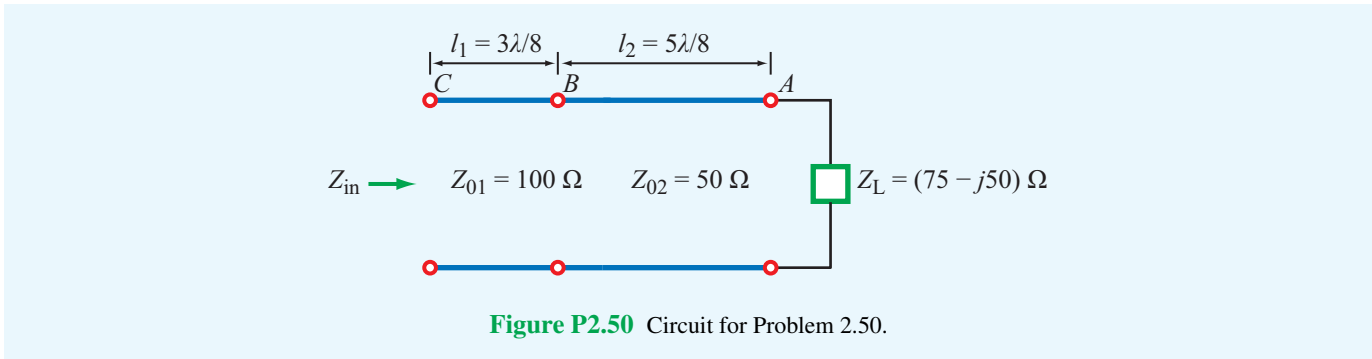


Figure P2.50 Circuit for Problem 2.50.

*2.50 Use the Smith chart to determine the input impedance Z_{in} of the two-line configuration shown in Fig. P2.50.

2.51 Repeat Problem 2.50 using Module 2.6.

*2.52 On a lossless transmission line terminated in a load $Z_L = 100 \Omega$, the standing-wave ratio was measured to be 2.5. Use the Smith chart to find the two possible values of Z_0 .

2.53 A lossless 50Ω transmission line is terminated in a load with $Z_L = (50 + j25) \Omega$. Use the Smith chart to find the following:

- (a) The reflection coefficient Γ .
- (b) The standing-wave ratio.
- (c) The input impedance at 0.35λ from the load.
- (d) The input admittance at 0.35λ from the load.
- (e) The shortest line length for which the input impedance is purely resistive.
- (f) The position of the first voltage maximum from the load.

2.54 Repeat Problem 2.53 using Module 2.6.

*2.55 A lossless 50Ω transmission line is terminated in a short circuit. Use the Smith chart to determine:

- (a) The input impedance at a distance 2.3λ from the load.
- (b) The distance from the load at which the input admittance is $Y_{in} = -j0.04 S$.

2.56 Repeat Problem 2.55 using Module 2.6.

*2.57 Use the Smith chart to find y_L if $z_L = 1.5 - j0.7$.

2.58 A lossless 100Ω transmission line $3\lambda/8$ in length is terminated in an unknown impedance. If the input impedance is $Z_{in} = -j2.5 \Omega$,

- (a) Use the Smith chart to find Z_L .
- (b) Verify your results using Module 2.6.

2.59 A 75Ω lossless line is 0.6λ long. If $S = 1.8$ and $\theta_r = -60^\circ$, use the Smith chart to find $|\Gamma|$, Z_L , and Z_{in} .

2.60 Repeat Problem 2.59 using Module 2.6.

*2.61 Using a slotted line on a 50Ω air-spaced lossless line, the following measurements were obtained: $S = 1.6$ and $|\tilde{V}|_{max}$ occurred only at 10 cm and 24 cm from the load. Use the Smith chart to find Z_L .

2.62 At an operating frequency of 5 GHz, a 50Ω lossless coaxial line with insulating material having a relative permittivity $\epsilon_r = 2.25$ is terminated in an antenna with an impedance $Z_L = 150 \Omega$. Use the Smith chart to find Z_{in} . The line length is 30 cm.

Section 2-11: Impedance Matching

*2.63 A 50Ω lossless line 0.6λ long is terminated in a load with $Z_L = (50 + j25) \Omega$. At 0.3λ from the load, a resistor with resistance $R = 30 \Omega$ is connected as shown in Fig. P2.63. Use the Smith chart to find Z_{in} .

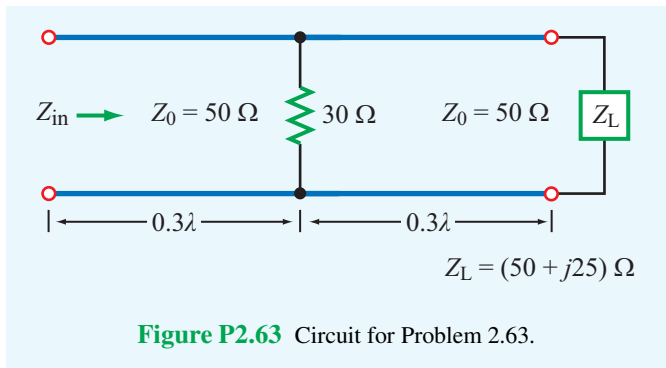


Figure P2.63 Circuit for Problem 2.63.

2.64 Use Module 2.7 to design a quarter-wavelength transformer to match a load with $Z_L = (100 - j200) \Omega$ to a 50Ω line.

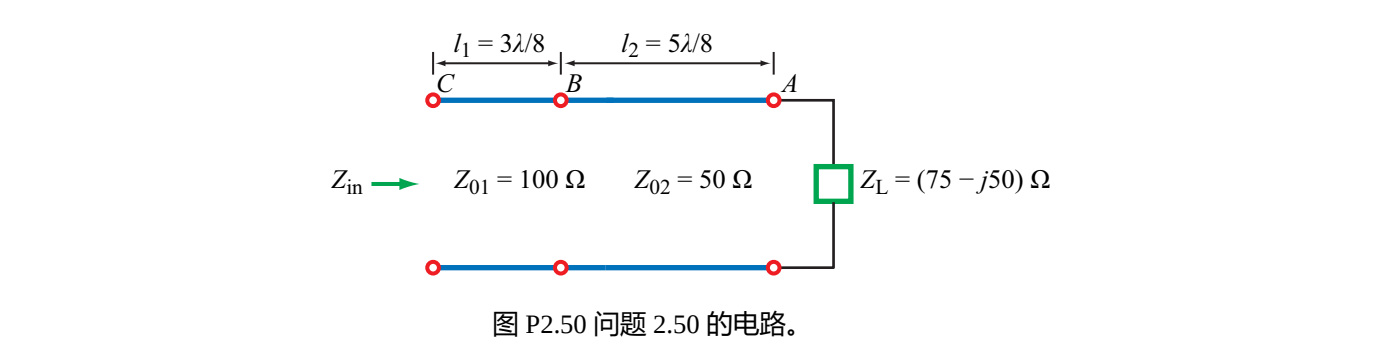


图 P2.50 问题 2.50 的电路。

*2.50 使用史密斯圆图确定图 P2.50 所示两线配置的输入阻抗 Z_{in} 。

2.51 使用模块 2.6 重复问题 2.50。

*2.52 在终止于负载 $Z_L = 100 \Omega$ 的无损传输线上，测得驻波比为 2.5。使用史密斯圆图找出 Z_0 的两个可能值。

2.53 无损 50Ω 传输线的端接负载为 $Z_L = (50 + j25) \Omega$ 。使用史密斯圆图查找以下内容：

- (a) 反射系数 Γ 。
- (b) 驻波比。
- (c) 负载 0.35λ 处的输入阻抗。
- (d) 负载 0.35λ 处的输入导纳。
- (e) 输入阻抗为纯电阻时的最短线路长度。
- (f) 负载的第一个电压最大值的位置。

2.54 使用模块 2.6 重复问题 2.53。

*2.55 无损 50Ω 传输线因短路而终止。使用史密斯圆图确定：

- (a) 距负载 2.3λ 处的输入阻抗。
- (b) 输入导纳为 $Y_{in} = -j0.04 S$ 时距负载的距离。

2.56 使用模块 2.6 重复问题 2.55。

*2.57 如果 $z_L = 1.5 - j0.7$ ，则使用史密斯圆图求 y_L 。

2.58 长度为 $3\lambda/8$ 的无损 100Ω 传输线以未知阻抗终止。如果输入阻抗为 $Z_{in} = -j2.5 \Omega$ ，

- (a) 使用史密斯圆图求 Z_L 。
- (b) 使用模块 2.6 验证您的结果。

2.59 75Ω 无损线的长度为 0.6λ 。如果 $S = 1.8$ 且 $\theta_r = -60^\circ$ ，则使用史密斯圆图求出 $|\Gamma|$ 、 Z_L 和 Z_{in} 。

2.60 使用模块 2.6 重复问题 2.59。

*2.61 使用 50Ω 空气间隔无损线路上的开缝线，获得以下测量结果： $S = 1.6$ 和 $|\tilde{V}|_{max}$ 仅发生在距离负载 10 厘米和 24 厘米处。使用史密斯圆图求出 Z_L 。

2.62 在 5 GHz 的工作频率下，采用相对介电常数 $\epsilon_r = 2.25$ 的绝缘材料的 50Ω 无损同轴线在阻抗 $Z_L = 150 \Omega$ 的天线中端接。使用史密斯圆图求出 Z_{in} 。线长为 30 厘米。

第 2-11 节：阻抗匹配

*2.63 一条 0.6λ 长的 50Ω 无损线路终止于 $Z_L = (50 + j25) \Omega$ 的负载。在距负载 0.3λ 处，连接一个电阻为 $R = 30 \Omega$ 的电阻，如图 P2.63 所示。使用史密斯圆图求出 Z_{in} 。

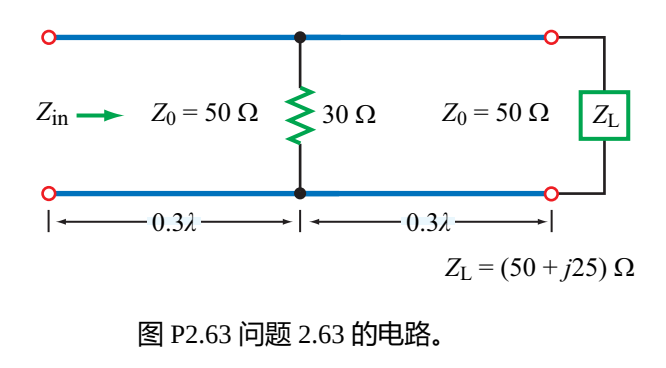


图 P2.63 问题 2.63 的电路。

2.64 使用模块 2.7 设计四分之一波长变压器，以将 $Z_L = (100 - j200) \Omega$ 的负载与 50Ω 线路相匹配。

2.65 Use Module 2.7 to design a quarter-wavelength transformer to match a load with $Z_L = (50 + j10) \Omega$ to a 100Ω line.

2.66 A 200Ω transmission line is to be matched to a computer terminal with $Z_L = (50 - j25) \Omega$ by inserting an appropriate reactance in parallel with the line. If $f = 800 \text{ MHz}$ and $\epsilon_r = 4$, determine the location nearest to the load at which inserting:

- (a) A capacitor can achieve the required matching, and the value of the capacitor.
- (b) An inductor can achieve the required matching, and the value of the inductor.

2.67 Repeat Problem 2.66 using Module 2.8.

2.68 A 50Ω lossless line is to be matched to an antenna with $Z_L = (75 - j20) \Omega$ using a shorted stub. Use the Smith chart to determine the stub length and distance between the antenna and stub.

***2.69** Repeat Problem 2.68 for a load with

$$Z_L = (100 + j50) \Omega.$$

2.70 Repeat Problem 2.68 using Module 2.9.

2.71 Repeat Problem 2.69 using Module 2.9.

2.72 Determine Z_{in} of the feed line shown in **Fig. P2.72**. All lines are lossless with $Z_0 = 50 \Omega$.

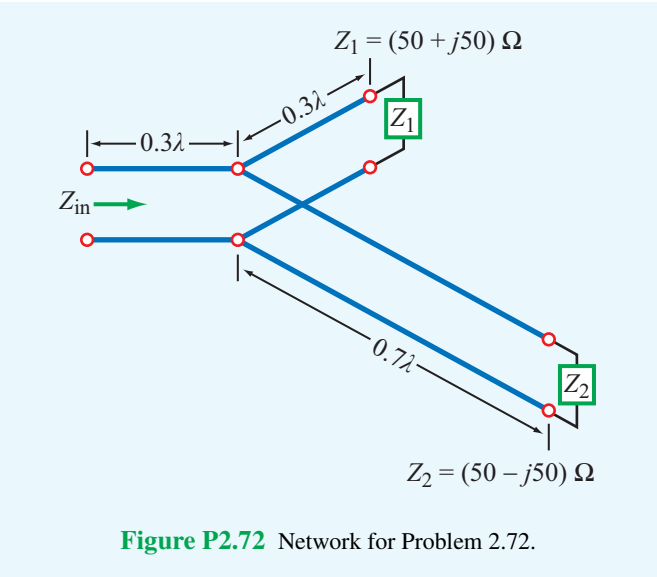


Figure P2.72 Network for Problem 2.72.

***2.73** Repeat Problem 2.72 for the case where all three transmission lines are $\lambda/4$ in length.

2.74 A 25Ω antenna is connected to a 75Ω lossless transmission line. Reflections back toward the generator can be eliminated by placing a shunt impedance Z at a distance l from the load (**Fig. P2.74**). Determine the values of Z and l .

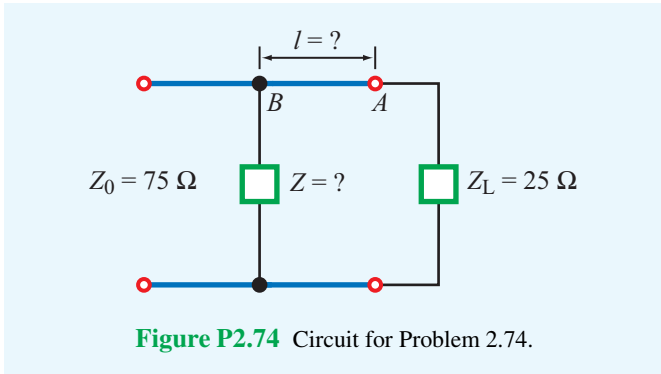


Figure P2.74 Circuit for Problem 2.74.

Section 2-12: Transients on Transmission Lines

2.75 Generate a bounce diagram for the voltage $V(z, t)$ for a 1-m long lossless line characterized by $Z_0 = 50 \Omega$ and $u_p = 2c/3$ (where c is the velocity of light) if the line is fed by a step voltage applied at $t = 0$ by a generator circuit with $V_g = 60 \text{ V}$ and $R_g = 100 \Omega$. The line is terminated in a load $R_L = 25 \Omega$. Use the bounce diagram to plot $V(t)$ at a point midway along the length of the line from $t = 0$ to $t = 25 \text{ ns}$.

2.76 Repeat Problem 2.75 for the current $I(z, t)$ on the line.

2.77 In response to a step voltage, the voltage waveform shown in **Fig. P2.77** was observed at the sending end of a lossless transmission line with $R_g = 50 \Omega$, $Z_0 = 50 \Omega$, and $\epsilon_r = 2.25$. Determine:

- (a) the generator voltage,
- (b) the length of the line, and
- (c) the load impedance.

***2.78** In response to a step voltage, the voltage waveform shown in **Fig. P2.78** was observed at the sending end of a shorted line with $Z_0 = 50 \Omega$ and $\epsilon_r = 4$. Determine V_g , R_g , and the line length.

2.65 使用模块 2.7 设计四分之一波长变压器，以将 $Z_L = (50 + j10) \Omega$ 的负载与 100Ω 线路相匹配。

2.66 一条 200Ω 传输线应通过与该线并联插入适当的电抗来与 $Z_L = (50 - j25) \Omega$ 的计算机终端相匹配。如果 $f = 800 \text{ MHz}$ 且 $\epsilon_r = 4$ ，则确定最接近负载的位置：

- (a) 一个电容器能否达到所需的匹配，以及电容器的值。
- (b) 电感能否达到所需的匹配，以及电感的值。

2.67 使用模块 2.8 重复问题 2.66。

2.68 50Ω 无损线路应使用短路短截线与 $Z_L = (75 - j20) \Omega$ 的天线相匹配。使用史密斯圆图确定短截线长度以及天线与短截线之间的距离。

*2.69 对负载重复问题 2.68

$$Z_L = (100 + j50) \Omega.$$

2.70 使用模块 2.9 重复问题 2.68。

2.71 使用模块 2.9 重复问题 2.69。

2.72 确定图P2.72所示进料线的 Z_{in} 。所有线都是无损的， $Z_0 = 50 \Omega$ 。

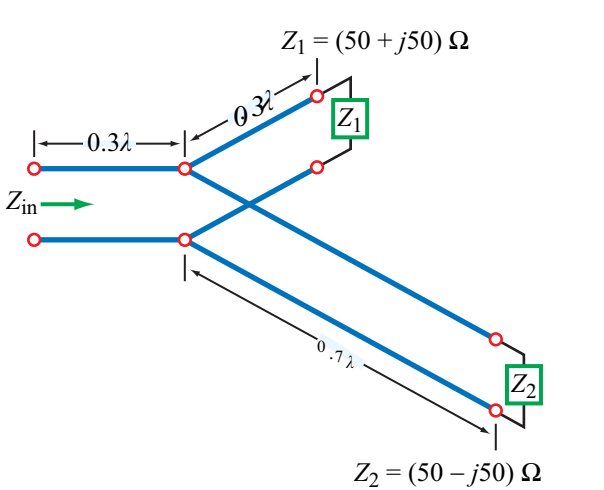


图 P2.72 问题 2.72 的网络。

*2.73 对于所有三个传输线长度均为 $\lambda/4$ 的情况，重复问题 2.72。

2.74 25Ω 天线连接到 75Ω 无损传输线。通过在距负载 l 处放置一个并联阻抗 Z 可以消除向发生器的反射（图 P2.74）。确定 Z 和 l 的值。

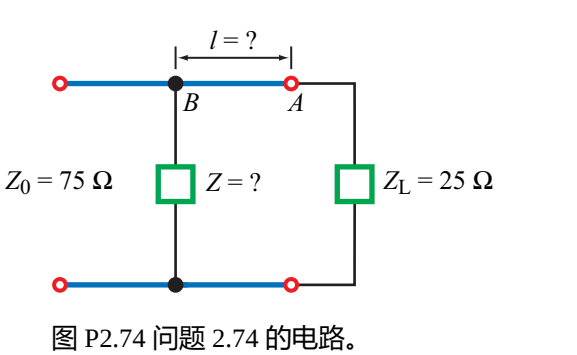


图 P2.74 问题 2.74 的电路。

第 2-12 节：传输线上的瞬变

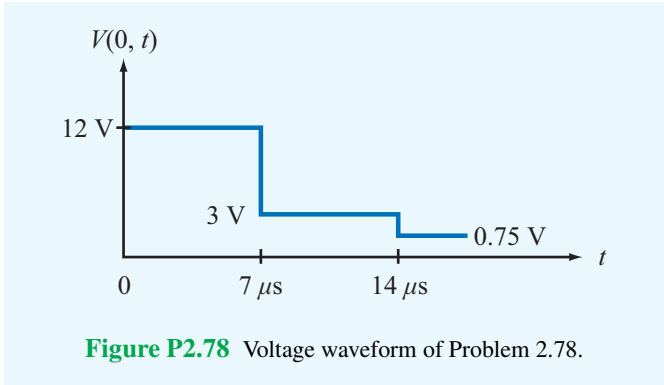
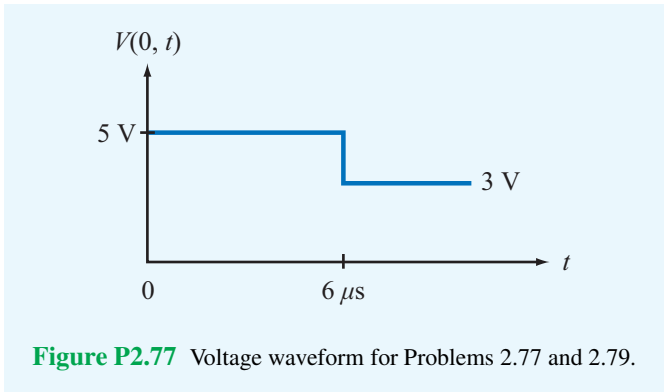
2.75 如果由 $V_g = 60 \text{ V}$ 和 $R_g = 100$ 的发生器电路在 $t = 0$ 处施加阶跃电压，为 1 m 长的无损线路生成电压 $V(z, t)$ 的反射图，其特征为 $Z_0 = 50 \Omega$ 和 $u_p = 2c/3$ （其中 c 是光速） Ω 。该线路终止于负载 $R_L = 25 \Omega$ 。使用反射图在从 $t = 0$ 到 $t = 25 \text{ ns}$ 的线长度中间的点处绘制 $V(t)$ 。

2.76 对线路上的电流 $I(z, t)$ 重复问题 2.75。

2.77 响应阶跃电压，在无损传输线的发送端观察到如图 P2.77 所示的电压波形，其中 $R_g = 50 \Omega$ 、 $Z_0 = 50 \Omega$ 和 $\epsilon_r = 2.25$ 。决定：

- (a) 发电机电压，(b) 线路长度，(c) 负载阻抗。

*2.78 响应阶跃电压，在 $Z_0 = 50 \Omega$ 和 $\epsilon_r =$ 短路线路的发送端观察到如图 P2.78 所示的电压波形。4. 确定 V_g 、 R_g 和线路长度。



2.79 Suppose the voltage waveform shown in **Fig. P2.77** was observed at the sending end of a $50\ \Omega$ transmission line in response to a step voltage introduced by a generator with $V_g = 15\text{ V}$ and an unknown series resistance R_g . The line is 1 km in length, its velocity of propagation is $1 \times 10^8\text{ m/s}$, and it is terminated in a load $R_L = 100\ \Omega$.

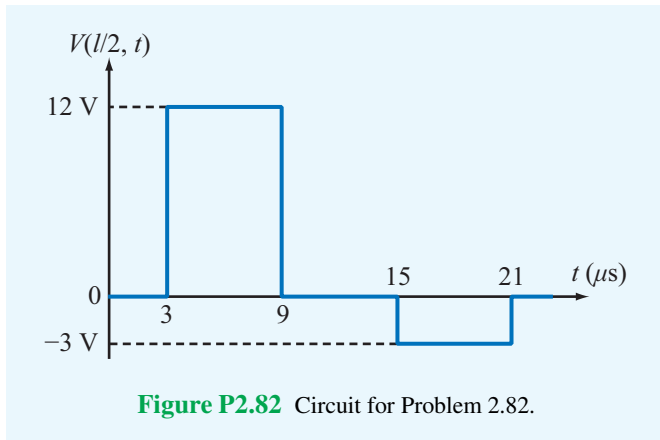
- (a) Determine R_g .
- (b) Explain why the drop in level of $V(0, t)$ at $t = 6\ \mu\text{s}$ cannot be due to reflection from the load.
- (c) Determine the shunt resistance R_f and location of the fault responsible for the observed waveform.

2.80 A generator circuit with $V_g = 200\text{ V}$ and $R_g = 25\ \Omega$ was used to excite a $75\ \Omega$ lossless line with a rectangular pulse of duration $\tau = 0.4\ \mu\text{s}$. The line is 200 m long, its $u_p = 2 \times 10^8\text{ m/s}$, and it is terminated in a load $R_L = 125\ \Omega$.

- (a) Synthesize the voltage pulse exciting the line as the sum of two step functions, $V_{g1}(t)$ and $V_{g2}(t)$.
- (b) For each voltage step function, generate a bounce diagram for the voltage on the line.
- (c) Use the bounce diagrams to plot the total voltage at the sending end of the line.
- (d) Confirm the result of part (c) by applying Module 2.10.

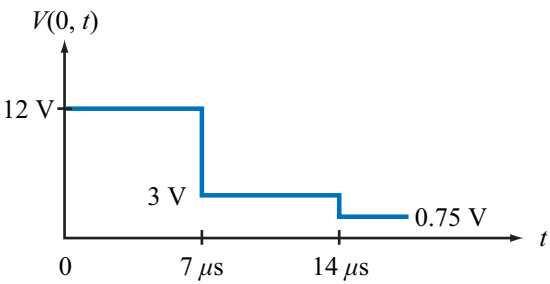
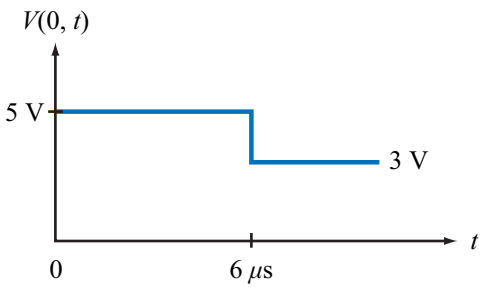
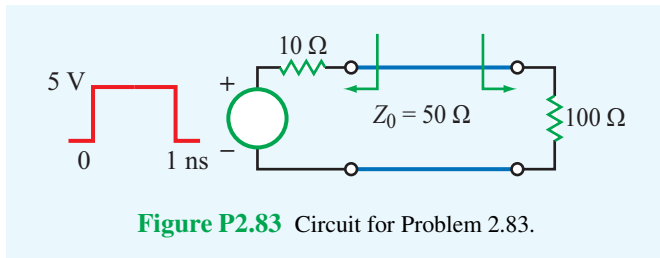
2.81 For the circuit of Problem 2.80, generate a bounce diagram for the current and plot its time history at the middle of the line.

***2.82** In response to a step voltage, the voltage waveform shown in **Fig. P2.82** was observed at the midpoint of a lossless transmission line with $Z_0 = 50\ \Omega$ and $u_p = 2 \times 10^8\text{ m/s}$. Determine: (a) the length of the line, (b) Z_L , (c) R_g , and (d) V_g .



2.83 The transmission-line circuit of **Fig. P2.83** is excited by a 1-ns long pulse that starts at $t = 0$. The line is 1 m long and the phase velocity of the line is $2 \times 10^8\text{ m/s}$. Generate:

- (a) a plot of the voltage across the line at $t = 6\text{ ns}$,
- (b) a plot of the voltage across the line at $t = 12\text{ ns}$,
- (c) a plot of the voltage as a function of time at the sending end of the line, and
- (d) a plot of the voltage as a function of time at the midpoint of the line.



2.79 假设在 $50\ \Omega$ 传输线的发送端观察到图 P2.77 所示的电压波形，以响应 $V_g = 15\text{ V}$ 和未知串联电阻 R_g 的发电机引入的阶跃电压。该线长 1 km，传播速度为 $1 \times 10^8\text{ m/s}$ ，终止于负载 $R_L = 100\ \Omega$ 。

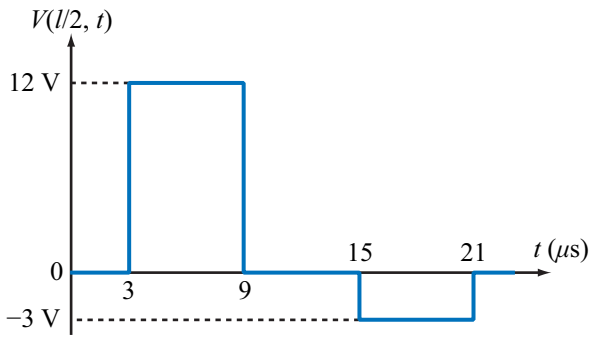
- (a) 确定 R_g 。(b) 解释为什么在 $t = 6\ \mu\text{s}$ 时 $V(0, t)$ 电平的下降不可能是由于负载的反射造成的。(c) 确定分流电阻 R_f 和导致观察到的波形的故障位置。

2.80 使用 $V_g = 200\text{ V}$ 和 $R_g = 25\ \Omega$ 的发电机电路，用持续时间为 $\tau = 0.4\ \mu\text{s}$ 的矩形脉冲激励 $75\ \Omega$ 无损线。该线路长 200 m，其 $u_p = 2 \times 10^8\text{ m/s}$ ，终止于负载 $R_L = 125\ \Omega$ 。

- (a) 将激励线路的电压脉冲合成为两个阶跃函数 $V_{g1}(t)$ 和 $V_{g2}(t)$ 之和。(b) 对于每个电压阶跃函数，生成线路上电压的反弹图。(c) 使用反弹图绘制线路发送端的总电压。(d) 通过应用模块 2.10 确认 (c) 部分的结果。

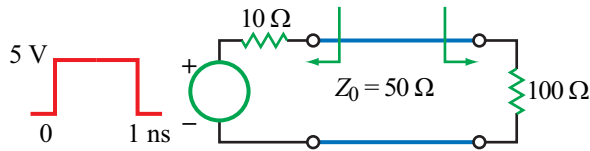
2.81 对于问题 2.80 的电路，生成电流的跳动图，并在线的中间绘制其时间历史。

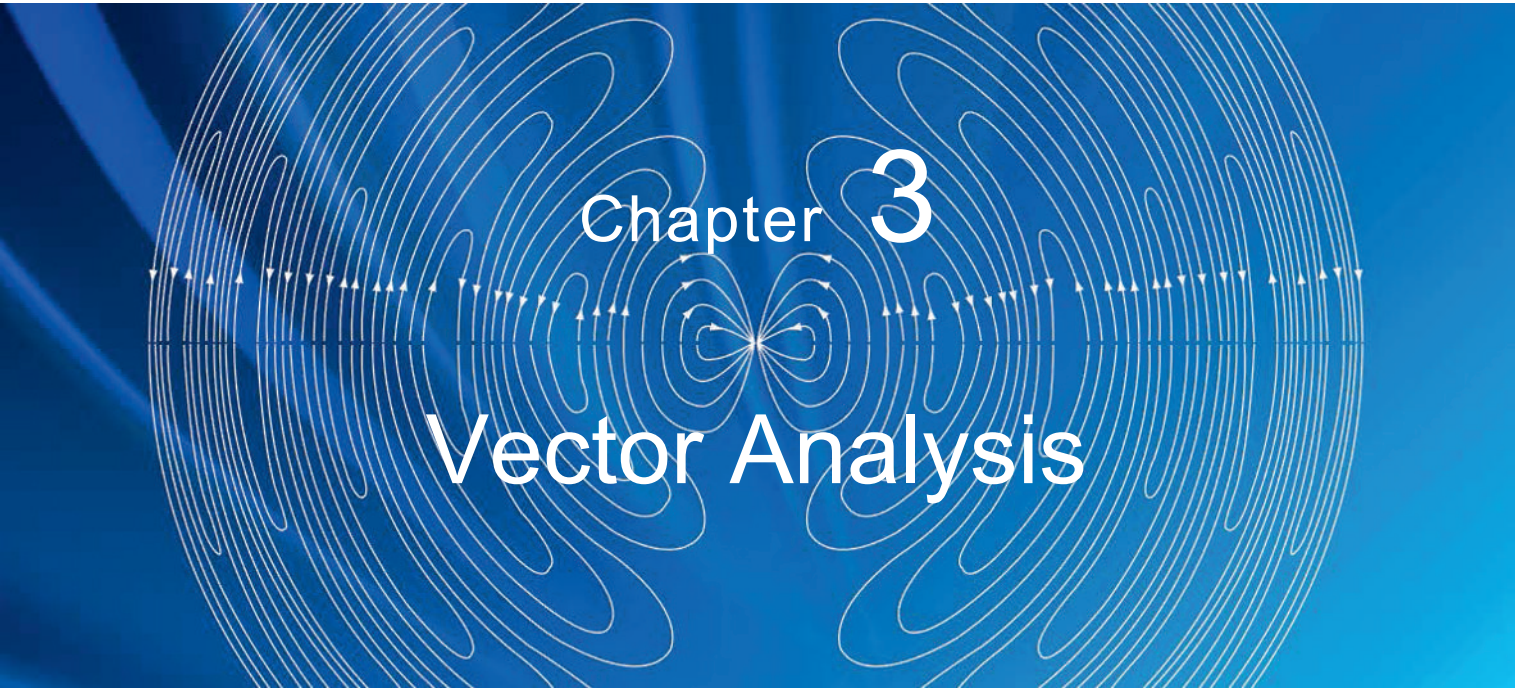
***2.82** 响应阶跃电压，在 $Z_0 = 50\ \Omega$ 和 $u_p = 2 \times 10^8\text{ m/s}$ 的无损传输线中点观察到图 P2.82 所示的电压波形。确定：(a) 线的长度、(b) Z_L 、(c) R_g 和 (d) V_g 。



2.83 图 P2.83 的传输线电路由从 $t = 0$ 开始的 1 ns 长脉冲激励。传输线长 1 m，相速度为 $2 \times 10^8\text{ m/s}$ 。产生：

- (a) $t = 6\text{ ns}$ 时线路两端电压的图，(b) $t = 12\text{ ns}$ 时线路两端电压的图，(c) 线路发送端电压随时间变化的图，(d) 线路中点处电压随时间变化的图。





Chapter Contents

	Overview, 131
3-1	Basic Laws of Vector Algebra, 131
3-2	Orthogonal Coordinate Systems, 137
3-3	Transformations between Coordinate Systems, 143
3-4	Gradient of a Scalar Field, 147
TB5	Global Positioning System, 151
3-5	Divergence of a Vector Field, 153
3-6	Curl of a Vector Field, 157
TB6	X-Ray Computed Tomography, 159
3-7	Laplacian Operator, 162
	Chapter 3 Summary, 163
	Problems, 164

Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Use vector algebra in Cartesian, cylindrical, and spherical coordinate systems.
2. Transform vectors between the three primary coordinate systems.
3. Calculate the gradient of a scalar function and the divergence and curl of a vector function in any of the three primary coordinate systems.
4. Apply the divergence theorem and Stokes’s theorem.



章节内容

概述, 131 3-1 矢量代数基本定律, 131 3-2 正交坐标系, 137 3-3 坐标系之间的变换, 143 3-4 标量场的梯度, 147 TB5 全球定位系统, 151 3-5 矢量场的散度, 153 3-6 矢量场的旋度, 157 TB6 X 射线计算机断层扫描, 159 3-7 拉普拉斯算子, 162 第 3 章总结, 163 问题, 164

目标

学习本章介绍的材料后, 您应该能够:

1. 在笛卡尔坐标系、圆柱坐标系和球坐标系中使用矢量代数。
2. 在三个主坐标系之间变换向量。
3. 计算三个主坐标系中任意一个标量函数的梯度以及向量函数的散度和旋度。
4. 应用散度定理和斯托克斯定理。

Overview

In our examination of wave propagation on a transmission line in Chapter 2, the primary quantities we worked with were voltage, current, impedance, and power. Each of these is a **scalar** quantity, meaning that it can be completely specified by its magnitude if it is a positive real number or by its magnitude and phase angle if it is a negative or a complex number (a negative number has a positive magnitude and a phase angle of π (rad)). This chapter is concerned with vectors. A **vector** has a magnitude and a direction. The speed of an object is a scalar, whereas its velocity is a vector.

Starting in the next chapter and throughout the succeeding chapters in this book, the primary electromagnetic quantities we deal with are the electric and magnetic fields, **E** and **H**. These, and many other related quantities, are vectors. Vector analysis provides the mathematical tools necessary for expressing and manipulating vector quantities in an efficient and convenient manner. To specify a vector in three-dimensional space, it is necessary to specify its components along each of the three directions.

► Several types of coordinate systems are used in the study of vector quantities, the most common being the Cartesian (or rectangular), cylindrical, and spherical systems. A particular coordinate system is usually chosen to best suit the geometry of the problem under consideration. ◀

Vector algebra governs the laws of addition, subtraction, and “multiplication” of vectors. The rules of vector algebra and vector representation in each of the aforementioned orthogonal coordinate systems (including vector transformation between them) are two of the three major topics treated in this chapter. The third topic is **vector calculus**, which encompasses the laws of differentiation and integration of vectors, the use of special vector operators (gradient, divergence, and curl), and the application of certain theorems that are particularly useful in the study of electromagnetics, most notably the divergence and Stokes’s theorems.

3-1 Basic Laws of Vector Algebra

A vector is a mathematical object that resembles an arrow. Vector **A** in Fig. 3-1 has **magnitude** (or length) $A = |\mathbf{A}|$ and **unit vector** $\hat{\mathbf{a}}$:

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{a}}A. \tag{3.1}$$

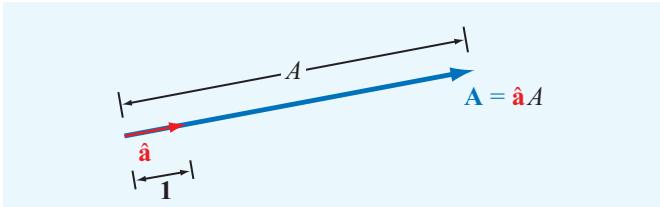


Figure 3-1 Vector $\mathbf{A} = \hat{\mathbf{a}}A$ has magnitude $A = |\mathbf{A}|$ and points in the direction of unit vector $\hat{\mathbf{a}} = \mathbf{A}/A$.

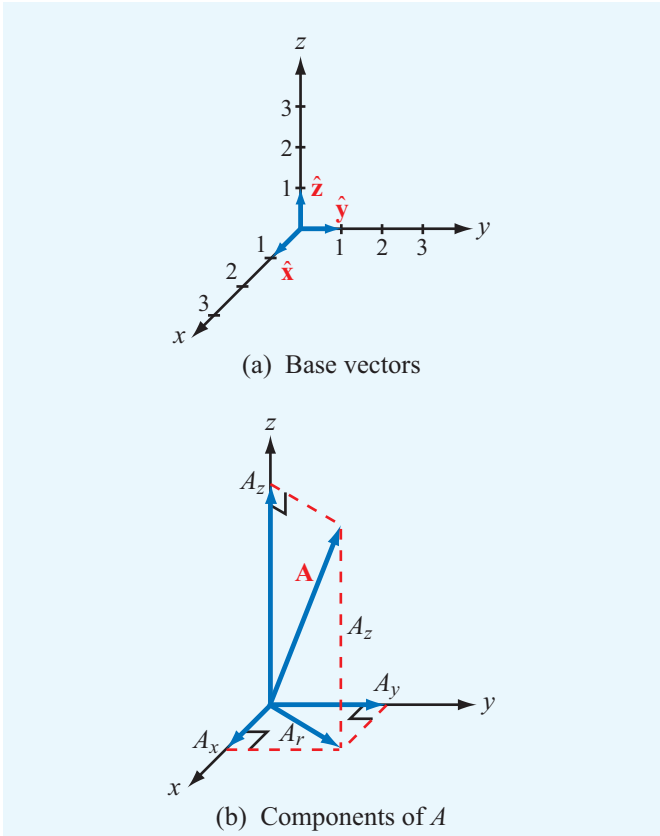


Figure 3-2 Cartesian coordinate system: (a) base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ and (b) components of vector **A**.

The unit vector $\hat{\mathbf{a}}$ has a magnitude of one ($|\hat{\mathbf{a}}| = 1$) and points from **A**’s tail or anchor to its head or tip. From Eq. (3.1),

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A}. \tag{3.2}$$

In the Cartesian (or rectangular) coordinate system shown in Fig. 3-2(a), the x , y , and z coordinate axes extend along

概述

在第 2 章对传输线上的波传播的研究中，我们使用的主要量是电压、电流、阻抗和功率。其中每一个都是标量，这意味着如果它是正实数，则可以完全通过其大小来指定；如果它是负数或复数，则可以通过其大小和相位角来完全指定（负数具有正大小和 π (rad) 的相位角）。本章涉及向量。向量具有大小和方向。物体的速度是标量，而速度是矢量。

从下一章开始以及本书的后续章节，我们处理的主要电磁量是电场和磁场，**E**和**H**。这些以及许多其他相关量都是矢量。矢量分析提供了以高效、便捷的方式表达和操作矢量所需的数学工具。要在三维空间中指定一个向量，必须指定其沿三个方向的分量。

► 在矢量研究中使用了几种类型的坐标系，最常见的是笛卡尔（或矩形）坐标系、圆柱坐标系和球坐标系。通常选择一个特定的坐标系来最适合所考虑问题的几何形状。 ◀

向量代数控制向量的加法、减法和“乘法”定律。上述每个正交坐标系中的向量代数和向量表示（包括它们之间的向量变换）的规则是本章讨论的三个主要主题中的两个。第三个主题是向量微积分，其中包括向量的微分和积分定律、特殊向量算子（梯度、散度和旋度）的使用，以及某些在电磁学研究中特别有用的定理的应用，尤其是散度和斯托克斯定理。

3-1 向量代数的基本定律

矢量是一个类似于箭头的数学对象。图 3-1 中的向量 **A** 具有大小（或长度） $A = |\mathbf{A}|$ 和单位向量 $\hat{\mathbf{a}}$:

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{a}}A. \tag{3.1}$$

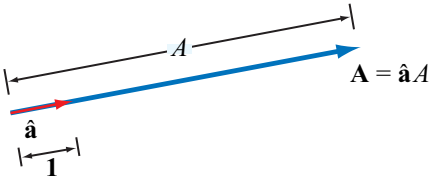


图 3-1 向量 $\mathbf{A} = \hat{\mathbf{a}}A$ 的大小为 $A = |\mathbf{A}|$ ，并指向单位向量 $\hat{\mathbf{a}} = \mathbf{A}/A$ 的方向。

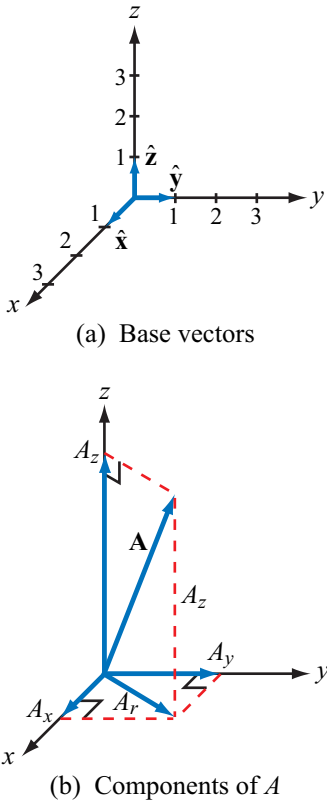


图 3-2 笛卡尔坐标系：(a) 基向量 $\hat{\mathbf{x}}$ 、 $\hat{\mathbf{y}}$ 和 $\hat{\mathbf{z}}$ 以及 (b) 向量 **A** 的分量。

单位向量 $\hat{\mathbf{a}}$ 的大小为 1 ($|\hat{\mathbf{a}}| = 1$)，并从 **A** 的尾部或锚点指向其头部或尖端。从方程。 (3.1),

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A}. \tag{3.2}$$

在图3-2(a)所示的笛卡尔（或直角）坐标系中， x 、 y 、 z 坐标轴沿

directions of the three mutually perpendicular unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$, which are also called **base vectors**. The vector $\mathbf{A}$ in Fig. 3-2(b) may be decomposed as

$$\mathbf{A} = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \quad (3.3)$$

where A_x , A_y , and A_z are $\mathbf{A}$'s scalar components along the x -, y -, and z axes, respectively. The component A_z is equal to the perpendicular projection of $\mathbf{A}$ onto the z axis, and similar definitions apply to A_x and A_y . Application of the Pythagorean theorem—first to the right triangle in the x - y plane to express the hypotenuse A_r in terms of A_x and A_y and then again to the vertical right triangle with sides A_r and A_z and hypotenuse A —yields the following expression for the magnitude of $\mathbf{A}$:

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}. \quad (3.4)$$

Since A is a nonnegative scalar, only the positive root applies. From Eq. (3.2), the unit vector $\hat{\mathbf{a}}$ is

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = \frac{\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}. \quad (3.5)$$

Occasionally, we use the shorthand notation $\mathbf{A} = (A_x, A_y, A_z)$ to denote a vector with components A_x , A_y , and A_z in a Cartesian coordinate system.

3-1.1 Equality of Two Vectors

Two vectors $\mathbf{A}$ and $\mathbf{B}$ are equal if they have equal magnitudes and identical unit vectors. Thus, if

$$\mathbf{A} = \hat{\mathbf{a}}A = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \quad (3.6a)$$

$$\mathbf{B} = \hat{\mathbf{b}}B = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \quad (3.6b)$$

then $\mathbf{A} = \mathbf{B}$ if and only if $A = B$ and $\hat{\mathbf{a}} = \hat{\mathbf{b}}$, which requires that $A_x = B_x$, $A_y = B_y$, and $A_z = B_z$.

► Equality of two vectors does not necessarily imply that they are identical; in Cartesian coordinates, two displaced parallel vectors of equal magnitude and pointing in the same direction are equal, but they are identical only if they lie on top of one another. ◀

3-1.2 Vector Addition and Subtraction

The sum of two vectors $\mathbf{A}$ and $\mathbf{B}$ is a vector

$$\mathbf{C} = \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z,$$

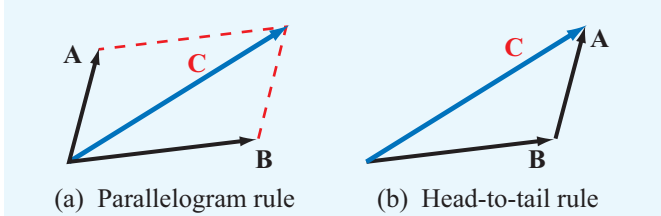


Figure 3-3 Vector addition by (a) the parallelogram rule and (b) the head-to-tail rule.

given by

$$\begin{aligned} \mathbf{C} = \mathbf{A} + \mathbf{B} &= (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z) + (\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z) \\ &= \hat{\mathbf{x}}(A_x + B_x) + \hat{\mathbf{y}}(A_y + B_y) + \hat{\mathbf{z}}(A_z + B_z) \\ &= \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z, \end{aligned} \quad (3.7)$$

with $C_x = A_x + B_x$, etc.

► Vector addition is commutative:

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}. \quad (3.8)$$

Graphically, vector addition can be accomplished by either the parallelogram or the head-to-tail rule (Fig. 3-3). Vector $\mathbf{C}$ is the diagonal of the parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$. With the head-to-tail rule, we may either add $\mathbf{A}$ to $\mathbf{B}$ or $\mathbf{B}$ to $\mathbf{A}$. When $\mathbf{A}$ is added to $\mathbf{B}$, it is repositioned so that its tail starts at the tip of $\mathbf{B}$ while keeping its length and direction unchanged. The sum vector $\mathbf{C}$ starts at the tail of $\mathbf{B}$ and ends at the tip of $\mathbf{A}$.

Subtraction of vector $\mathbf{B}$ from vector $\mathbf{A}$ is equivalent to the addition of $\mathbf{A}$ to negative $\mathbf{B}$. Thus,

$$\begin{aligned} \mathbf{D} = \mathbf{A} - \mathbf{B} &= \mathbf{A} + (-\mathbf{B}) \\ &= \hat{\mathbf{x}}(A_x - B_x) + \hat{\mathbf{y}}(A_y - B_y) + \hat{\mathbf{z}}(A_z - B_z). \end{aligned} \quad (3.9)$$

Graphically, the same rules used for vector addition are also applicable to vector subtraction; the only difference is that the arrowhead of $(-\mathbf{B})$ is drawn on the opposite end of the line segment representing the vector $\mathbf{B}$ (i.e., the tail and head are interchanged).

3-1.3 Position and Distance Vectors

The **position vector** of a point P in space is the vector from the origin to P . Assuming points P_1 and P_2 are at (x_1, y_1, z_1) and (x_2, y_2, z_2) in Fig. 3-4, their position vectors are

$$\mathbf{R}_1 = \overrightarrow{OP_1} = \hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1, \quad (3.10a)$$

三个相互垂直的单位向量 $\hat{x}$ 、 $\hat{y}$ 和 $\hat{z}$ 的方向，也称为基向量。图3-2(b)中的向量A可分解为

$$\mathbf{A} = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \quad (3.3)$$

其中 A_x 、 A_y 和 A_z 分别是 $\mathbf{A}$ 沿 x 、 y 和 z 轴的标量分量。分量 A_z 等于 $\mathbf{A}$ 在 z 轴上的垂直投影，类似的定义适用于 A_x 和 A_y 。应用毕达哥拉斯定理（首先应用到 x - y 平面中的直角三角形，以 A_x 和 A_y 表示斜边 A_r ，然后再次应用到边为 A_r 和 A_z 以及斜边 A 的垂直直角三角形，得出 A 大小的以下表达式：

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}. \quad (3.4)$$

由于 A 是非负标量，因此仅适用正根。从方程。 (3.2) 则单位向量 $\hat{a}$ 为

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = \frac{\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}. \quad (3.5)$$

有时，我们使用简写符号 $\mathbf{A} = (A_x, A_y, A_z)$ 来表示笛卡尔坐标系中具有分量 A_x 、 A_y 和 A_z 的向量。

3-1.1 两个向量相等

如果两个向量 $\mathbf{A}$ 和 $\mathbf{B}$ 具有相等的大小和相同的单位向量，则它们相等。因此，如果

$$\mathbf{A} = \hat{\mathbf{a}}A = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \quad (3.6a)$$

$$\mathbf{B} = \hat{\mathbf{b}}B = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \quad (3.6b)$$

则 $\mathbf{A} = \mathbf{B}$ 当且仅当 $A = B$ 和 $\hat{\mathbf{a}} = \hat{\mathbf{b}}$ ，这要求 $A_x = B_x$ 、 $A_y = B_y$ 和 $A_z = B_z$ 。

► 两个向量相等并不一定意味着它们相同；在笛卡尔坐标系中，两个大小相等且指向同一方向的位移平行向量相等，但仅当它们彼此重叠时才相同。 ◀

3-1.2 向量加法和减法

两个向量 $\mathbf{A}$ 和 $\mathbf{B}$ 的和是一个向量

$$\mathbf{C} = \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z,$$

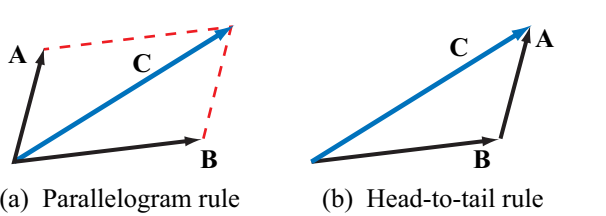


图 3-3 通过 (a) 平行四边形规则和 (b) 头尾规则进行向量加法。

给出的

$$\begin{aligned} \mathbf{C} = \mathbf{A} + \mathbf{B} &= (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z) + (\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z) \\ &= \hat{\mathbf{x}}(A_x + B_x) + \hat{\mathbf{y}}(A_y + B_y) + \hat{\mathbf{z}}(A_z + B_z) \\ &= \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z, \end{aligned} \quad (3.7)$$

与 $C_x = A_x + B_x$ 等

► 向量加法是可交换的：

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}. \quad (3.8)$$

从图形上看，向量加法可以通过平行四边形或头尾规则来完成（图3-3）。向量 $\mathbf{C}$ 是边为 $\mathbf{A}$ 和 $\mathbf{B}$ 的平行四边形的对角线。根据头尾规则，我们可以将 $\mathbf{A}$ 添加到 $\mathbf{B}$ 或将 $\mathbf{B}$ 添加到 $\mathbf{A}$ 。当 $\mathbf{A}$ 添加到 $\mathbf{B}$ 时，它会重新定位，使其尾部从 $\mathbf{B}$ 的尖端开始，同时保持其长度和方向不变。和向量 $\mathbf{C}$ 从 $\mathbf{B}$ 的尾部开始，到 $\mathbf{A}$ 的尖端结束。

从向量 $\mathbf{A}$ 中减去向量 $\mathbf{B}$ 相当于将 $\mathbf{A}$ 与负 $\mathbf{B}$ 相加。因此，

$$\begin{aligned} \mathbf{D} = \mathbf{A} - \mathbf{B} &= \mathbf{A} + (-\mathbf{B}) \\ &= \hat{\mathbf{x}}(A_x - B_x) + \hat{\mathbf{y}}(A_y - B_y) + \hat{\mathbf{z}}(A_z - B_z). \end{aligned} \quad (3.9)$$

从图形上看，用于矢量加法的相同规则也适用于矢量减法；唯一的区别是 $(-\mathbf{B})$ 的箭头画在表示向量 $\mathbf{B}$ 的线段的一端（即尾部和头部互换）。

3-1.3 位置和距离向量

空间中点 P 的位置向量是从原点到 P 的向量。假设图 3-4 中点 P_1 和 P_2 位于 (x_1, y_1, z_1) 和 (x_2, y_2, z_2) ，则它们的位置向量为

$$\mathbf{R}_1 = \overrightarrow{OP_1} = \hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1, \quad (3.10a)$$

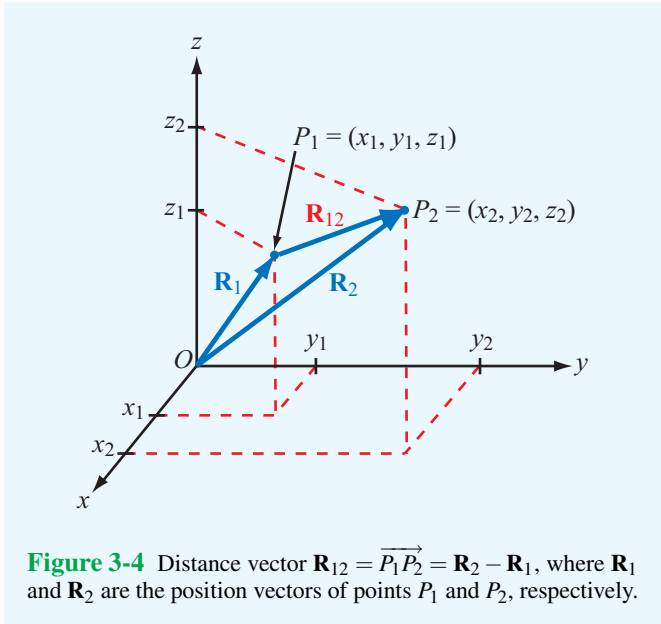


Figure 3-4 Distance vector $\mathbf{R}_{12} = \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1$, where $\mathbf{R}_1$ and $\mathbf{R}_2$ are the position vectors of points P_1 and P_2 , respectively.

$$\mathbf{R}_2 = \overrightarrow{OP_2} = \hat{x}x_2 + \hat{y}y_2 + \hat{z}z_2, \quad (3.10b)$$

where point O is the origin.
The **distance vector** from P_1 to P_2 is defined as

$$\begin{aligned} \mathbf{R}_{12} &= \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1 \\ &= \hat{x}(x_2 - x_1) + \hat{y}(y_2 - y_1) + \hat{z}(z_2 - z_1), \end{aligned} \quad (3.11)$$

and the distance d between P_1 and P_2 equals the magnitude of $\mathbf{R}_{12}$:

$$d = |\mathbf{R}_{12}| = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}. \quad (3.12)$$

Note that the first and second subscripts of $\mathbf{R}_{12}$ denote the locations of its tail and head, respectively (Fig. 3-4).

3-1.4 Vector Multiplication

There exist three types of products in vector calculus: the simple product, the scalar (or dot) product, and the vector (or cross) product.

Simple Product

The multiplication of a vector by a scalar is called a **simple product**. The product of the vector $\mathbf{A} = \hat{a}\mathbf{A}$ by a scalar k results in a vector $\mathbf{B}$ with magnitude $B = kA$ and direction the same as $\mathbf{A}$. That is, $\hat{b} = \hat{a}$. In Cartesian coordinates,

$$\begin{aligned} \mathbf{B} &= k\mathbf{A} = \hat{a}kA = \hat{x}(kA_x) + \hat{y}(kA_y) + \hat{z}(kA_z) \\ &= \hat{x}B_x + \hat{y}B_y + \hat{z}B_z. \end{aligned} \quad (3.13)$$

Scalar or Dot Product

The **scalar** (or **dot**) **product** of two co-anchored vectors $\mathbf{A}$ and $\mathbf{B}$, denoted $\mathbf{A} \cdot \mathbf{B}$ and pronounced “A dot B,” is defined geometrically as the product of the magnitude of $\mathbf{A}$ and the scalar component of $\mathbf{B}$ along $\mathbf{A}$, or vice versa. Thus,

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}, \quad (3.14)$$

where θ_{AB} is the angle between $\mathbf{A}$ and $\mathbf{B}$ (Fig. 3-5) measured from the tail of $\mathbf{A}$ to the tail of $\mathbf{B}$. Angle θ_{AB} is assumed to be in the range $0 \leq \theta_{AB} \leq 180^\circ$. The scalar product of $\mathbf{A}$ and $\mathbf{B}$ yields a scalar whose magnitude is less than or equal to the products of their magnitudes (equality holds when $\theta_{AB} = 0$) and whose sign is positive if $0 < \theta_{AB} < 90^\circ$ and negative if $90^\circ < \theta_{AB} < 180^\circ$. When $\theta_{AB} = 90^\circ$, $\mathbf{A}$ and $\mathbf{B}$ are orthogonal, and their dot product is zero. The quantity $A \cos \theta_{AB}$ is the scalar component of $\mathbf{A}$ along $\mathbf{B}$. Similarly, $B \cos \theta_{BA}$ is the scalar component of $\mathbf{B}$ along $\mathbf{A}$.

The dot product obeys both the commutative and distributive properties of multiplication:

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}, \quad (3.15a)$$

(commutative property)

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}. \quad (3.15b)$$

(distributive property)

The commutative property follows from Eq. (3.14) and the fact that $\theta_{AB} = \theta_{BA}$. The distributive property expresses the fact that the scalar component of the sum of two vectors along a third one equals the sum of their respective scalar components.

The dot product of a vector with itself gives

$$\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2 = A^2, \quad (3.16)$$

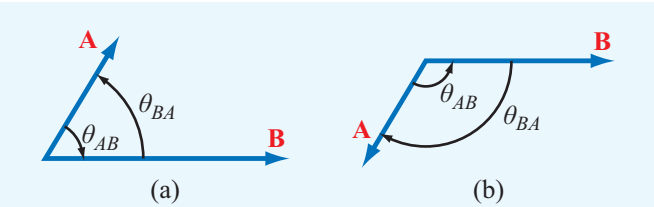


Figure 3-5 The angle θ_{AB} is the angle between $\mathbf{A}$ and $\mathbf{B}$, measured from $\mathbf{A}$ to $\mathbf{B}$ between vector tails. The dot product is positive if $0 \leq \theta_{AB} < 90^\circ$, as in (a), and it is negative if $90^\circ < \theta_{AB} \leq 180^\circ$, as in (b).

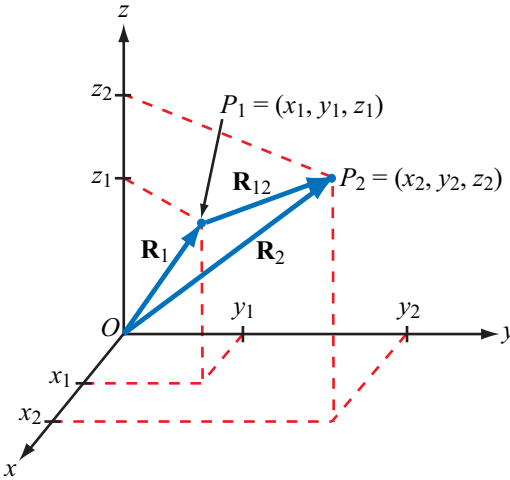


图3-4 距离向量 $\mathbf{R}_{12} = \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1$ ，其中 $\mathbf{R}_1$ 和 $\mathbf{R}_2$ 分别是点 P_1 和 P_2 的位置向量。

$$\mathbf{R}_2 = \overrightarrow{OP_2} = \hat{x}x_2 + \hat{y}y_2 + \hat{z}z_2, \quad (3.10b)$$

其中 O 点是原点。
从 P_1 到 P_2 的距离向量定义为

$$\begin{aligned} \mathbf{R}_{12} &= \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1 \\ &= \hat{x}(x_2 - x_1) + \hat{y}(y_2 - y_1) + \hat{z}(z_2 - z_1), \end{aligned} \quad (3.11)$$

P_1 和 P_2 之间的距离 d 等于 R_{12} 的大小：

$$d = |\mathbf{R}_{12}| = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}. \quad (3.12)$$

请注意， R_{12} 的第一个和第二个下标分别表示其尾部 and 头部的位 置（图 3-4）。

3-1.4 向量乘法

向量微积分中存在三种类型的乘积：简单乘积、标量（或点）乘积和向量（或叉）乘积。

标量与矢量相乘

矢量与实标量 k 相乘得到 $\mathbf{B} = k\mathbf{A}$ ，其模为 $|\mathbf{B}| = |k||\mathbf{A}|$ 。当 $\mathbf{A} \neq 0$ 时， $k > 0$ 则同向， $k < 0$ 则反向； $k = 0$ 时为零矢量，无确定方向。原书关于模和方向的表述仅适用于 $k > 0$ 。直角坐标下：

$$\begin{aligned} \mathbf{B} &= k\mathbf{A} = \hat{a}kA = \hat{x}(kA_x) + \hat{y}(kA_y) + \hat{z}(kA_z) \\ &= \hat{x}B_x + \hat{y}B_y + \hat{z}B_z. \end{aligned} \quad (3.13)$$

标量或点积

两个共同锚定向量 $\mathbf{A}$ 和 $\mathbf{B}$ 的标量（或点）积，表示为 $\mathbf{A} \cdot \mathbf{B}$ 并发音为 “A 点 B”，在几何上定义为 $\mathbf{A}$ 的大小与 $\mathbf{B}$ 沿 $\mathbf{A}$ 的标量分量的乘积，反之亦然。因此，

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}, \quad (3.14)$$

对非零矢量 $\mathbf{A}$ 、 $\mathbf{B}$ ，夹角 θ_{AB} 是将两矢量平移到共同起点后，两方向间的角，取值为 0° 至 180° 。点积是标量，且 $|\mathbf{A} \cdot \mathbf{B}| \leq |\mathbf{A}||\mathbf{B}|$ ；同向或反向平行时绝对值取等号。当 $0^\circ \leq \theta_{AB} < 90^\circ$ 时点积为正； $90^\circ < \theta_{AB} \leq 180^\circ$ 时为负； $\theta_{AB} = 90^\circ$ 时两矢量正交，点积为零。 $A \cos \theta_{AB}$ 是 $\mathbf{A}$ 沿 $\mathbf{B}$ 方向的标量分量， $B \cos \theta_{BA}$ 是 $\mathbf{B}$ 沿 $\mathbf{A}$ 方向的标量分量。

点积遵循乘法的交换律和分配律：

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}, \quad (3.15a)$$

(交换律)

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}. \quad (3.15b)$$

(分配财产)

交换律由等式得出。(3.14) 以及 $\theta_{AB} = \theta_{BA}$ 的事实。分配性质表达了这样一个事实：两个向量之和沿第三个向量的标量分量等于它们各自的标量分量之和。

向量与其自身的点积给出

$$\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2 = A^2, \quad (3.16)$$

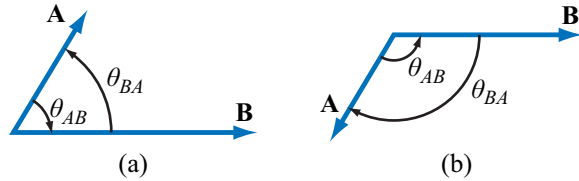


图3-5 矢量 $\mathbf{A}$ 、 $\mathbf{B}$ 的夹角。两矢量取共同起点后测量该角；(a) 夹角小于 90° ，点积为正；(b) 夹角大于 90° ，点积为负。

which implies that

$$A = |\mathbf{A}| = \sqrt[3]{\mathbf{A} \cdot \mathbf{A}}. \quad (3.17)$$

Also, θ_{AB} can be determined from

$$\theta_{AB} = \cos^{-1} \left[\frac{\mathbf{A} \cdot \mathbf{B}}{\sqrt{\mathbf{A} \cdot \mathbf{A}} \sqrt{\mathbf{B} \cdot \mathbf{B}}} \right]. \quad (3.18)$$

Since the base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ are each orthogonal to the other two, it follows that

$$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1, \quad (3.19a)$$

$$\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0. \quad (3.19b)$$

If $\mathbf{A} = (A_x, A_y, A_z)$ and $\mathbf{B} = (B_x, B_y, B_z)$, then

$$\mathbf{A} \cdot \mathbf{B} = (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z) \cdot (\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z). \quad (3.20)$$

Use of Eqs. (3.19a) and (3.19b) in Eq. (3.20) leads to

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z. \quad (3.21)$$

Vector or Cross Product

The **vector** (or **cross**) **product** of two vectors $\mathbf{A}$ and $\mathbf{B}$, denoted $\mathbf{A} \times \mathbf{B}$ and pronounced “A cross B,” yields a vector defined as

$$\mathbf{A} \times \mathbf{B} = \hat{\mathbf{n}} AB \sin \theta_{AB}, \quad (3.22)$$

where $\hat{\mathbf{n}}$ is a **unit vector normal to the plane containing $\mathbf{A}$ and $\mathbf{B}$** (Fig. 3-6(a)). The magnitude of the cross product, $AB \sin \theta_{AB}$, equals the area of the parallelogram defined by the two vectors. The direction of $\hat{\mathbf{n}}$ is governed by the **right-hand rule** (Fig. 3-6(b)): $\hat{\mathbf{n}}$ points in the direction of the right thumb when the fingers rotate from $\mathbf{A}$ to $\mathbf{B}$ through the angle θ_{AB} . Note that, since $\hat{\mathbf{n}}$ is perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$, $\mathbf{A} \times \mathbf{B}$ is perpendicular to both vectors $\mathbf{A}$ and $\mathbf{B}$.

The cross product is anticommutative and distributive:

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A} \quad (\text{anticommutative}). \quad (3.23a)$$

The anticommutative property follows from the application of the right-hand rule to determine $\hat{\mathbf{n}}$. The distributive property follows from the fact that the area of the parallelogram formed

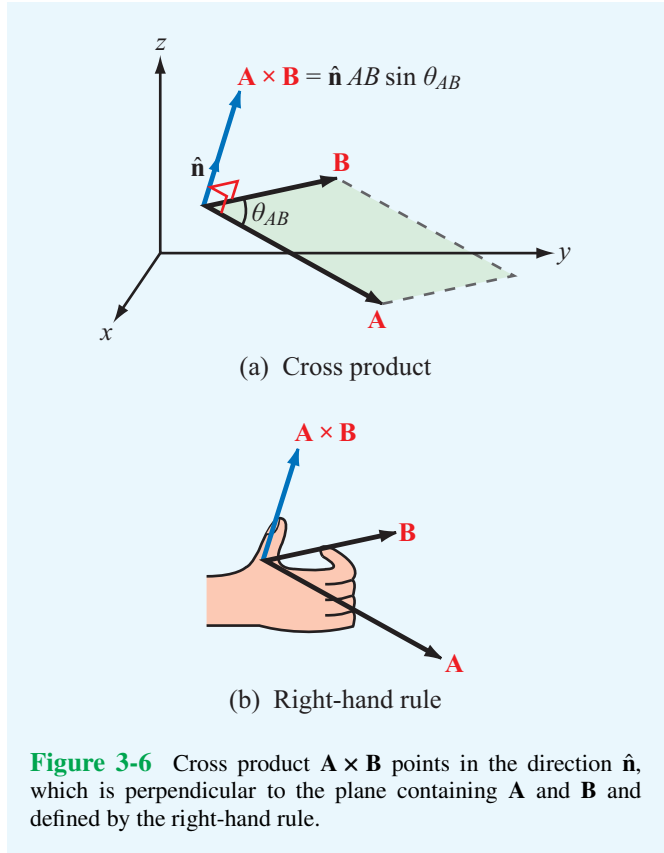


Figure 3-6 Cross product $\mathbf{A} \times \mathbf{B}$ points in the direction $\hat{\mathbf{n}}$, which is perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$ and defined by the right-hand rule.

by $\mathbf{A}$ and $(\mathbf{B} + \mathbf{C})$ equals the sum of those formed by $(\mathbf{A}$ and $\mathbf{B})$ and $(\mathbf{A}$ and $\mathbf{C})$:

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}, \quad (3.23b)$$

(distributive)

The cross product of a vector with itself vanishes. That is,

$$\mathbf{A} \times \mathbf{A} = 0. \quad (3.24)$$

From the definition of the cross product given by Eq. (3.22), it is easy to verify that the base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ of the Cartesian coordinate system obey the right-hand cyclic relations:

$$\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}, \quad \hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}, \quad \hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}. \quad (3.25)$$

Note the cyclic order ($xyzxyz \dots$). Also,

$$\hat{\mathbf{x}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0. \quad (3.26)$$

这意味着

$$A = |\mathbf{A}| = \sqrt[3]{\mathbf{A} \cdot \mathbf{A}}. \quad (3.17)$$

此外, θ_{AB} 可以由下式确定:

$$\theta_{AB} = \cos^{-1} \left[\frac{\mathbf{A} \cdot \mathbf{B}}{\sqrt{\mathbf{A} \cdot \mathbf{A}} \sqrt{\mathbf{B} \cdot \mathbf{B}}} \right]. \quad (3.18)$$

由于基向量 $\hat{\mathbf{x}}$ 、 $\hat{\mathbf{y}}$ 和 $\hat{\mathbf{z}}$ 均与其他两个正交, 因此可以得出:

$$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1, \quad (3.19a)$$

$$\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0. \quad (3.19b)$$

如果 $\mathbf{A} = (A_x, A_y, A_z)$ 和 $\mathbf{B} = (B_x, B_y, B_z)$, 则

$$\mathbf{A} \cdot \mathbf{B} = (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z) \cdot (\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z). \quad (3.20)$$

使用等式。式 (3.19a) 和 (3.19b) (3.20) 导致

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z. \quad (3.21)$$

向量或叉积

两个向量 $\mathbf{A}$ 和 $\mathbf{B}$ 的向量 (或叉) 乘积, 表示为 $\mathbf{A} \times \mathbf{B}$, 发音为 “A cross B”, 产生一个向量, 定义为

$$\mathbf{A} \times \mathbf{B} = \hat{\mathbf{n}} AB \sin \theta_{AB}, \quad (3.22)$$

其中 $\hat{\mathbf{n}}$ 是垂直于包含 $\mathbf{A}$ 和 $\mathbf{B}$ 的平面的单位向量 (图 3-6 (a))。叉积 $AB \sin \theta_{AB}$ 的大小等于由两个向量定义的平行四边形的面积。 $\hat{\mathbf{n}}$ 的方向遵循右手定则 (图 3-6 (b)) : 当手指从 $\mathbf{A}$ 旋转到 $\mathbf{B}$ 角度 θ_{AB} 时, $\hat{\mathbf{n}}$ 指向右手拇指的方向。请注意, 由于 $\hat{\mathbf{n}}$ 垂直于包含 $\mathbf{A}$ 和 $\mathbf{B}$ 的平面, 因此 $\mathbf{A} \times \mathbf{B}$ 垂直于向量 $\mathbf{A}$ 和 $\mathbf{B}$ 。

叉积是反交换的和分配的:

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A} \quad (\text{anticommutative}). \quad (3.23a)$$

反交换性质是根据右手定则的应用来确定的。分配律是根据平行四边形的面积形成的事实得出的

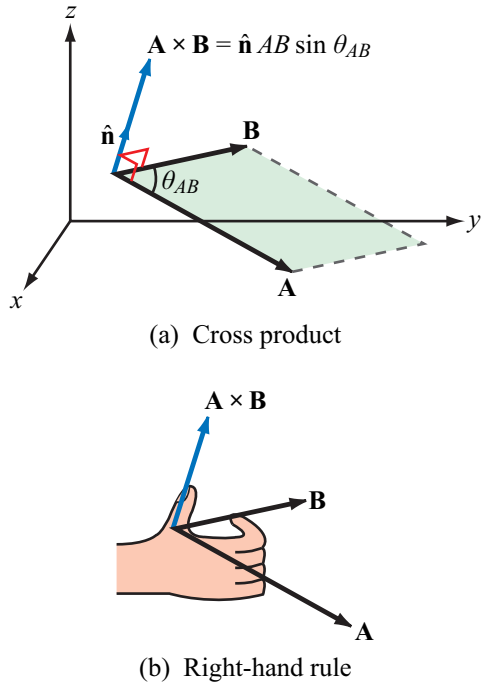


图 3-6 叉积 $\mathbf{A} \times \mathbf{B}$ 指向 $\hat{\mathbf{n}}$ 方向, 该方向垂直于包含 $\mathbf{A}$ 和 $\mathbf{B}$ 的平面, 并由右手定则定义。

由 $\mathbf{A}$ 和 $(\mathbf{B} + \mathbf{C})$ 等于由 $(\mathbf{A}$ 和 $\mathbf{B})$ 和 $(\mathbf{A}$ 和 $\mathbf{C})$ 形成的总和:

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}, \quad (3.23b)$$

(distributive)

向量与其自身的叉积消失。那是,

$$\mathbf{A} \times \mathbf{A} = 0. \quad (3.24)$$

根据方程给出的叉积的定义: 根据(3.22)式, 容易验证笛卡尔坐标系的基向量 $\hat{\mathbf{x}}$ 、 $\hat{\mathbf{y}}$ 和 $\hat{\mathbf{z}}$ 服从右手循环关系:

$$\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}, \quad \hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}, \quad \hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}. \quad (3.25)$$

注意循环顺序 ($xyzxyz \dots$)。还,

$$\hat{\mathbf{x}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0. \quad (3.26)$$

3-1 BASIC LAWS OF VECTOR ALGEBRA

135

If $\mathbf{A} = (A_x, A_y, A_z)$ and $\mathbf{B} = (B_x, B_y, B_z)$, then use of Eqs. (3.25) and (3.26) leads to

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= (\hat{x}A_x + \hat{y}A_y + \hat{z}A_z) \times (\hat{x}B_x + \hat{y}B_y + \hat{z}B_z) \\ &= \hat{x}(A_yB_z - A_zB_y) + \hat{y}(A_zB_x - A_xB_z) \\ &\quad + \hat{z}(A_xB_y - A_yB_x).\end{aligned}\quad (3.27)$$

The cyclical form of the result given by Eq. (3.27) allows us to express the cross product in the form of a determinant:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.\quad (3.28)$$

Example 3-1: Vectors and Angles

In Cartesian coordinates, vector $\mathbf{A}$ points from the origin to point $P_1 = (2, 3, 3)$, and vector $\mathbf{B}$ is directed from P_1 to point $P_2 = (1, -2, 2)$. Find:

- (a) vector $\mathbf{A}$, its magnitude A , and unit vector $\hat{\mathbf{a}}$,
- (b) the angle between $\mathbf{A}$ and the y axis,
- (c) vector $\mathbf{B}$,
- (d) the angle θ_{AB} between $\mathbf{A}$ and $\mathbf{B}$, and
- (e) perpendicular distance from the origin to vector $\mathbf{B}$.

Solution: (a) Vector $\mathbf{A}$ is given by the position vector of $P_1 = (2, 3, 3)$ (Fig. 3-7). Thus,

$$\mathbf{A} = \hat{x}2 + \hat{y}3 + \hat{z}3,$$

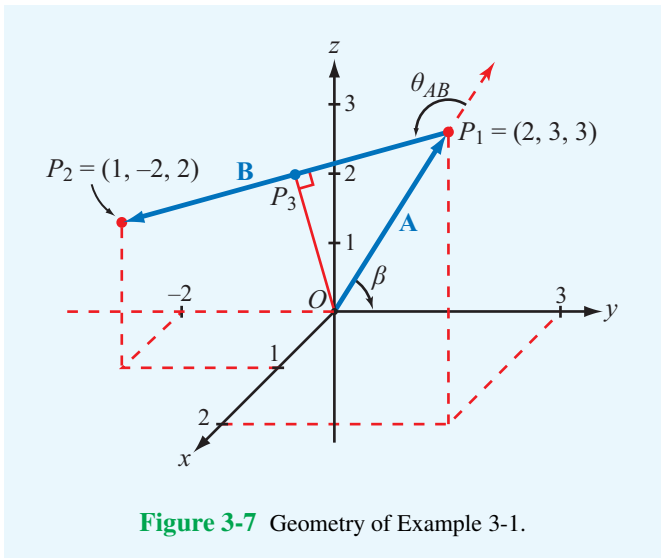


Figure 3-7 Geometry of Example 3-1.

$$A = |\mathbf{A}| = \sqrt{2^2 + 3^2 + 3^2} = \sqrt{22},$$

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = (\hat{x}2 + \hat{y}3 + \hat{z}3)/\sqrt{22}.$$

(b) The angle β between $\mathbf{A}$ and the y axis is obtained from

$$\mathbf{A} \cdot \hat{\mathbf{y}} = |\mathbf{A}||\hat{\mathbf{y}}|\cos\beta = A\cos\beta,$$

or

$$\beta = \cos^{-1}\left(\frac{\mathbf{A} \cdot \hat{\mathbf{y}}}{A}\right) = \cos^{-1}\left(\frac{3}{\sqrt{22}}\right) = 50.2^\circ.$$

(c)

$$\mathbf{B} = \hat{x}(1-2) + \hat{y}(-2-3) + \hat{z}(2-3) = -\hat{x} - \hat{y}5 - \hat{z}.$$

(d)

$$\theta_{AB} = \cos^{-1}\left[\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}\right] = \cos^{-1}\left[\frac{(-2-15-3)}{\sqrt{22}\sqrt{27}}\right] = 145.1^\circ.$$

(e) The perpendicular distance between the origin and vector $\mathbf{B}$ is the distance $|\overrightarrow{OP_3}|$ shown in Fig. 3-7. From right triangle OP_1P_3 ,

$$\begin{aligned}|\overrightarrow{OP_3}| &= |\mathbf{A}|\sin(180^\circ - \theta_{AB}) \\ &= \sqrt{22}\sin(180^\circ - 145.1^\circ) = 2.68.\end{aligned}$$

Example 3-2: Cross Product

Given vectors $\mathbf{A} = \hat{x}2 - \hat{y} + \hat{z}3$ and $\mathbf{B} = \hat{y}2 - \hat{z}3$, compute (a) $\mathbf{A} \times \mathbf{B}$, (b) $\hat{\mathbf{y}} \times \mathbf{B}$, and (c) $(\hat{\mathbf{y}} \times \mathbf{B}) \cdot \mathbf{A}$.

Solution: (a) Application of Eq. (3.28) gives

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & -1 & 3 \\ 0 & 2 & -3 \end{vmatrix} \\ &= \hat{x}((-1) \times (-3) - 3 \times 2) - \hat{y}(2 \times (-3) - 3 \times 0) \\ &\quad + \hat{z}(2 \times 2 - (-1 \times 0)) \\ &= -\hat{x}3 + \hat{y}6 + \hat{z}4.\end{aligned}$$

$$(b) \hat{\mathbf{y}} \times \mathbf{B} = \hat{\mathbf{y}} \times (\hat{\mathbf{y}}2 - \hat{\mathbf{z}}3) = -\hat{x}3.$$

$$(c) (\hat{\mathbf{y}} \times \mathbf{B}) \cdot \mathbf{A} = -\hat{x}3 \cdot (\hat{x}2 - \hat{y} + \hat{z}3) = -6.$$

Exercise 3-1: Find the distance vector between $P_1 = (1, 2, 3)$ and $P_2 = (-1, -2, 3)$ in Cartesian coordinates.

Answer: $\overrightarrow{P_1P_2} = -\hat{x}2 - \hat{y}4$. (See EM.)

3-1 BASIC LAWS OF VECTOR ALGEBRA

135

如果 $\mathbf{A} = (A_x, A_y, A_z)$ 和 $\mathbf{B} = (B_x, B_y, B_z)$, 则使用等式: (3.25) 和 (3.26) 导致

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= (\hat{x}A_x + \hat{y}A_y + \hat{z}A_z) \times (\hat{x}B_x + \hat{y}B_y + \hat{z}B_z) \\ &= \hat{x}(A_yB_z - A_zB_y) + \hat{y}(A_zB_x - A_xB_z) \\ &\quad + \hat{z}(A_xB_y - A_yB_x).\end{aligned}\quad (3.27)$$

方程给出的结果的循环形式。(3.27) 允许我们以行列式的形式表达叉积:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.\quad (3.28)$$

例 3-1: 向量和角度

在笛卡尔坐标系中, 矢量 $\mathbf{A}$ 从原点指向点 $P_1 = (2, 3, 3)$, 矢量 $\mathbf{B}$ 从 P_1 指向点 $P_2 = (1, -2, 2)$ 。寻找:

- (a) 向量 $\mathbf{A}$ 、其大小 A 和单位向量 $\hat{\mathbf{a}}$,
- (b) $\mathbf{A}$ 与 y 轴之间的角度,
- (c) 向量 $\mathbf{B}$,
- (d) $\mathbf{A}$ 与 $\mathbf{B}$ 之间的角度 θ_{AB} ,
- 以及 (e) 从原点到向量 $\mathbf{B}$ 的垂直距离。

解: (a) 向量 $\mathbf{A}$ 由 $P_1 = (2, 3, 3)$ (的位置向量给出。3-7)。因此,

$$\mathbf{A} = \hat{x}2 + \hat{y}3 + \hat{z}3,$$

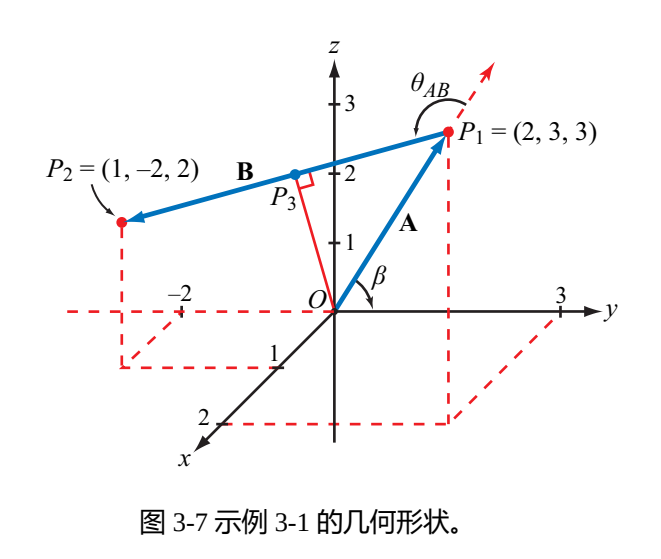


图 3-7 示例 3-1 的几何形状。

$$A = |\mathbf{A}| = \sqrt{2^2 + 3^2 + 3^2} = \sqrt{22},$$

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = (\hat{x}2 + \hat{y}3 + \hat{z}3)/\sqrt{22}.$$

(b) $\mathbf{A}$ 与 y 轴之间的角度 β 由下式获得

$$\mathbf{A} \cdot \hat{\mathbf{y}} = |\mathbf{A}||\hat{\mathbf{y}}|\cos\beta = A\cos\beta,$$

或者

$$\beta = \cos^{-1}\left(\frac{\mathbf{A} \cdot \hat{\mathbf{y}}}{A}\right) = \cos^{-1}\left(\frac{3}{\sqrt{22}}\right) = 50.2^\circ.$$

(c)

$$\mathbf{B} = \hat{x}(1-2) + \hat{y}(-2-3) + \hat{z}(2-3) = -\hat{x} - \hat{y}5 - \hat{z}.$$

(d)

$$\theta_{AB} = \cos^{-1}\left[\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}\right] = \cos^{-1}\left[\frac{(-2-15-3)}{\sqrt{22}\sqrt{27}}\right] = 145.1^\circ.$$

(e) 原点与矢量 $\mathbf{B}$ 之间的垂直距离为图 3-7 所示的距离 $|\overrightarrow{OP_3}|$ 。由直角三角形 OP_1P_3 ,

$$\begin{aligned}|\overrightarrow{OP_3}| &= |\mathbf{A}|\sin(180^\circ - \theta_{AB}) \\ &= \sqrt{22}\sin(180^\circ - 145.1^\circ) = 2.68.\end{aligned}$$

例 3-2: 叉积

给定向量 $\mathbf{A} = \hat{x}2 - \hat{y} + \hat{z}3$ 和 $\mathbf{B} = \hat{y}2 - \hat{z}3$, 计算 (a) $\mathbf{A} \times \mathbf{B}$ 、(b) $\hat{\mathbf{y}} \times \mathbf{B}$ 和 (c) $(\hat{\mathbf{y}} \times \mathbf{B}) \cdot \mathbf{A}$ 。

解决方案: (a) 应用方程。(3.28) 给出

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & -1 & 3 \\ 0 & 2 & -3 \end{vmatrix} \\ &= \hat{x}((-1) \times (-3) - 3 \times 2) - \hat{y}(2 \times (-3) - 3 \times 0) \\ &\quad + \hat{z}(2 \times 2 - (-1 \times 0)) \\ &= -\hat{x}3 + \hat{y}6 + \hat{z}4.\end{aligned}$$

$$(b) \hat{\mathbf{y}} \times \mathbf{B} = \hat{\mathbf{y}} \times (\hat{\mathbf{y}}2 - \hat{\mathbf{z}}3) = -\hat{x}3.$$

$$(c) (\hat{\mathbf{y}} \times \mathbf{B}) \cdot \mathbf{A} = -\hat{x}3 \cdot (\hat{x}2 - \hat{y} + \hat{z}3) = -6.$$

练习 3-1: 求笛卡尔坐标中 $P_1 = (1, 2, 3)$ 和 $P_2 = (-1, -2, 3)$ 之间的距离向量。

Answer: $\overrightarrow{P_1P_2} = -\hat{x}2 - \hat{y}4$. (See EM.)

Exercise 3-2: Find the angle θ_{AB} between vectors **A** and **B** of Example 3-1 from the cross product between them.

Answer: $\theta_{AB} = 145.1^\circ$. (See [EM](#).)

Exercise 3-3: Find the angle between vector **B** of Example 3-1 and the z axis.

Answer: 101.1° . (See [EM](#).)

Exercise 3-4: Vectors **A** and **B** lie in the y - z plane and both have the same magnitude of 2 (**Fig. E3.4**). Determine (a) $\mathbf{A} \cdot \mathbf{B}$ and (b) $\mathbf{A} \times \mathbf{B}$.

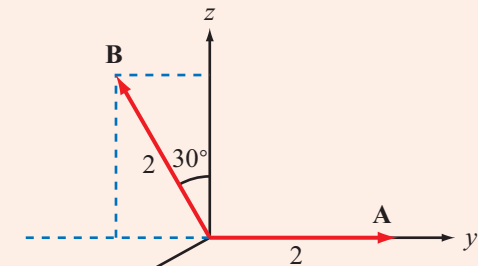


Figure E3.4

Answer: (a) $\mathbf{A} \cdot \mathbf{B} = -2$; (b) $\mathbf{A} \times \mathbf{B} = \hat{x} 3.46$. (See [EM](#).)

Exercise 3-5: If $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{C}$, does it follow that $\mathbf{B} = \mathbf{C}$?

Answer: No. (See [EW](#).)

3-1.5 Scalar and Vector Triple Products

When three vectors are multiplied, not all combinations of dot and cross products are meaningful. For example, the product

$$\mathbf{A} \times (\mathbf{B} \cdot \mathbf{C})$$

does not make sense because $\mathbf{B} \cdot \mathbf{C}$ is a scalar, and the cross product of the vector **A** with a scalar is not defined under the rules of vector algebra. Other than the product of the form $\mathbf{A}(\mathbf{B} \cdot \mathbf{C})$, the only two meaningful products of three vectors are the scalar triple product and the vector triple product.

Scalar Triple Product

The dot product of a vector with the cross product of two other vectors is called a scalar triple product, so named because the

result is a scalar. A scalar triple product obeys the cyclic order:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}). \quad (3.29)$$

The equalities hold as long as the cyclic order ($ABCABC \dots$) is preserved. The scalar triple product of vectors $\mathbf{A} = (A_x, A_y, A_z)$, $\mathbf{B} = (B_x, B_y, B_z)$, and $\mathbf{C} = (C_x, C_y, C_z)$ can be expressed in the form of a 3×3 determinant:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}. \quad (3.30)$$

The validity of Eqs. (3.29) and (3.30) can be verified by expanding **A**, **B**, and **C** in component form and carrying out the multiplications.

Vector Triple Product

The vector triple product involves the cross product of a vector with the cross product of two others, such as

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}). \quad (3.31)$$

Since each cross product yields a vector, the result of a vector triple product is also a vector. The vector triple product does not obey the associative law. That is,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}, \quad (3.32)$$

which means that it is important to specify which cross multiplication is to be performed first. By expanding the vectors **A**, **B**, and **C** in component form, it can be shown that

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}), \quad (3.33)$$

which is known as the “bac-cab” rule.

Example 3-3: Vector Triple Product

Given $\mathbf{A} = \hat{x} - \hat{y} + \hat{z}2$, $\mathbf{B} = \hat{y} + \hat{z}$, and $\mathbf{C} = -\hat{x}2 + \hat{z}3$, find $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$ and compare it with $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

Solution:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -\hat{x}3 - \hat{y} + \hat{z}$$

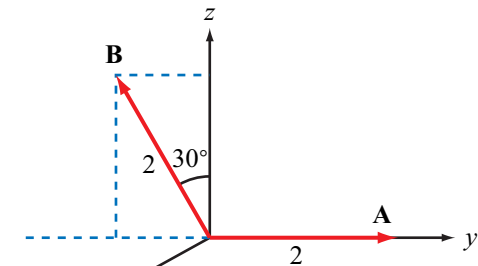
练习 3-2: 根据例 3-1 中的向量 A 和 B 之间的叉积求出它们之间的角度 θ_{AB} 。

Answer: $\theta_{AB} = 145.1^\circ$. (See [EM](#).)

练习 3-3: 求例 3-1 中的向量 B 与 z 轴之间的角度。

答案: 101.1° 。(参见 [EM](#)。)

练习 3-4: 向量 A 和 B 位于 y - z 平面上, 并且大小相同, 均为 2 (图 E3.4)。确定 (a) $\mathbf{A} \cdot \mathbf{B}$ 和 (b) $\mathbf{A} \times \mathbf{B}$ 。



图E3.4

答案: (a) $\mathbf{A} \cdot \mathbf{B} = -2$; (b) $\mathbf{A} \times \mathbf{B} = \hat{x} 3.46$ 。(参见 [EM](#)。)

练习3-5: 如果 $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{C}$, 是否可以得出 $\mathbf{B} = \mathbf{C}$?

答案: 不。(参见 [EM](#)。)

3-1.5 标量和向量的三重积

当三个向量相乘时, 并非所有点积和叉积的组合都有意义。例如, 产品

$$\mathbf{A} \times (\mathbf{B} \cdot \mathbf{C})$$

没有意义, 因为 $\mathbf{B} \cdot \mathbf{C}$ 是标量, 并且向量 **A** 与标量的叉积没有根据向量代数规则定义。除了 $\mathbf{A}(\mathbf{B} \cdot \mathbf{C})$ 形式的乘积之外, 三个向量的唯一两个有意义的乘积是标量三重乘积和向量三重乘积。

标量三重积

一个向量与另外两个向量的叉积的点积称为标量三重积, 如此命名是因为

结果是一个标量。标量三重积遵循循环顺序:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}). \quad (3.29)$$

只要保留循环顺序 ($ABCABC \dots$), 这些等式就成立。向量 $\mathbf{A} = (A_x, A_y, A_z)$ 、 $\mathbf{B} = (B_x, B_y, B_z)$ 和 $\mathbf{C} = (C_x, C_y, C_z)$ 的标量三重积可以表示为 3×3 行列式的形式:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}. \quad (3.30)$$

方程的有效性。(3.29)和(3.30)可以通过将A、B和C以分量形式展开并进行乘法来验证。

向量三重积

向量三重积涉及一个向量与另外两个向量的叉积, 例如

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}). \quad (3.31)$$

由于每个叉积都会产生一个向量, 因此向量三重积的结果也是一个向量。向量三重积不遵守结合律。那是,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}, \quad (3.32)$$

这意味着指定首先执行哪个叉乘非常重要。将向量A、B、C展开分量形式, 可以证明:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}), \quad (3.33)$$

这就是所谓的 “bac-cab” 规则。

例 3-3: 向量三重积

给定 $\mathbf{A} = \hat{x} - \hat{y} + \hat{z}2$ 、 $\mathbf{B} = \hat{y} + \hat{z}$ 和 $\mathbf{C} = -\hat{x}2 + \hat{z}3$, 找到 $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$ 并将其与 $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ 进行比较。

解决方案:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -\hat{x}3 - \hat{y} + \hat{z}$$

and

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ -3 & -1 & 1 \\ -2 & 0 & 3 \end{vmatrix} = -\hat{\mathbf{x}}3 + \hat{\mathbf{y}}7 - \hat{\mathbf{z}}2.$$

A similar procedure gives $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}4 + \hat{\mathbf{z}}$. The fact that the results of two vector triple products are different demonstrates the inequality stated in Eq. (3.32).

Concept Question 3-1: When are two vectors equal and when are they identical?

Concept Question 3-2: When is the position vector of a point identical to the distance vector between two points?

Concept Question 3-3: If $\mathbf{A} \cdot \mathbf{B} = 0$, what is θ_{AB} ?

Concept Question 3-4: If $\mathbf{A} \times \mathbf{B} = 0$, what is θ_{AB} ?

Concept Question 3-5: Is $\mathbf{A}(\mathbf{B} \cdot \mathbf{C})$ a vector triple product?

Concept Question 3-6: If $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{C}$, does it follow that $\mathbf{B} = \mathbf{C}$?

3-2 Orthogonal Coordinate Systems

A three-dimensional coordinate system allows us to uniquely specify locations of points in space and the magnitudes and directions of vectors. Coordinate systems may be orthogonal or nonorthogonal.

► An **orthogonal coordinate system** is one in which coordinates are measured along locally mutually perpendicular axes. ◀

Nonorthogonal systems are very specialized and seldom used in solving practical problems. Many orthogonal coordinate systems have been devised, but the most commonly used are

- the Cartesian (also called rectangular),
- the cylindrical, and
- the spherical coordinate system.

Why do we need more than one coordinate system? Whereas a point in space has the same location and an object has the same shape regardless of which coordinate system is used to describe them, the solution of a practical problem can be greatly facilitated by the choice of a coordinate system that best fits the geometry under consideration. The following subsections examine the properties of each of the aforementioned orthogonal systems, and Section 3-3 describes how a point or vector may be transformed from one system to another.

3-2.1 Cartesian Coordinates

The Cartesian coordinate system was introduced in Section 3-1 to illustrate the laws of vector algebra. Instead of repeating these laws for the Cartesian system, we summarize them in **Table 3-1**. Differential calculus involves the use of differential lengths, areas, and volumes. In Cartesian coordinates, a **differential length vector** (**Fig. 3-8**) is expressed as

$$d\mathbf{l} = \hat{\mathbf{x}} dl_x + \hat{\mathbf{y}} dl_y + \hat{\mathbf{z}} dl_z = \hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz, \quad (3.34)$$

where $dl_x = dx$ is a differential length along $\hat{\mathbf{x}}$, and similar interpretations apply to $dl_y = dy$ and $dl_z = dz$.

A **differential area vector** $d\mathbf{s}$ is a vector with magnitude ds equal to the product of two differential lengths (such as dl_y and dl_z) and direction specified by a unit vector along the third

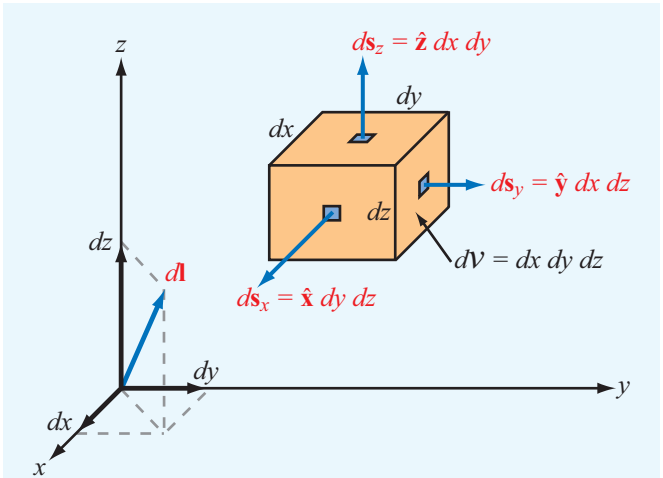


Figure 3-8 Differential length, area, and volume in Cartesian coordinates.

和

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ -3 & -1 & 1 \\ -2 & 0 & 3 \end{vmatrix} = -\hat{\mathbf{x}}3 + \hat{\mathbf{y}}7 - \hat{\mathbf{z}}2.$$

类似的过程给出 $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}4 + \hat{\mathbf{z}}$ 。两个向量三元乘积的结果不同这一事实证明了等式所述的 inequality。 (3.32)。

Concept Question 3-1: When are two vectors equal and when are they identical?

Concept Question 3-2: When is the position vector of a point identical to the distance vector between two points?

Concept Question 3-3: If $\mathbf{A} \cdot \mathbf{B} = 0$, what is θ_{AB} ?

Concept Question 3-4: If $\mathbf{A} \times \mathbf{B} = 0$, what is θ_{AB} ?

Concept Question 3-5: Is $\mathbf{A}(\mathbf{B} \cdot \mathbf{C})$ a vector triple product?

Concept Question 3-6: If $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{C}$, does it follow that $\mathbf{B} = \mathbf{C}$?

3-2 正交坐标系

三维坐标系允许我们唯一地指定空间中点的位置以及向量的大小和方向。坐标系可以是正交的或非正交的。

► 正交坐标系是一种沿局部相互垂直的轴测量坐标的坐标系。 ◀

非正交系统非常专业，很少用于解决实际问题。已经设计了许多正交坐标系，但最常用的是

- 笛卡尔（也称为矩形），
- 圆柱形，并且
- 球坐标系。

为什么我们需要多个坐标系？尽管无论使用哪种坐标系来描述，空间中的点都具有相同的位置，物体也具有相同的形状，但通过选择最适合所考虑的几何形状的坐标系，可以极大地促进实际问题的解决。以下小节检查了上述每个正交系统的属性，第 3-3 节描述了如何将点或向量从一个系统变换到另一个系统。

3-2.1 笛卡尔坐标

第 3-1 节介绍了笛卡尔坐标系，以说明向量代数的定律。我们没有在笛卡尔系统中重复这些定律，而是将它们总结在表 3-1 中。微分学涉及不同长度、面积和体积的使用。在笛卡尔坐标系中，微分长度向量（图3-8）表示为

$$d\mathbf{l} = \hat{\mathbf{x}} dl_x + \hat{\mathbf{y}} dl_y + \hat{\mathbf{z}} dl_z = \hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz, \quad (3.34)$$

其中 $dl_x = dx$ 是沿 $\hat{\mathbf{x}}$ 的微分长度，类似的解释适用于 $dl_y = dy$ 和 $dl_z = dz$ 。微分面积向量 $d\mathbf{s}$ 是一个向量，其大小 ds 等于两个微分长度（例如 dl_y 和 dl_z ）和由沿第三个单位向量指定的方向的乘积。

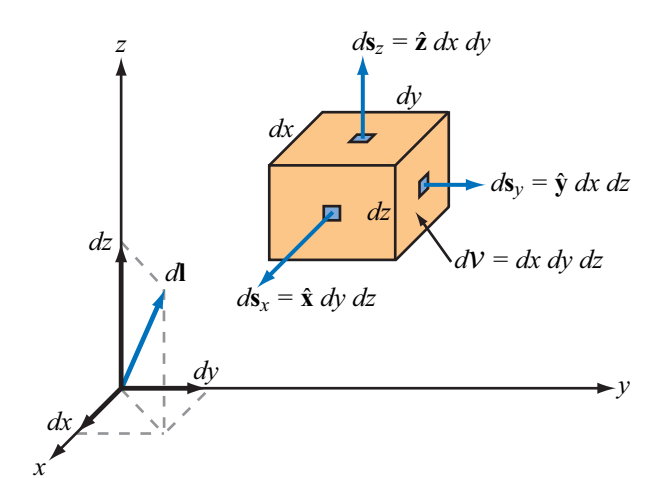


图 3-8 笛卡尔坐标系中的长度、面积和体积微分。

Table 3-1 Summary of vector relations.

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of $\mathbf{A}$ $ \mathbf{A} =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1,$ for $P(x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1,$ for $P(r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1,$ for $P(R_1, \theta_1, \phi_1)$
Base vector properties	$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}} \cdot \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A} \cdot \mathbf{B} =$	$A_xB_x + A_yB_y + A_zB_z$	$A_rB_r + A_\phi B_\phi + A_zB_z$	$A_RB_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A} \times \mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz$	$\hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz$	$\hat{\mathbf{R}} dR + \hat{\boldsymbol{\theta}} R d\theta + \hat{\boldsymbol{\phi}} R \sin \theta d\phi$
Differential surface areas	$ds_x = \hat{\mathbf{x}} dy dz$ $ds_y = \hat{\mathbf{y}} dx dz$ $ds_z = \hat{\mathbf{z}} dx dy$	$ds_r = \hat{\mathbf{r}} r d\phi dz$ $ds_\phi = \hat{\boldsymbol{\phi}} dr dz$ $ds_z = \hat{\mathbf{z}} r dr d\phi$	$ds_R = \hat{\mathbf{R}} R^2 \sin \theta d\theta d\phi$ $ds_\theta = \hat{\boldsymbol{\theta}} R \sin \theta dR d\phi$ $ds_\phi = \hat{\boldsymbol{\phi}} R dR d\theta$
Differential volume $d\mathcal{V} =$	$dx dy dz$	$r dr d\phi dz$	$R^2 \sin \theta dR d\theta d\phi$

direction (such as $\hat{\mathbf{x}}$). Thus, for a differential area vector in the y – z plane,

$$ds_x = \hat{\mathbf{x}} dl_y dl_z = \hat{\mathbf{x}} dy dz \quad (\text{y–}z \text{ plane}), \tag{3.35a}$$

with the subscript on ds denoting its direction. Similarly,

$$ds_y = \hat{\mathbf{y}} dx dz \quad (\text{x–}z \text{ plane}), \tag{3.35b}$$

$$ds_z = \hat{\mathbf{z}} dx dy \quad (\text{x–}y \text{ plane}). \tag{3.35c}$$

A **differential volume** equals the product of all three differential lengths:

$$d\mathcal{V} = dx dy dz. \tag{3.36}$$

3-2.2 Cylindrical Coordinates

The cylindrical coordinate system is useful for solving problems involving structures with cylindrical symmetry, such as calculating the capacitance per unit length of a coaxial transmission line. In the cylindrical coordinate system, the location of a point in space is defined by three variables: r ,

ϕ , and z (**Fig. 3-9**). The coordinate r is the **radial distance** in the x – y the **azimuth angle** measured from the positive x axis, and z is as previously defined in the Cartesian coordinate system. Their ranges are $0 \leq r < \infty$, $0 \leq \phi < 2\pi$, and $-\infty < z < \infty$. Point $P(r_1, \phi_1, z_1)$ in **Fig. 3-9** is located at the intersection of three surfaces. These are the cylindrical surface defined by $r = r_1$, the vertical half-plane defined by $\phi = \phi_1$ (which extends outwardly from the z axis), and the horizontal plane defined by $z = z_1$.

► The mutually perpendicular base vectors are $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\phi}}$, and $\hat{\mathbf{z}}$ with $\hat{\mathbf{r}}$ pointing away from the origin along r , $\hat{\boldsymbol{\phi}}$ pointing in a direction tangential to the cylindrical surface, and $\hat{\mathbf{z}}$ pointing along the vertical. Unlike the Cartesian system, where base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ are independent of the location of P , both $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$ are functions of ϕ in the cylindrical system. ◀

表 3-1 向量关系汇总。

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of $\mathbf{A}$ $ \mathbf{A} =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1,$ for $P(x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1,$ for $P(r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1,$ for $P(R_1, \theta_1, \phi_1)$
Base vector properties	$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}} \cdot \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A} \cdot \mathbf{B} =$	$A_xB_x + A_yB_y + A_zB_z$	$A_rB_r + A_\phi B_\phi + A_zB_z$	$A_RB_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A} \times \mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz$	$\hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz$	$\hat{\mathbf{R}} dR + \hat{\boldsymbol{\theta}} R d\theta + \hat{\boldsymbol{\phi}} R \sin \theta d\phi$
Differential surface areas	$ds_x = \hat{\mathbf{x}} dy dz$ $ds_y = \hat{\mathbf{y}} dx dz$ $ds_z = \hat{\mathbf{z}} dx dy$	$ds_r = \hat{\mathbf{r}} r d\phi dz$ $ds_\phi = \hat{\boldsymbol{\phi}} dr dz$ $ds_z = \hat{\mathbf{z}} r dr d\phi$	$ds_R = \hat{\mathbf{R}} R^2 \sin \theta d\theta d\phi$ $ds_\theta = \hat{\boldsymbol{\theta}} R \sin \theta dR d\phi$ $ds_\phi = \hat{\boldsymbol{\phi}} R dR d\theta$
Differential volume $d\mathcal{V} =$	$dx dy dz$	$r dr d\phi dz$	$R^2 \sin \theta dR d\theta d\phi$

方向（例如 $\hat{\mathbf{x}}$ ）。因此，对于 y - z 平面上的微分面积矢量，

$$ds_x = \hat{\mathbf{x}} dl_y dl_z = \hat{\mathbf{x}} dy dz \quad (\text{y–}z \text{ 平面}), \tag{3.35a}$$

ds 上的下标表示其方向。相似地，

$$ds_y = \hat{\mathbf{y}} dx dz \quad (\text{x–}z \text{ plane}), \tag{3.35b}$$

$$ds_z = \hat{\mathbf{z}} dx dy \quad (\text{x–}y \text{ plane}). \tag{3.35c}$$

微分体积等于所有三个微分长度的乘积：

$$d\mathcal{V} = dx dy dz. \tag{3.36}$$

3-2.2 圆柱坐标

柱坐标系可用于解决涉及柱对称结构的问题，例如计算同轴传输线单位长度的电容。在柱坐标系中，空间中点的位置由三个变量定义： r 、

ϕ 、 z （图3-9）。坐标 r 是 x - y 方向上的径向距离，即从正 x 轴测量的方位角， z 是先前在笛卡尔坐标系中定义的。它们的范围是 $0 \leq r < \infty$ 、 $0 \leq \phi < 2\pi$ 和 $-\infty < z < \infty$ 。图3-9中的点 $P(r_1, \phi_1, z_1)$ 位于三个曲面的交点处。它们是由 $r = r_1$ 定义的圆柱面，由从 z 轴) 向外延伸的 $\phi = \phi_1$ (定义的垂直半平面，以及由 $z = z_1$ 定义的水平面。

► 相互垂直的基向量为 $\hat{\mathbf{r}}$ 、 $\hat{\boldsymbol{\phi}}$ 和 $\hat{\mathbf{z}}$ ，其中 $\hat{\mathbf{r}}$ 沿 r 远离原点， $\hat{\boldsymbol{\phi}}$ 指向与圆柱面相切的方向， $\hat{\mathbf{z}}$ 沿垂直方向指向。与笛卡尔坐标系不同，在笛卡尔坐标系中，基向量 $\hat{\mathbf{x}}$ 、 $\hat{\mathbf{y}}$ 和 $\hat{\mathbf{z}}$ 与 P 的位置无关，而 $\hat{\mathbf{r}}$ 和 $\hat{\boldsymbol{\phi}}$ 都是圆柱坐标系中 ϕ 的函数。 ◀

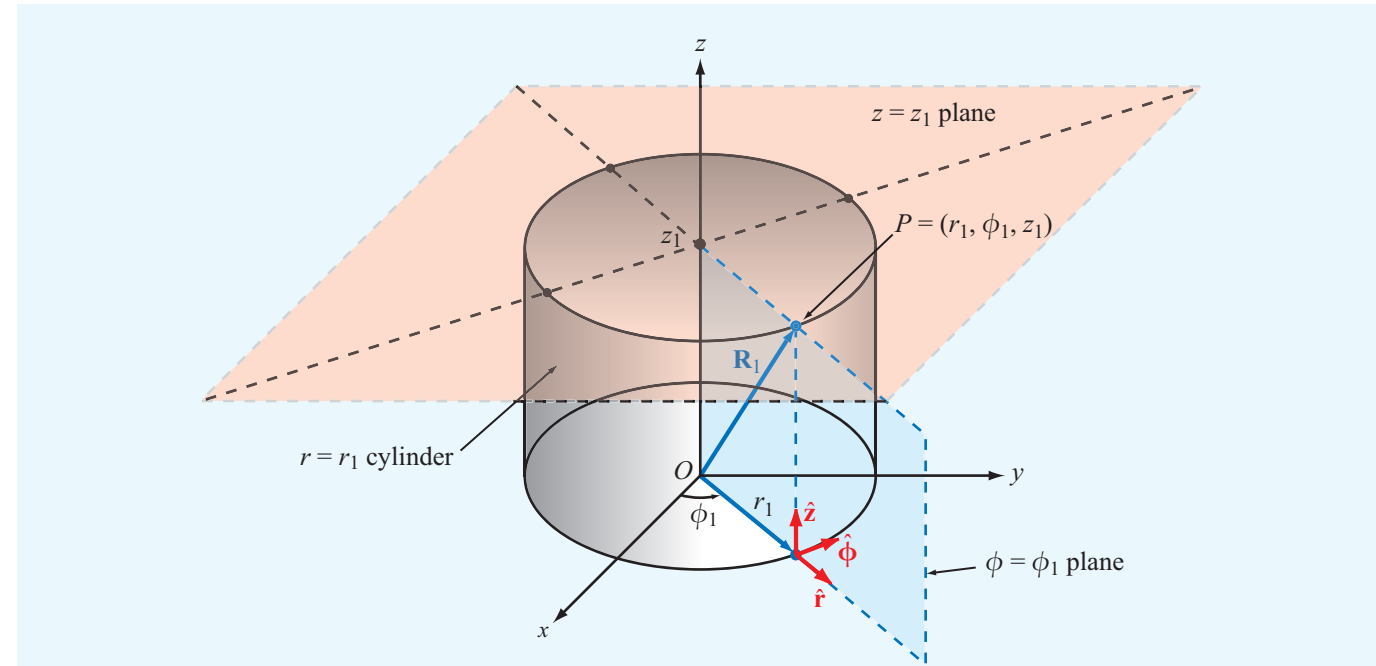


Figure 3-9 Point $P(r_1, \phi_1, z_1)$ in cylindrical coordinates; r_1 is the radial distance from the origin in the x - y plane, ϕ_1 is the angle in the x - y plane measured from the x axis toward the y axis, and z_1 is the vertical distance from the x - y plane.

The base unit vectors obey the following right-hand cyclic relations:

$$\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}, \quad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}, \quad \hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}, \quad (3.37)$$

and like all unit vectors, $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$, and $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0$.

In cylindrical coordinates, a vector is expressed as

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z, \quad (3.38)$$

where A_r , A_ϕ , and A_z are the components of $\mathbf{A}$ along the $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\phi}}$, and $\hat{\mathbf{z}}$ directions. The magnitude of $\mathbf{A}$ is obtained by applying Eq. (3.17), which gives

$$|\mathbf{A}| = \sqrt{\mathbf{A} \cdot \mathbf{A}} = \sqrt{A_r^2 + A_\phi^2 + A_z^2}. \quad (3.39)$$

The position vector $\vec{OP}$ shown in **Fig. 3-9** has components along r and z only. Thus,

$$\mathbf{R}_1 = \vec{OP} = \hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1. \quad (3.40)$$

The dependence of $\mathbf{R}_1$ on ϕ_1 is implicit through the dependence of $\hat{\mathbf{r}}$ on ϕ_1 . Hence, when using Eq. (3.40) to denote the

position vector of point $P(r_1, \phi_1, z_1)$, it is necessary to specify that $\hat{\mathbf{r}}$ is at ϕ_1 .

Figure 3-10 shows a differential volume element in cylindrical coordinates. The differential lengths along $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\phi}}$, and $\hat{\mathbf{z}}$ are

$$dl_r = dr, \quad dl_\phi = r d\phi, \quad dl_z = dz. \quad (3.41)$$

Note that the differential length along $\hat{\boldsymbol{\phi}}$ is $r d\phi$, not just $d\phi$. The differential length $d\mathbf{l}$ in cylindrical coordinates is given by

$$d\mathbf{l} = \hat{\mathbf{r}} dl_r + \hat{\boldsymbol{\phi}} dl_\phi + \hat{\mathbf{z}} dl_z = \hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz. \quad (3.42)$$

As was stated previously for the Cartesian coordinate system, the product of any pair of differential lengths is equal to the magnitude of a vector differential surface area with a surface normal pointing along the direction of the third coordinate. Thus,

$$ds_r = \hat{\mathbf{r}} dl_\phi dl_z = \hat{\mathbf{r}} r d\phi dz \quad (\phi\text{-}z \text{ cylindrical surface}), \quad (3.43a)$$

$$ds_\phi = \hat{\boldsymbol{\phi}} dl_r dl_z = \hat{\boldsymbol{\phi}} dr dz \quad (r\text{-}z \text{ plane}), \quad (3.43b)$$

$$ds_z = \hat{\mathbf{z}} dl_r dl_\phi = \hat{\mathbf{z}} r dr d\phi \quad (r\text{-}\phi \text{ plane}). \quad (3.43c)$$

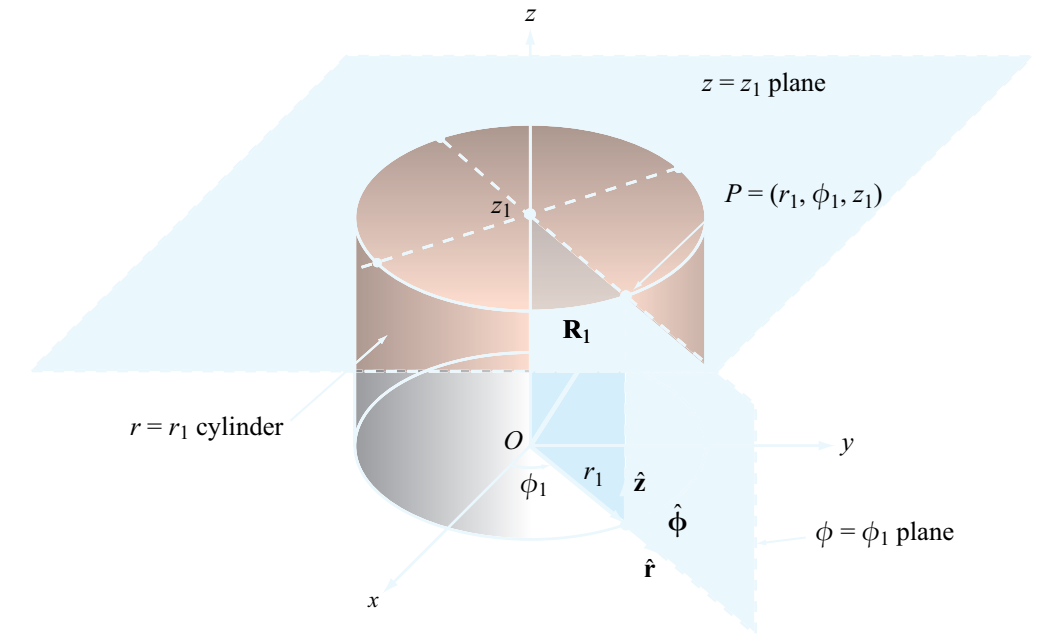


图3-9 柱坐标中的点 $P(r_1, \phi_1, z_1)$; r_1 是 x - y 平面中距原点的径向距离, ϕ_1 是 x - y 平面中从 x 轴朝向 y 轴测量的角度, z_1 是距 x - y 平面的垂直距离。

基本单位向量遵循以下右手循环关系:

$$\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}, \quad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}, \quad \hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}, \quad (3.37)$$

与所有单位向量一样, $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ 和 $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0$ 。

在柱坐标系中, 向量表示为

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z, \quad (3.38)$$

其中 A_r 、 A_ϕ 和 A_z 是 $\mathbf{A}$ 沿 $\hat{\mathbf{r}}$ 、 $\hat{\boldsymbol{\phi}}$ 和 $\hat{\mathbf{z}}$ 方向的分量。 $\mathbf{A}$ 的大小可通过应用方程式获得。(3.17), 给出

$$|\mathbf{A}| = \sqrt{\mathbf{A} \cdot \mathbf{A}} = \sqrt{A_r^2 + A_\phi^2 + A_z^2}. \quad (3.39)$$

图 3-9 所示的位置向量 $\vec{OP}$ 仅具有沿 r 和 z 的分量。因此,

$$\mathbf{R}_1 = \vec{OP} = \hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1. \quad (3.40)$$

$\mathbf{R}_1$ 对 ϕ_1 的依赖是通过 $\hat{\mathbf{r}}$ 对 ϕ_1 的依赖来隐含的。因此, 当使用方程式时, (3.40) 表示

点 $P(r_1, \phi_1, z_1)$ 的位置向量, 需要指定 $\hat{\mathbf{r}}$ 在 ϕ_1 处。

图 3-10 显示了柱坐标系中的微分体积单元。沿 $\hat{\mathbf{r}}$ 、 $\hat{\boldsymbol{\phi}}$ 和 $\hat{\mathbf{z}}$ 的差分长度为

$$dl_r = dr, \quad dl_\phi = r d\phi, \quad dl_z = dz. \quad (3.41)$$

请注意, 沿 $\hat{\boldsymbol{\phi}}$ 的差分长度是 $r d\phi$, 而不仅仅是 $d\phi$ 。柱坐标系中的微分长度 $d\mathbf{l}$ 由下式给出

$$d\mathbf{l} = \hat{\mathbf{r}} dl_r + \hat{\boldsymbol{\phi}} dl_\phi + \hat{\mathbf{z}} dl_z = \hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz. \quad (3.42)$$

正如前面针对笛卡尔坐标系所述, 任何一对微分长度的乘积都等于向量微分表面积的大小, 其中表面法线指向第三坐标的方向。因此,

$$ds_r = \hat{\mathbf{r}} dl_\phi dl_z = \hat{\mathbf{r}} r d\phi dz \quad (\phi\text{-}z \text{ cylindrical surface}), \quad (3.43a)$$

$$ds_\phi = \hat{\boldsymbol{\phi}} dl_r dl_z = \hat{\boldsymbol{\phi}} dr dz \quad (r\text{-}z \text{ plane}), \quad (3.43b)$$

$$ds_z = \hat{\mathbf{z}} dl_r dl_\phi = \hat{\mathbf{z}} r dr d\phi \quad (r\text{-}\phi \text{ plane}). \quad (3.43c)$$

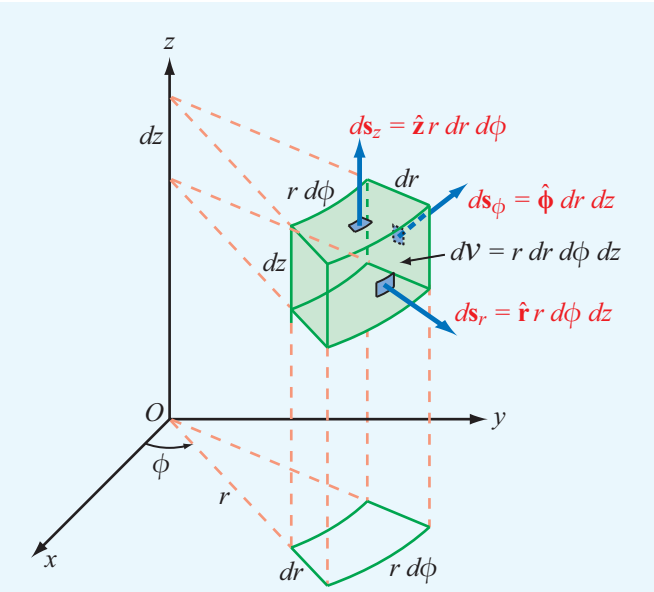


Figure 3-10 Differential areas and volume in cylindrical coordinates.

The differential volume is the product of the three differential lengths,

$$dV = dl_r dl_\phi dl_z = r dr d\phi dz. \quad (3.44)$$

These properties of the cylindrical coordinate system are summarized in Table 3-1.

Example 3-4: Distance Vector in Cylindrical Coordinates

Find an expression for the unit vector of vector **A** shown in Fig. 3-11 in cylindrical coordinates.

Solution: In triangle OP_1P_2 ,

$$\overrightarrow{OP_2} = \overrightarrow{OP_1} + \mathbf{A}.$$

Hence,

$$\mathbf{A} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = \hat{\mathbf{r}}r_0 - \hat{\mathbf{z}}h,$$

and

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\hat{\mathbf{r}}r_0 - \hat{\mathbf{z}}h}{\sqrt{r_0^2 + h^2}}.$$

We note that the expression for **A** is independent of ϕ_0 . This implies that all vectors from point P_1 to any point on the circle defined by $r = r_0$ in the x - y plane are equal in the cylindrical

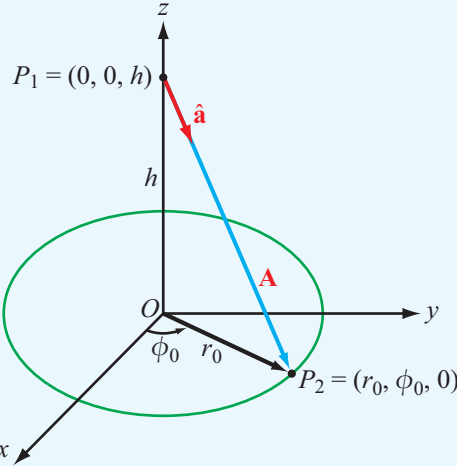


Figure 3-11 Geometry of Example 3-4.

coordinate system, which is not true. The ambiguity can be resolved by specifying that **A** passes through a point whose $\phi = \phi_0$.

Example 3-5: Cylindrical Area

Find the area of a cylindrical surface described by $r = 5$, $30^\circ \leq \phi \leq 60^\circ$, and $0 \leq z \leq 3$ (Fig. 3-12).

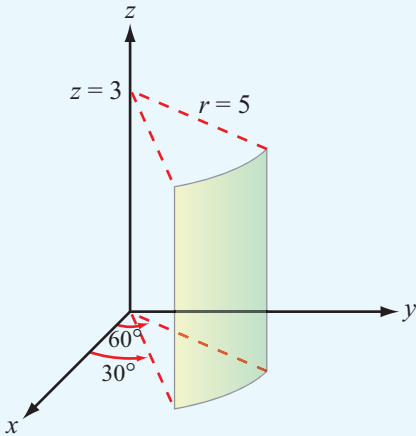


Figure 3-12 Cylindrical surface of Example 3-5.

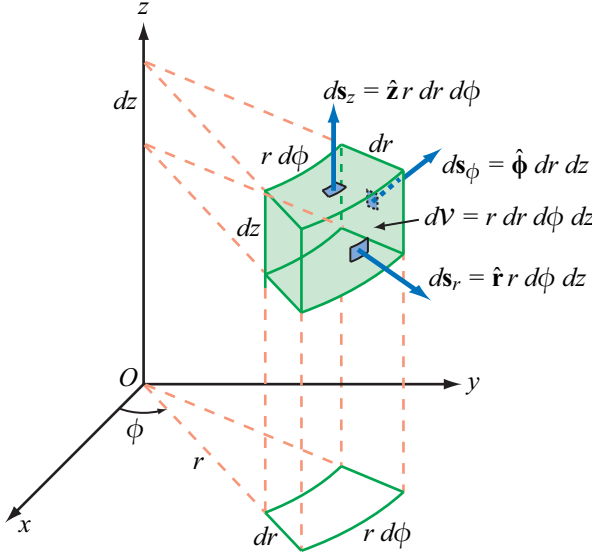


图 3-10 柱坐标系中的微分面积和体积。

微分体积是三个微分长度的乘积，

$$dV = dl_r dl_\phi dl_z = r dr d\phi dz. \quad (3.44)$$

表 3-1 总结了柱坐标系的这些属性。

示例 3-4：柱坐标系中的距离向量

求图 3-11 所示矢量 **A** 的单位矢量在柱坐标系中的表达式。

解：在三角形 OP_1P_2 中，

$$\overrightarrow{OP_2} = \overrightarrow{OP_1} + \mathbf{A}.$$

因此，

$$\mathbf{A} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = \hat{\mathbf{r}}r_0 - \hat{\mathbf{z}}h,$$

和

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\hat{\mathbf{r}}r_0 - \hat{\mathbf{z}}h}{\sqrt{r_0^2 + h^2}}.$$

我们注意到 **A** 的表达式独立于 ϕ_0 。这意味着从点 P_1 到 x - y 平面中由 $r = r_0$ 定义的任何点的所有向量在圆柱中都相等

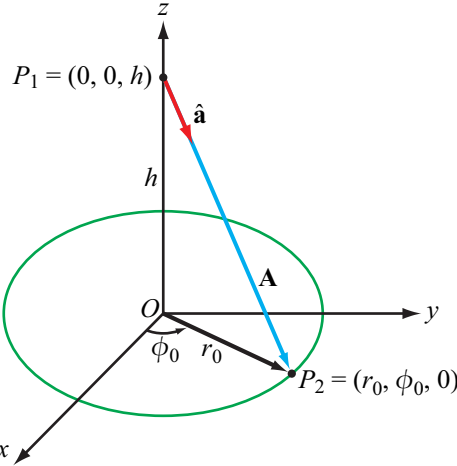


图 3-11 示例 3-4 的几何形状。

坐标系，这是不正确的。可以通过指定 **A** 经过其 $\phi = \phi_0$ 的点来解决歧义。

例 3-5：圆柱面积

求 $r = 5$ 、 $30^\circ \leq \phi \leq 60^\circ$ 和 $0 \leq z \leq 3$ 描述的圆柱表面的面积（图 3-12）。

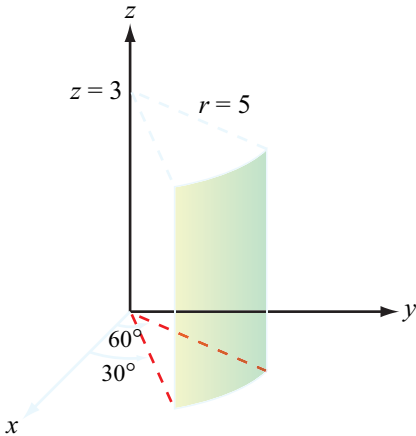
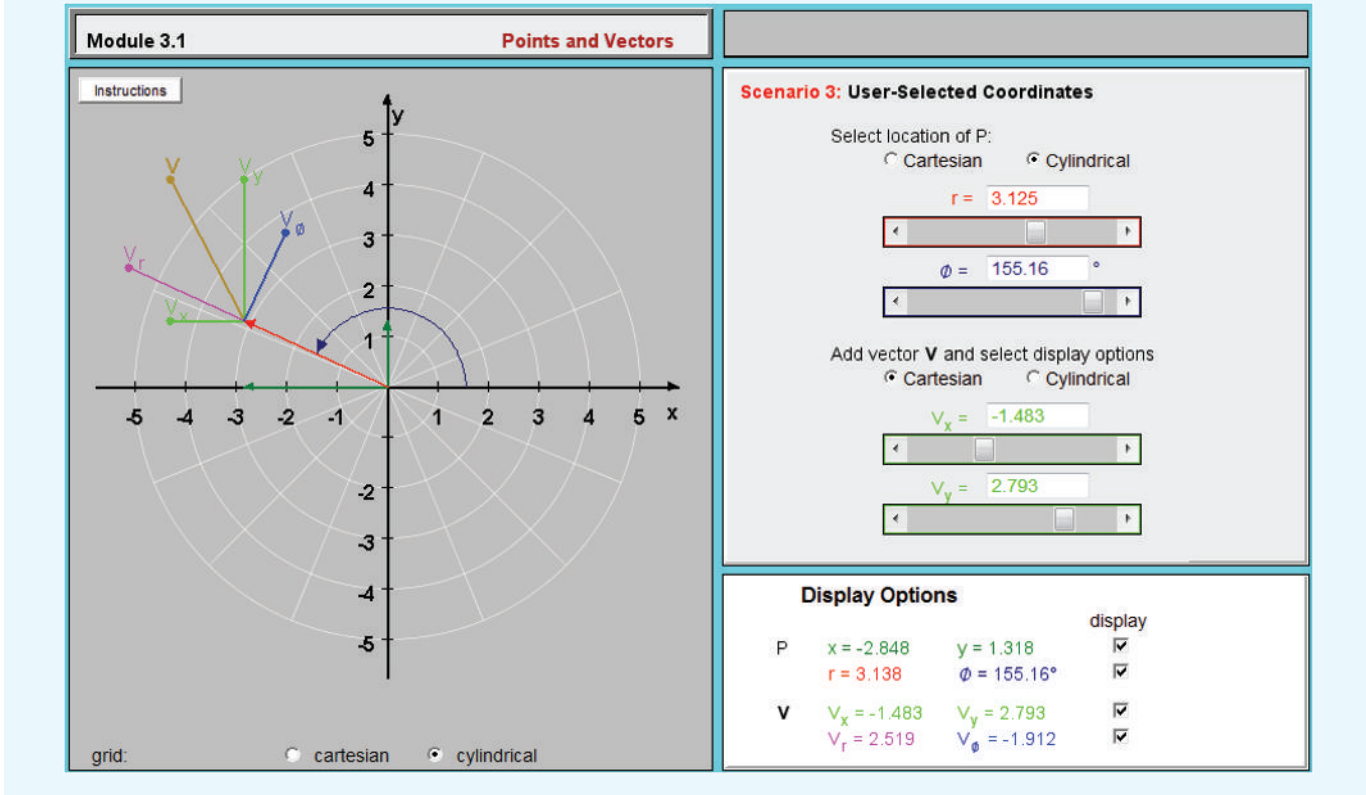


图3-12 例3-5的圆柱面。

Module 3.1 Points and Vectors Examine the relationships between Cartesian coordinates (x, y) and cylindrical coordinates (r, ϕ) for points and vectors.



Solution: The prescribed surface is shown in Fig. 3-12. Use of Eq. (3.43a) for a surface element with constant r gives

$$S = r \int_{\phi=30^\circ}^{60^\circ} d\phi \int_{z=0}^3 dz = 5\phi \Big|_{\pi/6}^{\pi/3} z \Big|_0^3 = \frac{5\pi}{2}.$$

Note that ϕ had to be converted to radians before evaluating the integration limits.

Exercise 3-6: A circular cylinder of radius $r = 5$ cm is concentric with the z axis and extends between $z = -3$ cm and $z = 3$ cm. Use Eq. (3.44) to find the cylinder's volume.
Answer: 471.2 cm³. (See [EM](#).)

3-2.3 Spherical Coordinates

In the spherical coordinate system, the location of a point in space is uniquely specified by the variables R , θ , and ϕ (Fig. 3-13). The range coordinate R , which measures the

distance from the origin to the point, describes a sphere of radius R centered at the origin. The **zenith angle** θ is measured from the positive z axis and it describes a conical surface with its apex at the origin, and the azimuth angle ϕ is the same as in cylindrical coordinates. The ranges of R , θ , and ϕ are $0 \leq R < \infty$, $0 \leq \theta \leq \pi$, and $0 \leq \phi < 2\pi$. The base vectors $\hat{\mathbf{R}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ obey the right-hand cyclic relations:

$$\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}, \quad \hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}, \quad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}. \quad (3.45)$$

A vector with components A_R , A_θ , and A_ϕ is written as

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi, \quad (3.46)$$

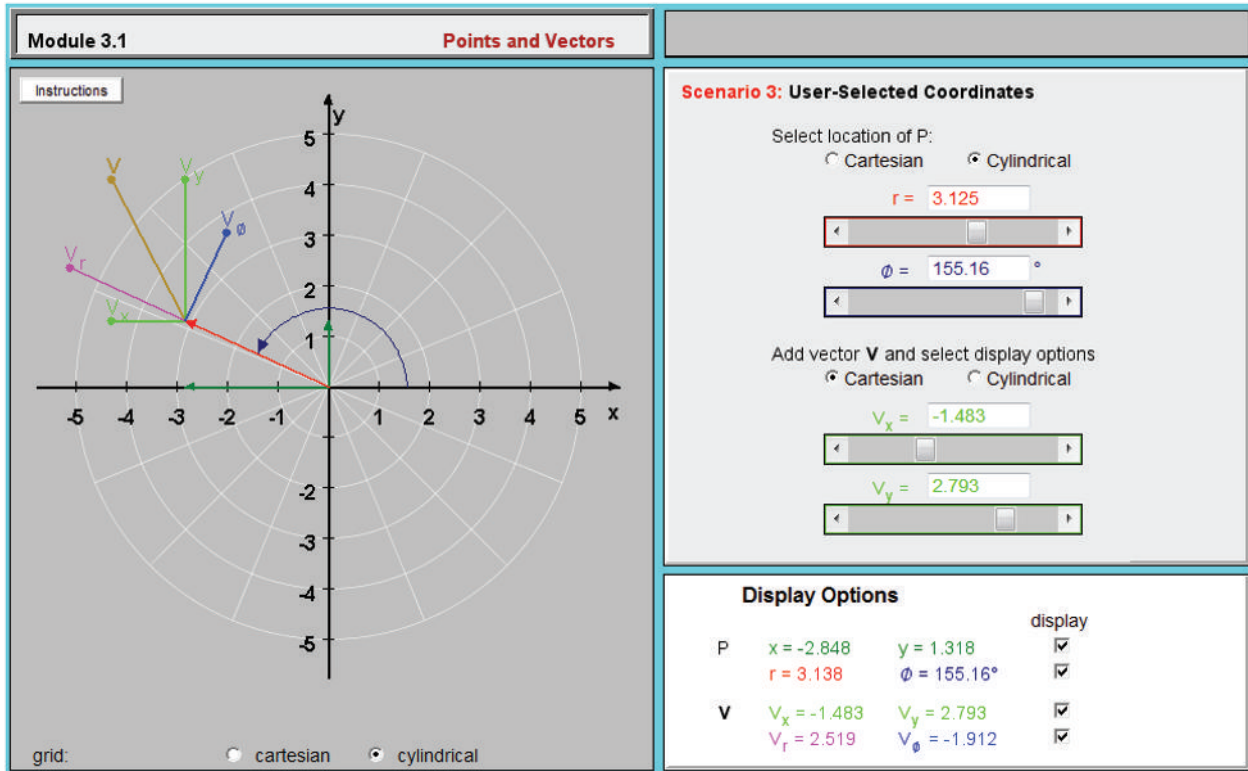
and its magnitude is

$$|\mathbf{A}| = \sqrt{\mathbf{A} \cdot \mathbf{A}} = \sqrt{A_R^2 + A_\theta^2 + A_\phi^2}. \quad (3.47)$$

The position vector of point $P(R_1, \theta_1, \phi_1)$ is simply

$$\mathbf{R}_1 = \overrightarrow{OP} = \hat{\mathbf{R}}R_1, \quad (3.48)$$

模块 3.1 点和向量 检查点和向量的笛卡尔坐标 (x, y) 和柱坐标 (r, ϕ) 之间的关系。



解：规定曲面如图3-12所示。使用等式。(3.43a) 对于具有常数 r 的表面元素，给出

$$S = r \int_{\phi=30^\circ}^{60^\circ} d\phi \int_{z=0}^3 dz = 5\phi \Big|_{\pi/6}^{\pi/3} z \Big|_0^3 = \frac{5\pi}{2}.$$

请注意，在评估积分限制之前，必须将 ϕ 转换为弧度。

练习 3-6：半径为 $r = 5$ cm 的圆柱体与 z 轴同心，并在 $z = -3$ cm 和 $z = 3$ cm 之间延伸。使用方程式。(3.44) 求圆柱体的体积。

答案：471.2 厘米³。（参见^{EM}。）

3-2.3 球坐标

在球坐标系中，空间中点的位置由变量 R 、 θ 和 ϕ （唯一指定。3-13）。范围坐标 R ，测量

从原点到该点的距离，描述了以原点为中心、半径为 R 的球体。天顶角 θ 是从 z 轴正方向测量的，它描述了一个以原点为顶点的圆锥面，而方位角 ϕ 与柱坐标系中的相同。 R 、 θ 和 ϕ 的范围是 $0 \leq R < \infty$ 、 $0 \leq \theta \leq \pi$ 和 $0 \leq \phi < 2\pi$ 。基向量 $\hat{\mathbf{R}}$ 、 $\hat{\boldsymbol{\theta}}$ 和 $\hat{\boldsymbol{\phi}}$ 服从右手循环关系：

$$\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}, \quad \hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}, \quad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}. \quad (3.45)$$

具有分量 A_R 、 A_θ 和 A_ϕ 的向量可写为

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi, \quad (3.46)$$

它的大小是

$$|\mathbf{A}| = \sqrt{\mathbf{A} \cdot \mathbf{A}} = \sqrt{A_R^2 + A_\theta^2 + A_\phi^2}. \quad (3.47)$$

点 $P(R_1, \theta_1, \phi_1)$ 的位置向量就是

$$\mathbf{R}_1 = \overrightarrow{OP} = \hat{\mathbf{R}}R_1, \quad (3.48)$$

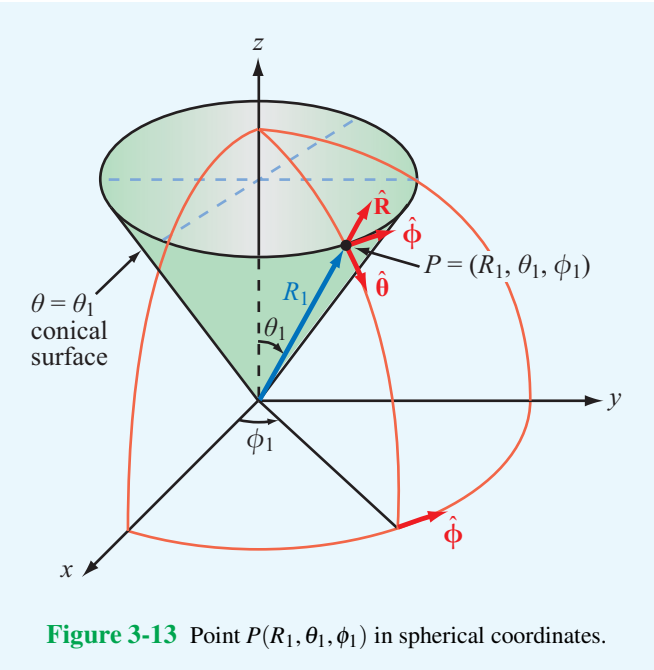


Figure 3-13 Point $P(R_1, \theta_1, \phi_1)$ in spherical coordinates.

As shown in Fig. 3-14, the differential lengths along $\hat{\mathbf{R}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ are

$$dl_R = dR, \quad dl_\theta = R d\theta, \quad dl_\phi = R \sin \theta d\phi. \quad (3.49)$$

Hence, the expressions for the vector differential length $d\mathbf{l}$, the vector differential surface $d\mathbf{s}$, and the differential volume dV are

$$\begin{aligned} d\mathbf{l} &= \hat{\mathbf{R}} dl_R + \hat{\boldsymbol{\theta}} dl_\theta + \hat{\boldsymbol{\phi}} dl_\phi \\ &= \hat{\mathbf{R}} dR + \hat{\boldsymbol{\theta}} R d\theta + \hat{\boldsymbol{\phi}} R \sin \theta d\phi, \end{aligned} \quad (3.50a)$$

$$\begin{aligned} d\mathbf{s}_R &= \hat{\mathbf{R}} dl_\theta dl_\phi = \hat{\mathbf{R}} R^2 \sin \theta d\theta d\phi \\ &\quad (\theta\text{-}\phi \text{ spherical surface}), \end{aligned} \quad (3.50b)$$

$$\begin{aligned} d\mathbf{s}_\theta &= \hat{\boldsymbol{\theta}} dl_R dl_\phi = \hat{\boldsymbol{\theta}} R \sin \theta dR d\phi \\ &\quad (R\text{-}\phi \text{ conical surface}), \end{aligned} \quad (3.50c)$$

$$d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}} dl_R dl_\theta = \hat{\boldsymbol{\phi}} R dR d\theta \quad (R\text{-}\theta \text{ plane}), \quad (3.50d)$$

$$dV = dl_R dl_\theta dl_\phi = R^2 \sin \theta dR d\theta d\phi. \quad (3.50e)$$

These relations are summarized in Table 3-1.

Example 3-6: Surface Area in Spherical Coordinates

The spherical strip shown in Fig. 3-15 is a section of a sphere of radius 3 cm. Find the area of the strip.

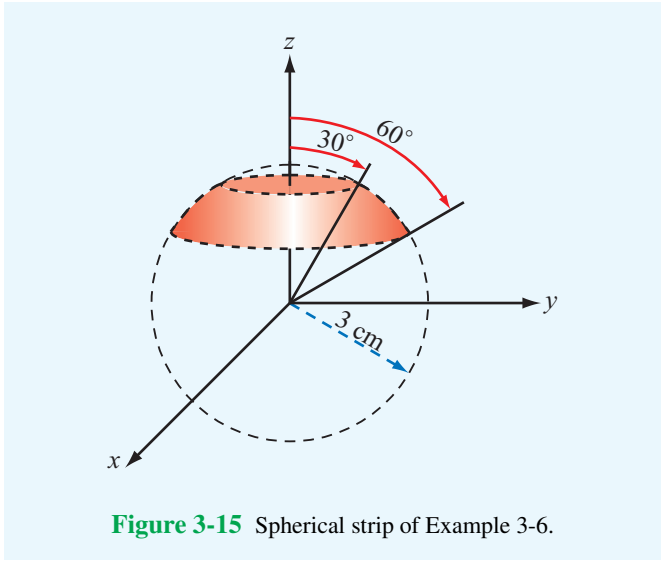


Figure 3-15 Spherical strip of Example 3-6.

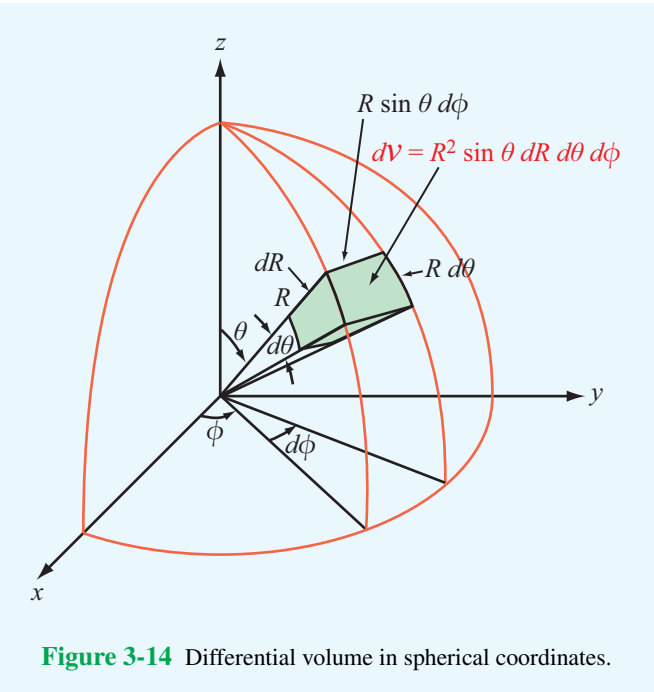


Figure 3-14 Differential volume in spherical coordinates.

while keeping in mind that $\hat{\mathbf{R}}$ is implicitly dependent on θ_1 and ϕ_1 .

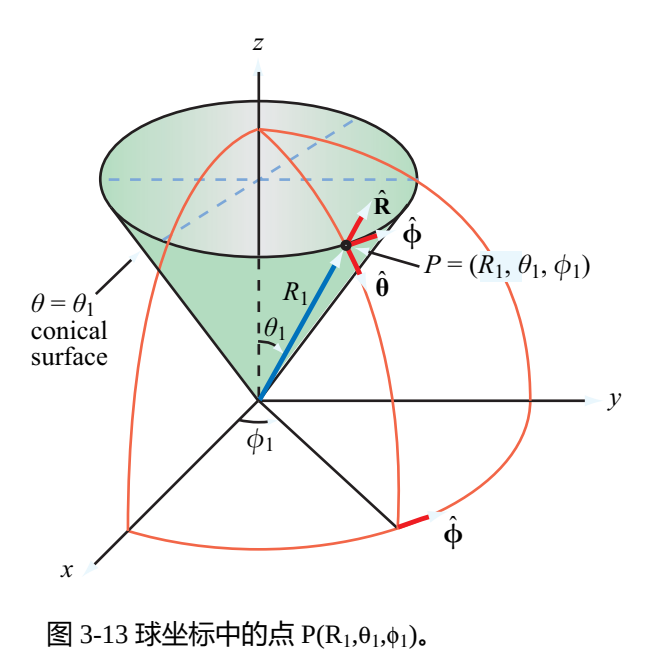


图 3-13 球坐标中的点 $P(R_1, \theta_1, \phi_1)$ 。

如图3-14所示，沿 $\hat{\mathbf{R}}$ 、 $\hat{\boldsymbol{\theta}}$ 和 $\hat{\boldsymbol{\phi}}$ 的差分长度为

$$dl_R = dR, \quad dl_\theta = R d\theta, \quad dl_\phi = R \sin \theta d\phi. \quad (3.49)$$

因此，矢量微分长度 $d\mathbf{l}$ 、矢量微分面 $d\mathbf{s}$ 和微分体积 dV 的表达式为

$$\begin{aligned} d\mathbf{l} &= \hat{\mathbf{R}} dl_R + \hat{\boldsymbol{\theta}} dl_\theta + \hat{\boldsymbol{\phi}} dl_\phi \\ &= \hat{\mathbf{R}} dR + \hat{\boldsymbol{\theta}} R d\theta + \hat{\boldsymbol{\phi}} R \sin \theta d\phi, \end{aligned} \quad (3.50a)$$

$$\begin{aligned} d\mathbf{s}_R &= \hat{\mathbf{R}} dl_\theta dl_\phi = \hat{\mathbf{R}} R^2 \sin \theta d\theta d\phi \\ &\quad (\theta\text{-}\phi \text{ spherical surface}), \end{aligned} \quad (3.50b)$$

$$\begin{aligned} d\mathbf{s}_\theta &= \hat{\boldsymbol{\theta}} dl_R dl_\phi = \hat{\boldsymbol{\theta}} R \sin \theta dR d\phi \\ &\quad (R\text{-}\phi \text{ conical surface}), \end{aligned} \quad (3.50c)$$

$$d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}} dl_R dl_\theta = \hat{\boldsymbol{\phi}} R dR d\theta \quad (R\text{-}\theta \text{ plane}), \quad (3.50d)$$

$$dV = dl_R dl_\theta dl_\phi = R^2 \sin \theta dR d\theta d\phi. \quad (3.50e)$$

表 3-1 总结了这些关系。

示例 3-6：球坐标中的表面积

图3-15所示的球形条是半径为3厘米的球体的截面。求该条带的面积。

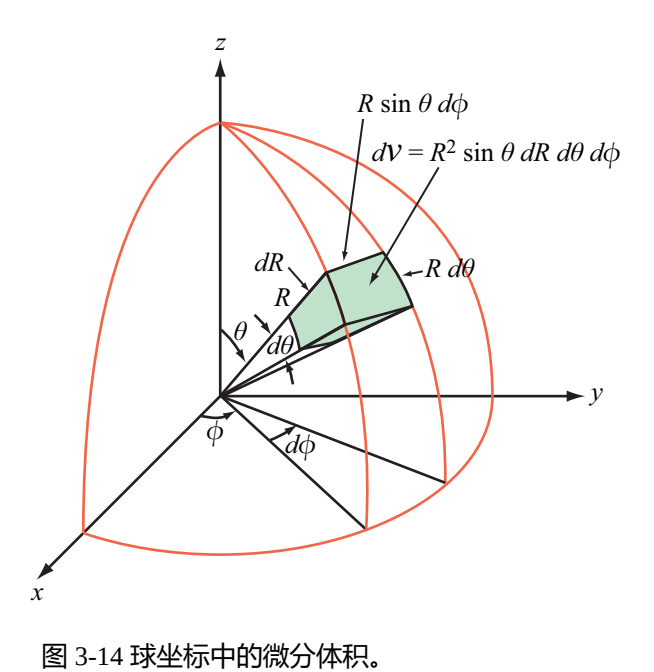


图 3-14 球坐标中的微分体积。

同时请记住， $\hat{\mathbf{R}}$ 隐式依赖于 θ_1 和 ϕ_1 。

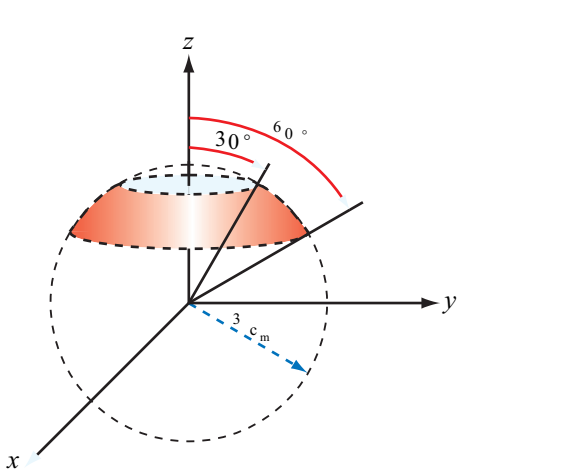


图3-15 实施例3-6的球形条。

Solution: Use of Eq. (3.50b) for the area of an elemental spherical area with constant radius R gives

$$\begin{aligned} S &= R^2 \int_{\theta=30^\circ}^{60^\circ} \sin \theta \, d\theta \int_{\phi=0}^{2\pi} d\phi \\ &= 9(-\cos \theta) \Big|_{30^\circ}^{60^\circ} \phi \Big|_0^{2\pi} \quad (\text{cm}^2) \\ &= 18\pi(\cos 30^\circ - \cos 60^\circ) = 20.7 \text{ cm}^2. \end{aligned}$$

Example 3-7: Charge in a Sphere

A sphere of radius 2 cm contains a volume charge density ρ_v given by

$$\rho_v = 4 \cos^2 \theta \quad (\text{C/m}^3).$$

Find the total charge Q contained in the sphere.

Solution:

$$\begin{aligned} Q &= \int_v \rho_v \, dV \\ &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{R=0}^{2 \times 10^{-2}} (4 \cos^2 \theta) R^2 \sin \theta \, dR \, d\theta \, d\phi \\ &= 4 \int_0^{2\pi} \int_0^{\pi} \left(\frac{R^3}{3} \right) \Big|_0^{2 \times 10^{-2}} \sin \theta \cos^2 \theta \, d\theta \, d\phi \\ &= \frac{32}{3} \times 10^{-6} \int_0^{2\pi} \left(-\frac{\cos^3 \theta}{3} \right) \Big|_0^{\pi} d\phi \\ &= \frac{64}{9} \times 10^{-6} \int_0^{2\pi} d\phi \\ &= \frac{128\pi}{9} \times 10^{-6} = 44.68 \quad (\mu\text{C}). \end{aligned}$$

Note that the limits on R were converted to meters prior to evaluating the integral on R .

3-3 Transformations between Coordinate Systems

The position of a given point in space of course does not depend on the choice of coordinate system. That is, its location is the same irrespective of which specific coordinate system is used to represent it. The same is true for vectors. Nevertheless, certain coordinate systems may be more useful than others in solving a given problem, so it is essential that we have the tools to “translate” the problem from one system to another. In this

section, we shall establish the relations between the variables (x, y, z) of the Cartesian system, (r, ϕ, z) of the cylindrical system, and (R, θ, ϕ) of the spherical system. These relations will then be used to transform expressions for vectors expressed in any one of the three systems into expressions applicable in the other two.

3-3.1 Cartesian to Cylindrical Transformations

Point P in **Fig. 3-16** has Cartesian coordinates (x, y, z) and cylindrical coordinates (r, ϕ, z) . Both systems share the coordinate z , and the relations between the other two pairs of coordinates can be obtained from the geometry in **Fig. 3-16**. They are

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \left(\frac{y}{x} \right), \quad (3.51)$$

and the inverse relations are

$$x = r \cos \phi, \quad y = r \sin \phi. \quad (3.52)$$

Next, with the help of **Fig. 3-17**, which shows the directions of the unit vectors $\hat{x}$, $\hat{y}$, $\hat{r}$, and $\hat{\phi}$ in the x - y plane, we obtain the relations:

$$\hat{r} \cdot \hat{x} = \cos \phi, \quad \hat{r} \cdot \hat{y} = \sin \phi, \quad (3.53a)$$

$$\hat{\phi} \cdot \hat{x} = -\sin \phi, \quad \hat{\phi} \cdot \hat{y} = \cos \phi. \quad (3.53b)$$

To express $\hat{r}$ in terms of $\hat{x}$ and $\hat{y}$, we write $\hat{r}$ as

$$\hat{r} = \hat{x}a + \hat{y}b, \quad (3.54)$$

where a and b are unknown transformation coefficients. The dot product $\hat{r} \cdot \hat{x}$ gives

$$\hat{r} \cdot \hat{x} = \hat{x} \cdot \hat{x}a + \hat{y} \cdot \hat{x}b = a. \quad (3.55)$$

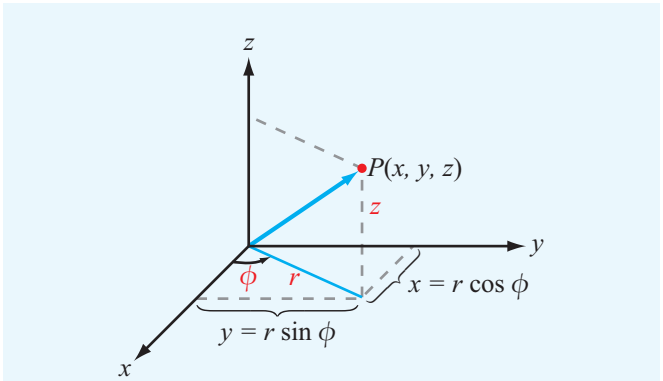


Figure 3-16 Interrelationships between Cartesian coordinates (x, y, z) and cylindrical coordinates (r, ϕ, z) .

解决方案：使用方程。(3.50b) 对于具有恒定半径 R 的基本球形区域的面积，给出

$$\begin{aligned} S &= R^2 \int_{\theta=30^\circ}^{60^\circ} \sin \theta \, d\theta \int_{\phi=0}^{2\pi} d\phi \\ &= 9(-\cos \theta) \Big|_{30^\circ}^{60^\circ} \phi \Big|_0^{2\pi} \quad (\text{cm}^2) \\ &= 18\pi(\cos 30^\circ - \cos 60^\circ) = 20.7 \text{ cm}^2. \end{aligned}$$

例 3-7：球体中的电荷

半径为 2 cm 的球体包含的体积电荷密度 ρ_v 为

$$\rho_v = 4 \cos^2 \theta \quad (\text{C/m}^3).$$

求球体中所含的总电荷 Q 。

解决方案：

$$\begin{aligned} Q &= \int_v \rho_v \, dV \\ &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{R=0}^{2 \times 10^{-2}} (4 \cos^2 \theta) R^2 \sin \theta \, dR \, d\theta \, d\phi \\ &= 4 \int_0^{2\pi} \int_0^{\pi} \left(\frac{R^3}{3} \right) \Big|_0^{2 \times 10^{-2}} \sin \theta \cos^2 \theta \, d\theta \, d\phi \\ &= \frac{32}{3} \times 10^{-6} \int_0^{2\pi} \left(-\frac{\cos^3 \theta}{3} \right) \Big|_0^{\pi} d\phi \\ &= \frac{64}{9} \times 10^{-6} \int_0^{2\pi} d\phi \\ &= \frac{128\pi}{9} \times 10^{-6} = 44.68 \quad (\mu\text{C}). \end{aligned}$$

请注意，在评估 R 的积分之前， R 的限值已转换为米。

3-3 坐标系之间的变换

当然，空间中给定点的位置不取决于坐标系的选择。也就是说，无论使用哪种特定坐标系来表示它，它的位置都是相同的。对于向量也是如此。然而，在解决给定问题时，某些坐标系可能比其他坐标系更有用，因此我们拥有将问题从一个系统“转换”到另一个系统的工具至关重要。在这个

在本节中，我们将建立笛卡尔系统的变量 (x, y, z) 、圆柱系统的变量 (r, ϕ, z) 和球面系统的变量 (R, θ, ϕ) 之间的关系。然后，这些关系将用于将三个系统中任何一个系统中表达的向量表达式转换为适用于其他两个系统的表达式。

3-3.1 笛卡尔到圆柱的变换

图3-16中的点P有笛卡尔坐标 (x, y, z) 和柱坐标 (r, ϕ, z) 。两个系统共享坐标 z ，另外两对坐标之间的关系可以从图3-16的几何结构中得到。他们是

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \left(\frac{y}{x} \right), \quad (3.51)$$

反比关系是

$$x = r \cos \phi, \quad y = r \sin \phi. \quad (3.52)$$

接下来，借助图 3-17（显示 x - y 平面中单位向量 $\hat{x}$ 、 $\hat{y}$ 、 $\hat{r}$ 和 $\hat{\phi}$ 的方向），我们得到以下关系：

$$\hat{r} \cdot \hat{x} = \cos \phi, \quad \hat{r} \cdot \hat{y} = \sin \phi, \quad (3.53a)$$

$$\hat{\phi} \cdot \hat{x} = -\sin \phi, \quad \hat{\phi} \cdot \hat{y} = \cos \phi. \quad (3.53b)$$

为了用 $\hat{x}$ 和 $\hat{y}$ 表示 $\hat{r}$ ，我们将 $\hat{r}$ 写为

$$\hat{r} = \hat{x}a + \hat{y}b, \quad (3.54)$$

其中 a 和 b 是未知的变换系数。点积 $\hat{r} \cdot \hat{x}$ 给出

$$\hat{r} \cdot \hat{x} = \hat{x} \cdot \hat{x}a + \hat{y} \cdot \hat{x}b = a. \quad (3.55)$$

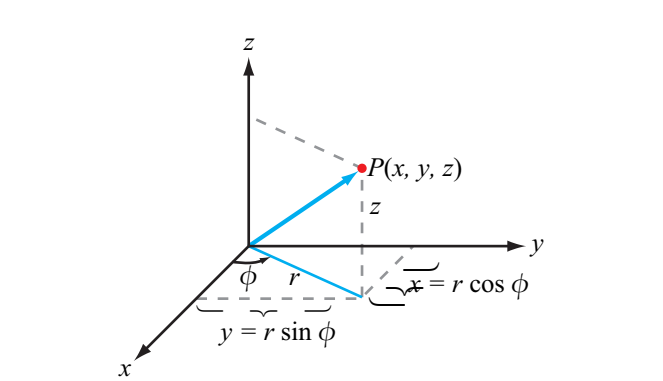


图 3-16 笛卡尔坐标 (x, y, z) 和柱坐标 (r, ϕ, z) 之间的相互关系。

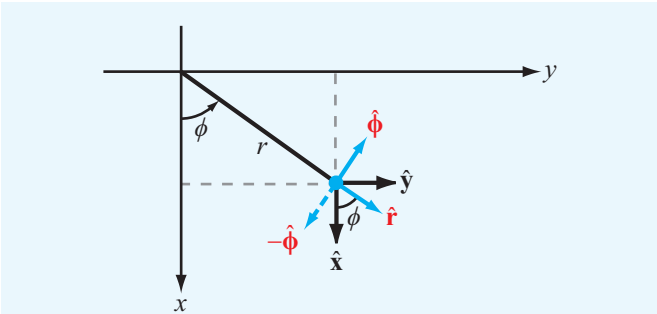


Figure 3-17 Interrelationships between base vectors $(\hat{x}, \hat{y})$ and $(\hat{r}, \hat{\phi})$.

Comparison of Eq. (3.55) with Eq. (3.53a) yields $a = \cos \phi$. Similarly, application of the dot product $\hat{r} \cdot \hat{y}$ to Eq. (3.54) gives $b = \sin \phi$. Hence,

$$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi. \quad (3.56a)$$

Repetition of the procedure for $\hat{\phi}$ leads to

$$\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi. \quad (3.56b)$$

The third base vector $\hat{z}$ is the same in both coordinate systems. By solving Eqs. (3.56a) and (3.56b) simultaneously for $\hat{x}$ and $\hat{y}$, we obtain the following inverse relations:

$$\hat{x} = \hat{r} \cos \phi - \hat{\phi} \sin \phi, \quad (3.57a)$$

$$\hat{y} = \hat{r} \sin \phi + \hat{\phi} \cos \phi. \quad (3.57b)$$

The relations given by Eqs. (3.56a) to (3.57b) are not only useful for transforming the base vectors $(\hat{x}, \hat{y})$ into $(\hat{r}, \hat{\phi})$, and vice versa, they also can be used to transform the components of a vector expressed in either coordinate system into its corresponding components expressed in the other system. For example, a vector $\mathbf{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$ in Cartesian coordinates can be described by $\mathbf{A} = \hat{r}A_r + \hat{\phi}A_\phi + \hat{z}A_z$ in cylindrical coordinates by applying Eqs. (3.56a) and (3.56b). That is,

$$A_r = A_x \cos \phi + A_y \sin \phi, \quad (3.58a)$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi, \quad (3.58b)$$

and, conversely,

$$A_x = A_r \cos \phi - A_\phi \sin \phi, \quad (3.59a)$$

$$A_y = A_r \sin \phi + A_\phi \cos \phi. \quad (3.59b)$$

The transformation relations given in this and the following two subsections are summarized in **Table 3-2**.

Example 3-8: Cartesian to Cylindrical Transformations

Given point $P_1 = (3, -4, 3)$ and vector $\mathbf{A} = \hat{x}2 - \hat{y}3 + \hat{z}4$ defined in Cartesian coordinates, express P_1 and $\mathbf{A}$ in cylindrical coordinates and evaluate $\mathbf{A}$ at P_1 .

Solution: For point P_1 , $x = 3$, $y = -4$, and $z = 3$. Using Eq. (3.51), we have

$$r = \sqrt{x^2 + y^2} = 5, \quad \phi = \tan^{-1} \frac{y}{x} = -53.1^\circ = 306.9^\circ,$$

and z remains unchanged. Hence, $P_1 = P_1(5, 306.9^\circ, 3)$ in cylindrical coordinates.

The cylindrical components of vector $\mathbf{A} = \hat{r}A_r + \hat{\phi}A_\phi + \hat{z}A_z$ can be determined by applying Eqs. (3.58a) and (3.58b):

$$A_r = A_x \cos \phi + A_y \sin \phi = 2 \cos \phi - 3 \sin \phi,$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi = -2 \sin \phi - 3 \cos \phi,$$

$$A_z = 4.$$

Hence,

$$\mathbf{A} = \hat{r}(2 \cos \phi - 3 \sin \phi) - \hat{\phi}(2 \sin \phi + 3 \cos \phi) + \hat{z}4.$$

At point P , $\phi = 306.9^\circ$, which gives

$$\mathbf{A} = \hat{r} 3.60 - \hat{\phi} 0.20 + \hat{z} 4.$$

3-3.2 Cartesian to Spherical Transformations

From **Fig. 3-18**, we obtain the following relations between the Cartesian coordinates (x, y, z) and the spherical coordinates (R, θ, ϕ) :

$$R = \sqrt{x^2 + y^2 + z^2}, \quad (3.60a)$$

$$\theta = \tan^{-1} \left[\frac{\sqrt{x^2 + y^2}}{z} \right], \quad (3.60b)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right). \quad (3.60c)$$

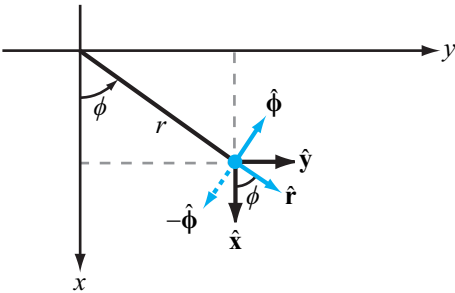


图 3-17 基向量 $(\hat{x}, \hat{y})$ 和 $(\hat{r}, \hat{\phi})$ 之间的相互关系。

等式的比较(3.55) 与等式。 (3.53a) 产生 $a = \cos \phi$ 。类似地，将点积 $\hat{r} \cdot \hat{y}$ 应用于等式： (3.54) 给出 $b = \sin \phi$ 。因此，

$$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi. \quad (3.56a)$$

重复 $\hat{\phi}$ 的过程会导致

$$\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi. \quad (3.56b)$$

第三基向量 $\hat{z}$ 在两个坐标系中是相同的。通过求解方程。(3.56a)和(3.56b)同时对于 $\hat{x}$ 和 $\hat{y}$ ，我们得到以下逆关系：

$$\hat{x} = \hat{r} \cos \phi - \hat{\phi} \sin \phi, \quad (3.57a)$$

$$\hat{y} = \hat{r} \sin \phi + \hat{\phi} \cos \phi. \quad (3.57b)$$

等式给出的关系。(3.56a) 至 (3.57b) 不仅可用于将基向量 $(\hat{x}, \hat{y})$ 转换为 $(\hat{r}, \hat{\phi})$ ，反之亦然，它们还可用于将在任一坐标系中表达的向量分量转换为在另一坐标系中表达的相应分量。例如，笛卡尔坐标中的向量 $\mathbf{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$ 可以通过应用等式 1 来由柱坐标中的 $\mathbf{A} = \hat{r}A_r + \hat{\phi}A_\phi + \hat{z}A_z$ 来描述。(3.56a) 和 (3.56b)。那是，

$$A_r = A_x \cos \phi + A_y \sin \phi, \quad (3.58a)$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi, \quad (3.58b)$$

反之，

$$A_x = A_r \cos \phi - A_\phi \sin \phi, \quad (3.59a)$$

$$A_y = A_r \sin \phi + A_\phi \cos \phi. \quad (3.59b)$$

表 3-2 总结了本节和以下两小节中给出的变换关系。

示例 3-8：笛卡尔坐标到圆柱坐标变换

给定笛卡尔坐标中定义的点 $P_1 = (3, -4, 3)$ 和向量 $\mathbf{A} = \hat{x}2 - \hat{y}3 + \hat{z}4$ ，用柱坐标表示 P_1 和 $\mathbf{A}$ ，并在 P_1 处计算 $\mathbf{A}$ 。

解：对于点 P_1 、 $x = 3$ 、 $y = -4$ 和 $z = 3$ 。(3.51)，我们有

$$r = \sqrt{x^2 + y^2} = 5, \quad \phi = \tan^{-1} \frac{y}{x} = -53.1^\circ = 306.9^\circ,$$

且 z 保持不变。因此，柱坐标系中的 $P_1 = P_1(5, 306.9^\circ, 3)$ 。

矢量 $\mathbf{A} = \hat{r}A_r + \hat{\phi}A_\phi + \hat{z}A_z$ 的圆柱分量可以通过应用等式来确定。(3.58a) 和 (3.58b)：

$$A_r = A_x \cos \phi + A_y \sin \phi = 2 \cos \phi - 3 \sin \phi,$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi = -2 \sin \phi - 3 \cos \phi,$$

$$A_z = 4.$$

因此，

$$\mathbf{A} = \hat{r}(2 \cos \phi - 3 \sin \phi) - \hat{\phi}(2 \sin \phi + 3 \cos \phi) + \hat{z}4.$$

在点 P ， $\phi = 306.9^\circ$ ，这给出

$$\mathbf{A} = \hat{r} 3.60 - \hat{\phi} 0.20 + \hat{z} 4.$$

3-3.2 笛卡尔到球面的变换

从图3-18，我们得到笛卡尔坐标 (x, y, z) 和球坐标 (R, θ, ϕ) 之间的关系：

$$R = \sqrt{x^2 + y^2 + z^2}, \quad (3.60a)$$

$$\theta = \tan^{-1} \left[\frac{\sqrt{x^2 + y^2}}{z} \right], \quad (3.60b)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right). \quad (3.60c)$$

Table 3-2 Coordinate transformation relations.

Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{\mathbf{r}} = \hat{\mathbf{x}} \cos \phi + \hat{\mathbf{y}} \sin \phi$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{\mathbf{x}} = \hat{\mathbf{r}} \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi$ $\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{\mathbf{R}} = \hat{\mathbf{x}} \sin \theta \cos \phi + \hat{\mathbf{y}} \sin \theta \sin \phi + \hat{\mathbf{z}} \cos \theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}} \cos \theta \cos \phi + \hat{\mathbf{y}} \cos \theta \sin \phi - \hat{\mathbf{z}} \sin \theta$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{\mathbf{x}} = \hat{\mathbf{R}} \sin \theta \cos \phi + \hat{\boldsymbol{\theta}} \cos \theta \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi$ $\hat{\mathbf{y}} = \hat{\mathbf{R}} \sin \theta \sin \phi + \hat{\boldsymbol{\theta}} \cos \theta \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{\mathbf{R}} = \hat{\mathbf{r}} \sin \theta + \hat{\mathbf{z}} \cos \theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{r}} \cos \theta - \hat{\mathbf{z}} \sin \theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{\mathbf{r}} = \hat{\mathbf{R}} \sin \theta + \hat{\boldsymbol{\theta}} \cos \theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$ $\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$

表3-2 坐标变换关系。

Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{\mathbf{r}} = \hat{\mathbf{x}} \cos \phi + \hat{\mathbf{y}} \sin \phi$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{\mathbf{x}} = \hat{\mathbf{r}} \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi$ $\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{\mathbf{R}} = \hat{\mathbf{x}} \sin \theta \cos \phi + \hat{\mathbf{y}} \sin \theta \sin \phi + \hat{\mathbf{z}} \cos \theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}} \cos \theta \cos \phi + \hat{\mathbf{y}} \cos \theta \sin \phi - \hat{\mathbf{z}} \sin \theta$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{\mathbf{x}} = \hat{\mathbf{R}} \sin \theta \cos \phi + \hat{\boldsymbol{\theta}} \cos \theta \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi$ $\hat{\mathbf{y}} = \hat{\mathbf{R}} \sin \theta \sin \phi + \hat{\boldsymbol{\theta}} \cos \theta \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{\mathbf{R}} = \hat{\mathbf{r}} \sin \theta + \hat{\mathbf{z}} \cos \theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{r}} \cos \theta - \hat{\mathbf{z}} \sin \theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{\mathbf{r}} = \hat{\mathbf{R}} \sin \theta + \hat{\boldsymbol{\theta}} \cos \theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$ $\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$

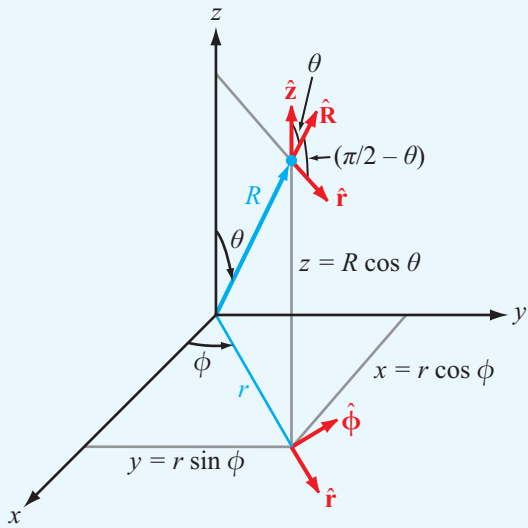


Figure 3-18 Interrelationships between (x, y, z) and (R, θ, ϕ) .

The converse relations are

$$x = R \sin \theta \cos \phi, \quad (3.61a)$$

$$y = R \sin \theta \sin \phi, \quad (3.61b)$$

$$z = R \cos \theta. \quad (3.61c)$$

The unit vector $\hat{\mathbf{R}}$ lies in the $\hat{\mathbf{r}}\text{--}\hat{\mathbf{z}}$ plane. Hence, it can be expressed as a linear combination of $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$ as

$$\hat{\mathbf{R}} = \hat{\mathbf{r}}a + \hat{\mathbf{z}}b, \quad (3.62)$$

where a and b are transformation coefficients. Since $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$ are mutually perpendicular,

$$\hat{\mathbf{R}} \cdot \hat{\mathbf{r}} = a, \quad (3.63a)$$

$$\hat{\mathbf{R}} \cdot \hat{\mathbf{z}} = b. \quad (3.63b)$$

From Fig. 3-18, the angle between $\hat{\mathbf{R}}$ and $\hat{\mathbf{r}}$ is the complement of θ and that between $\hat{\mathbf{R}}$ and $\hat{\mathbf{z}}$ is θ . Hence, $a = \hat{\mathbf{R}} \cdot \hat{\mathbf{r}} = \sin \theta$

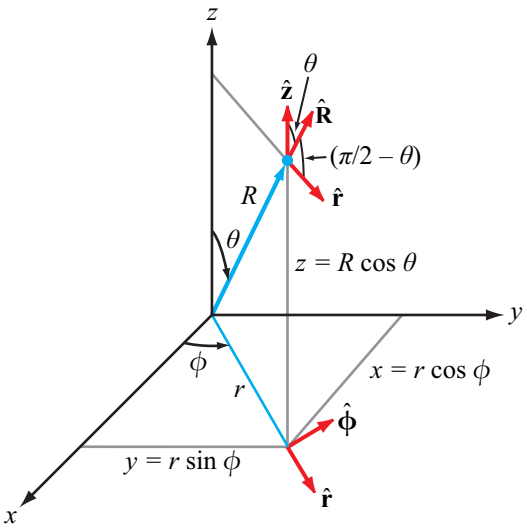


图 3-18 (x, y, z) 和 (R, θ, ϕ) 之间的相互关系。

其逆关系为

$$x = R \sin \theta \cos \phi, \quad (3.61a)$$

$$y = R \sin \theta \sin \phi, \quad (3.61b)$$

$$z = R \cos \theta. \quad (3.61c)$$

单位向量 $\hat{\mathbf{R}}$ 位于 $\hat{\mathbf{r}}\text{--}\hat{\mathbf{z}}$ 平面。因此，它可以表示为 $\hat{\mathbf{r}}$ 和 $\hat{\mathbf{z}}$ 的线性组合：

$$\hat{\mathbf{R}} = \hat{\mathbf{r}}a + \hat{\mathbf{z}}b, \quad (3.62)$$

其中 a 和 b 是变换系数。由于 $\hat{\mathbf{r}}$ 和 $\hat{\mathbf{z}}$ 相互垂直，

$$\hat{\mathbf{R}} \cdot \hat{\mathbf{r}} = a, \quad (3.63a)$$

$$\hat{\mathbf{R}} \cdot \hat{\mathbf{z}} = b. \quad (3.63b)$$

从图3-18可知， $\hat{\mathbf{R}}$ 和 $\hat{\mathbf{r}}$ 之间的角度是 θ 的补角， $\hat{\mathbf{R}}$ 和 $\hat{\mathbf{z}}$ 之间的角度是 θ 。因此， $a = \hat{\mathbf{R}} \cdot \hat{\mathbf{r}} = \sin \theta$